3D-PRINTED PROSTHETIC ARM WITH EMG-BASED CONTROL AND SENSORY FEEDBACK

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DECLARATION OF ORIGINALITY

I hereby declare that this project report is based on my original work except for citations and references which have been duly acknowledged. I also declare that it had not been previously and concurrently submitted for any other degree of award at APU or other institution.

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APPROVAL FORM

I certify that this project report entitled **3D-PRINTED PROSTHETIC ARM WITH EMG-BASED CONTROL AND SENSORY FEEDBACK** was prepared by **ISAMELDIN AHMED HUSSEIN AHMED** with **TP069561** has met the required standards for submission in partial fulfilment of the requirements for the award of Bachelor of Engineering (Hons) in Mechatronics Engineering at the Asia Pacific University of Technology and Innovation.

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3D-PRINTED PROSTHETIC ARM WITH EMG-BASED CONTROL AND SENSORY FEEDBACK

Abstract

This project aims to improve the quality of life of upper-limb amputees. It presents the design and implementation of an electromyography (EMG)-controlled prosthetic arm that is integrated with sensory feedback for improved functional performance. The system is made to capture the users EMG signals from their muscles, processes them and then maps them to servo motor movement commands to achieve finger flexion and extension for grasping and releasing actions. To develop a multi-servo prosthetic that can replicate basic human motions and grasp patterns the EXIII's Hackberry prosthetic arm is used a base design and its design is adapted and improved to provide wrist and elbow movement as added features. The methodology combined the use of control, mechanical and electrical systems and main control of the system is handled by the ESP32 to control actuator motion and sensory feedback. Results show that the prosthetic system works effectively and is aligned with high accuracy and repeatability as well as feedback given from sensory data to the user. Overall, the project achieved its main objectives in alignment with EMG control, through structural and mechanical improvements for stronger structural rigidity are necessary for a fully capable system to take out into the real world.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

SYMBOL, ABBREVIATION OR NOMENCLATURE	FULL MEANING
A	Ampere
ABS	Acrylonitrile Butadiene Styrene
AC	Alternating Current
ADC	Analog-to-Digital Converter
BEM	Board of Engineers Malaysia
DC	Direct Current
EMG	Electromyography
ESP32	Microcontroller used for control and signal processing
GND	Ground
GPIO	General Purpose Input/Output
Hz	Hertz
mV	Millivolt
μV	Microvolt
NTC	Negative Temperature Coefficient (Thermistor)
PETG	Polyethylene Terephthalate Glycol
PLA+	Polylactic Acid Plus
PWM	Pulse Width Modulation
RMS	Root Mean Square
sEMG	Surface Electromyography

Tg	Glass Transition Temperature
TPU	Thermoplastic Polyurethane
Vcc	Supply Voltage
W	Watt
WHO	World Health Organization

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CHAPTER 1
INTRODUCTION TO STUDY

1.1 Introduction

As the world advances further and further with technology, the aims globally have remained the same with the main one being to make people's lives easier to live and more convenient. This cannot be more relevant when it comes to those with upper limb amputations. In this case we will specifically be looking at the human arm, one of the most important and key elements for any human to function, survive and go about their daily lives. Many prosthetic solutions exist; however, few strike a balance between cost, ease of use and durability. Populations in developing countries are victims of this due to a lack of access to affordable, robust and user-friendly prosthetics. Furthermore, modern prosthetics rarely offer any sensory feedback to users.

1.2 Research Problems

Many available prosthetics in today's modern world are out of reach to the majority of individuals who need them the most. Developing countries' populations are most notoriously struck by this problem as they are battling disease and a lack of accessible healthcare. To further amplify the problem, most individuals in developing countries lack the financial capability to purchase a prosthetic arm let alone the added maintenance costs and rehabilitation costs as well as added psychological adjustments.

1.3 Aim and Objectives

Aim:

The main aim of the project is to develop a 3D-printed prosthetic arm controlled by electromyographic signals (EMG) and that provides the user with environmental feedback from sensors.

Objectives:

1. Process EMG signals to control the five fingers, palm, and elbow with at least 70% accuracy.
2. Develop a feedback system that allows the user to identify surface roughness (smooth, medium, rough) and temperature (hot, warm, cold).
3. Adapt existing prosthetic arm designs to create a low-cost, 3D-printed model using affordable, off-the-shelf components.
4. Evaluate the prosthetic arm's movement accuracy, feedback performance, and task success rate during simple daily activities.

1.4 Justification for This Research

This project aims to study and improve upon the limitations of current prosthetic arms controlled by EMG signals. Many Amputees cannot afford any form of prosthesis due to their high upfront cost and high maintenance costs, 3D printing significantly reduces this cost and makes it more accessible to such individuals. Furthermore, many prosthetic arms neglect the added psychological benefits and ease of use brought upon with added sensory feedback, users will have the added benefit of the arm feeling as close as possible to a natural limb. The use of EMG signals to control the arm also gives the user more intuitive control and natural motion of the arm as compared to mechanical or switch-based systems.

This research also helps contribute to advancements in more affordable solutions for amputees to help them lead more normal lives. The approach will help fill the large gap in affordable amputation solutions for people from all walks of all regardless of their monetary and financial standing. Lastly it helps address the practical needs of amputees and promotes technology inclusiveness in healthcare.

1.5 Organisation for The Rest of The Chapters

This report contains three chapters that comprise of "Project Phase 1" of my final year project namely:

1. Chapter 1 (Introduction to the study): This chapter outlines the overall structure and skeleton of the project by introducing the topic with its aim and objectives as well as the problem statement and the reasoning for carrying out this project in the first place.
2. Chapter 2 (Literature Review): This chapter consists of the bulk of phase 1. It outlines a review of literature findings from existing research and relevant literature from similar or related projects. The literature in question is extracted and gotten from various published sources to ensure legitimacy and credibility. It is also here that will enable
3. Chapter 3 (Concept Design & Research Methodology): This will act as the project's road map. Here we will proceed with developing the approach and methodology that will be used in the development of the arm and its corresponding systems. The principles and concepts that will be used as well will be highlighted here with the aim of ensuring that the projects objectives are met while adhering to professional engineering practices.

1.6 Summary

Many active prostheses are out of financial reach for most individuals who need them. While low-cost solutions exist, they often fail to mimic human sensory activities making them feel less natural for users. The main objectives of this project are to make a prosthetic arm capable of moving all 5 fingers, palm movement and elbow movement while still having sensory feedback to allow users to get information from their environment and use it like a regular limb. This will use EMG signals to control the arm, the main hurdle here being filtering and processing said signals as EMG signals are only up to a few millivolts. Furthermore, gaining synchronization and low latency within the sensory feedback system will also be a challenge. With all this being said, the prospect of pushing the prosthesis industry forward to gain new knowledge on how to improve amputee's daily lives with such implementations is the main motivator for carrying out this project.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In the biomedical field new advancements in 3D printing have transformed the industry by allowing for lightweight, user customizable and -most importantly- affordable prosthetic designs. In an effort to bridge the gap between life like limbs and robotics, embedded systems have been implemented to prosthetics. These systems allow for a multitude of possibilities for users to interact with said prosthetics. The main methodology and theory I will be exploring here is the use of EMG signals for control of the prosthetic arm. The following is a review that analyses current design methodologies and embedded systems implementations as well as the direction the industry is heading towards. This will be in terms of 3D printed Prosthetics controlled by EMG signals as well as how sensory feedback systems have been implemented. The goal for this research is to carefully analyse findings that can be used to better implement previous solutions for a more natural and intuitive prosthetic arm.

User dissatisfaction and abandonment of upper limb prostheses is significant in todays modern world. These rates are up to 20% in adults and higher in children, often due to poor prosthesis functionality, comfort and usability. Prostheses can be categorized into three main types namely: passive, active and hybrid devices. Each category has its own trade-offs, functions, control method and user experience. Passive devices are comfortable but lack functionality, body powered prosthetics are often reliable and low cost but at the detriment to user health with higher levels of fatigue and limited dexterity. Hybrid devices on the other hand offer more natural movement and power but are expensive and require frequent charging and maintenance. As the human anatomy is complex solutions to it often prove to be the same. Multiple joints and degrees of freedom all add up to complexity with prosthesis further increasing the likely of failure and thus user dissatisfaction. Thus a need for a solution to these problems is paramount to success and retention among users (Brack & Amalu, 2021)

2.2 3D Printing in Prosthetics

3D printing can be a gateway for communities with little to no industrial access to medical facilities and industries capable of producing prosthetics. Through the use of 3D printing waiting times for moulds and their added high cost can significantly be reduced as compared to traditional prosthetic manufacturing while still allowing for the use of modern digital tools. Furthermore, this still allows for the use of advanced materials which help prosthetics lighter, stronger and most notably significantly more affordable as compared to said traditional options. Open-source designs and affordable printers also play a part in this affordability by allowing communities to produce prosthetics locally, even in low-resource areas that don't have access to more advanced manufacturing centres (Borthakur, 2023).

Further studies also indicate the same patterns of an advanced technology, 3D printing via FDM with synthetic plastic materials and filaments can reduce fabrication time, costs and weight compared to laminated methods. The same principles apply here as well, where rapid prototyping of hand or forearm components such as in the InMoov prosthetic models is possible. The patterns at which the print is configured to also play a huge role in the arms overall structural rigidity. Lattice infills help to strike a balance between strength and weight within a prosthetic. Flexible materials like TPU allow for the creation of flexible sockets and interfaces. Computer aided design plays a role in this technology with support for personalization of prosthetics. These aspects of 3D printable prosthetics help extend access to remote areas. However, challenges and doubts still remain over their biocompatibility and durability under humid conditions (Beygi & Wong, 2023).

This trend continues with more studies indicating that 3D Printing is an effective manufacturing method for producing low-cost robotic arm prosthesis prototypes. It enables the rapid fabrication of complex prototype geometries that would be difficult or expensive to produce with traditional manufacturing methods. Material costs are a major factor in any manufacturing industry and the same applies here. Studies have emphasized that material usage is significantly reduced through the use of 3D printing in turn reducing overall production costs which may be then passed down to the user. This method also allows for designs optimized using CAD tools. This enables faster design iterations and customization to user requirements (Salazar et al., 2024).

Filament fabrication parameters for 3D printing have been shown to influence tensile strength and structural rigidity of the final components. Material selection aside, the next most significant variable which plays a part in mechanical performance is simply filament fabrication parameters. These are settings a user inputs into their printer's software right before printing. They dictate the layer bonds, print patterns and infill percentages. These parameters also play a part in printers service life and material costs. As with any manufacturing process, wasted material can significantly drive-up costs. All these aspects play a part in the final mechanical and financial performance of 3D-Printed components (Hamat et al., 2023).

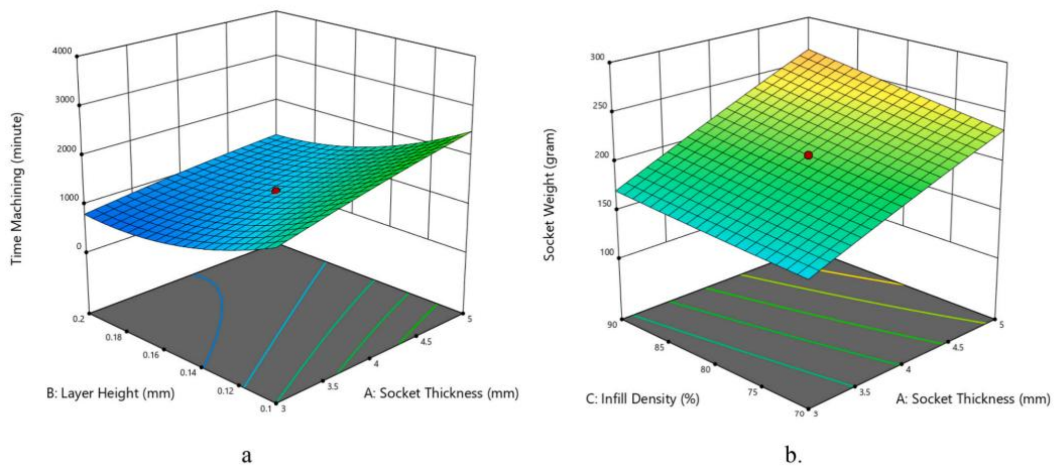


Figure 2.1: Response surface plots of printing time (a) and socket weight (b). (Lestari et al., 2024)

A study focused on fused deposition modelling 3D printing showed how the effects on process parameters effect the efficiency and material usage when manufacturing a Prosthetic socket using PLA as the filament. The main objective here was to analyse 5 specific FDM printing parameters: socket thickness, layer heigh, infill density, print speed and the nozzles temperature. These parameters were chosen to see how they would affect the printing time and socket weight. The parameters and levels tested were:

- Socket thickness: 3 mm, 4 mm, 5 mm
- Layer height: 0.1 mm, 0.15 mm, 0.3 mm
- Infill density: 80%, 90%, 100%

- Print speed: 70 mm/s, 80 mm/s, 90 mm/s
- Nozzle temperature: 190 °C, 200 °C, 210 °C

The findings from this study revealed that an optimized combination of these printing parameters would significantly reduce printing times and socket weight without a comprise to any fabrication reliability. This showed that an optimization of these parameters improved the manufacturing efficiency in FDM. **Figure 2.1** highlights the relationship between the two. (Lestari et al., 2024)

Material selection plays a key role in prosthesis manufacturing, in the case of 3D printed Manufacturing, thermoplastic and PLA-based filaments are suitable here. Their low cost, ease of printing and adequate mechanical performance is ideal for non-industrial use. However, studies have noted that 3D-printing is suitable for low-load applications and prototyping but for more long-term applications and solutions, material and structural optimizations are necessary for long-term durability and higher load conditions (Salazar et al., 2024).

2.3 EMG Signal acquisition methods

A comparison between different methods of EMG signal acquisition reveals limitation and constraints for each method. A comparison between fine-wire EMG and surface EMG carried out for leg muscles at different walking speeds had 10 healthy adults participate. Different walking speeds were set at slow, user preferred, fast and maximum. The study revealed that surface EMG was reliable mainly for surface level muscles especially at slower speed. On the other hand, fine wire EMG was found to reliable work on deeper muscles within the human body. The study further revealed this results with the EMG signal activity in **Figure 2.2**. (Harrison et al., 2021).

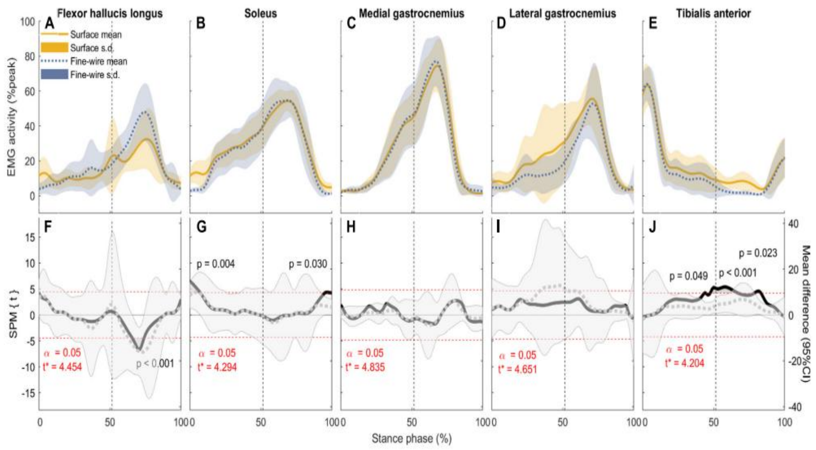


Figure 2.2: EMG activity of each muscle

(Harrison et al., 2021)

Figure 2.2 illustrates the fact that surface EMG presents a key limitation with noise and cross-talk from neighbouring muscles being among them. Surface EMG signals also show poor detection of deep muscles while fine-wire methods convey more accurate information. However, fine-wire presents itself as an invasive procedure with limitations on its use with approved physicians. (Harrison et al., 2021).

Method	Advantages	Disadvantages
Surface EMG	Non-invasive, low-cost, easy setup	Crosstalk, artifacts, lower selectivity
Invasive EMG	High SNR, selective, deep access	Surgical risk, discomfort, time-intensive

Table 2.1: EMG Signal acquisition methods comparison

Further studies used different approaches to test this problem. One study on the review of EMG signals and how they are collected and processed. It outlined 3 methods to acquire EMG signals. Surface EMG, fine wire and a high-density configuration of surface EMG. The results for the latter two were similar as those found in other studies, however the addition of multiple electrodes for a high-density configuration showed an increase in spatial resolution. The same challenges persisted however with noise and cross talk being present. (Sul et al., 2025)

To combat apparent and lingering issues in EMG signal acquisition, researchers created a signal preprocessing flow. This included the use of filters, namely high-pass and low-pass as well as the addition of notch filters to remove noise and interference. Rectification and smoothing were also applied to the signal. Feature extraction was also carried out to the signals using root mean square and zero crossing and the extracted featured were used to control and exoskeleton and the prediction of motor control. The key takeaways from this that proper acquisition of EMG signals, filtering and feature extraction were critical for accurate and rapid response exoskeleton control. These methods can then in turn be applied to prosthetics as both work on the same basic principle of EMG control. (Sul et al., 2025)

2.4 EMG Control Systems

A review of EMG signal process highlights many constraints. The main being the core processor of said signals. The embedded system of an EMG based prosthetic will have to carry out various signal conditioning stages before any meaningful analysis can be done on the signals. The signals are often noisy and non-stationary requiring amplification, filtering, rectification and normalization. The next critical step upon signal conditioning completion is feature extraction. This creates meaning from raw signals into compact and discriminative representations which are suitable for embedded processing.

Common time-domain methods such as zero crossing and root mean square are used in EMG control systems due to their low computational complexity and fast execution time. However, these methods do not provide information rich signals. Instead, frequency-domain and time-frequency features such as those derived from Fourier transforms provide much richer signal information, however, these methods require higher computational power. (Castruita-López et al., 2025)

No signal classifier is deemed to be optimal, instead a trade-off must be made between classification accuracy, computational load and power consumption. A balance must be struck between these three for any successful embedded system prosthesis. Therefore, the choice of the embedded hardware platform becomes paramount to the systems success. Embedded hardware platforms such as microcontrollers, Digital Signal Processors and System-on-Chip architecture are the main options here. **Table 2.2** portrays the various hardware platforms and their suitability for this project. Priority is given to hardware that can strike a balance between cost, compact size and energy efficiency, in this case microcontrollers are the most suitable. A review of EMG signal processing highlighted the importance of real-time processing capabilities. It was noted that excessive latency in EMG signal classification may lead to unintuitive real-world performance of prosthetics in terms of their control. Effective EMG-based embedded systems require a balanced approach, where all aspects such as signal processing, classification algorithms and hardware platforms are chosen with each other in mind rather than each independently. (Castruita-López et al., 2025)

Hardware Platform	**Key Features**	**Accuracy**	**Latency**	**Suitability for Projects**
Microcontrollers	Low-cost, time-domain features (RMS, MAV, WL)	0-95%	0-300 ms	Basic EMG + simple feedback
FPGAs	Parallel SVM/CNN, high-channel support	5-98%	1 ms	Multi-DOF control + sensors
SoCs	Python MLPs/CNNs, portable	4-99%	5-11 ms	Feedback fusion prototypes

Tabel 2.2: Comparison of Hardware Platforms

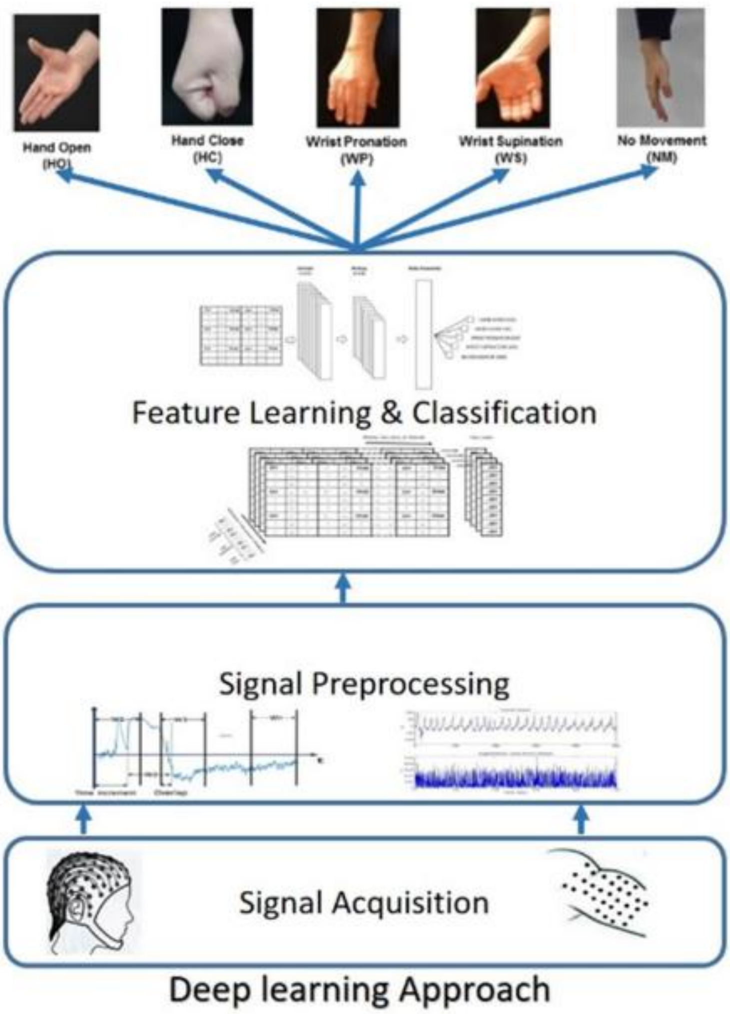


Figure 2.3: Deep learning Approach model for EMG Signal Processing (Aly & Youssef, 2021)

Methods on processing EMG signals have lead researchers to the implementation of deep learning models. Figure 2.3 portrays this with a combination of EEG brain signals with EMG signals to decode hand and wrist movements. It addresses challenges in traditional machine learning approaches. The main challenge with traditional approaches is the need for handcrafted features, however the proposed method in figure 2.3 using deep learning models to extract features directly from raw bioelectrical signals. The dataset used in this case was a mixture of surface EMG signals and multichannel EEG signals where were tied down to hand and wrist motion tasks. This combined approach showed significant improvements in classification accuracy compared to traditional machine learning methods with up to a ~3.5% gain. (Aly & Youssef, 2021)

Another study reviewing surface electromyography monitoring systems in rehabilitation with an emphasis on signal acquisition. This outlined the fundamental biophysics of sEMG signals and the challenges associated with accurate data extraction and processing. These signals capture electrical activity generated from muscle contraction and how signal quality is influenced by noise, electrode placement and anatomical factors. The review highlighted major issues with electrode motion artifacts, signal noise and the lack of any standardized procedures for acquisition of these signals. (Al-Ayyad et al., 2023)

2.5 Lack of Sensory and Haptic Feedback in Prosthetic Hands

The control of Prosthetic limbs and arms is still very challenging even with advanced designs that can replicate finger movements. The main thorn in this is the lack of any haptic feedback or sense of touch. Without these key elements, users cannot feel how rough a surface is or how hot the surface may be for their safety. The absence of these feedback systems limits the dexterity of and natural control that a human arm would normally have. Prosthetic users mainly rely on their vision to control their hands, therefore their movements are less precise and slower than normal limbs. Tasks that usually require great motor skills and dexterity can become nearly impossible to do without some form of sensory feedback. Providing even the simplest form of feedback can improve accuracy, control and confidence within prosthetic users which allows them to perform their daily tasks effectively and safely. With this in mind, improving sensory feedback is a key step towards making prosthetics feel more functional and natural to use (Abd et al., 2022).

Advancements in technology saw researchers implement a hybrid multi model sensory feedback system. This system relied on both haptic force and position feedback for hand prosthesis users. The system enhanced perception beyond user's vision. It integrated mechanotactile feedback, this is where force applied to the user's arm corresponds to prosthetic grip force. The system also integrated vibrotactile feedback where vibration patterns represented specific grip positions. For mechanotactile feedback, force is proportionally applied through a controller using force sensors. This allowed participants to perceive to how much force was necessary to hold an object. In terms of vibrotactile feedback, four vibration motors were added here. The motors would encode four discrete hand positions; open wide grip, narrow grip and closed this enabled accurate recognition of prosthetic arm position (Sariyildiz et al., 2023).

Figure 2.4: Experimental setup of hybrid sensory feedback system (Sariyildiz et al., 2023)

Experiments with participants involved evaluating their ability to perceive force applied varied with force level. The results showed that participants tended to underestimate higher force levels and force perception accuracy was highest at the bicep when it is in a relaxed state. Next, participants were given the task to identify all four grip positions based on vibrational cues with the vibrotactile system. This bolstered good results with participants display strong proprioceptive feedback reliability. The next experiment involved combining both in a hybrid setup. This showed the best results with participants achieving a ~90.5% success rate in identifying objects varying in stiffness and shape. This demonstrated that combined haptic and proprioceptive cues significantly enhanced object property recognition among participants. However, the study also found that no significant effect was caused by the location of said feedback on the accuracy of object recognition accuracy. (Sariyildiz et al., 2023)

Furthermore, implementation of sensory feedback is not without its challenges. Research and design challenges pose a threat to their implementation as developing effective sensory feedback systems that allow amputees to regain tactile perception is not fully realised in commercial prosthetics. Actuator design remains a thorn in the technologies advancement. A lighter actuator often lacks sufficient power while more powerful actuators flip the coin with increased weight and user discomfort. (Brack & Amalu, 2021)

Studies aimed to evaluate this issue and its implications forced user to rely on visual cues for confirmation to grasp objects. This in turn would reduce their dexterity and the arms useability. The solution they proposed was a low-cost, non-invasive vibrotactile sensory feedback system integrated into a 3D-printed myoelectric prosthetic arm. Its design consisted of a model in SolidWorks and 3D-printed parts to match participants with left forearm amputation. Control was provided using a muscle sensor, Arduino microcontroller and servo motors for hand actuation. Vibrotactile feedback implemented via a touch sensor the index finger of the prosthetic were used to activate a vibrating motor on a residual limb. Testing was performed with and without vibrotactile feedback with user blindfolded and unblindfolded. The results showed that the feedback provided from vibrotactile feedback was paramount for task success. However no significant improvement was shown when vision was available (Nguyen & Malmberg, 2022).

2.6 Manufacturing, Environmental Impact and Sustainability

A Study done by researchers from IIT Bombay with an emphasis on the development of a 3D-printed prosthetic with EMG based control. The study emphasized the use of additive manufacturing techniques to create lightweight and customizable prosthetic structures. The system they developed utilized the use of surface EMG signals and a microcontroller-based processing flow. This approach to embedded electronic control systems contributes to Sustainable Development Goal 3 (Good Health & well-being) by enhancing the technologies for rehabilitation of individuals with limb loss and particularly in resource restricted regions with no access to technology advanced industries. The study portrayed that while the systems offer significant advantages in cost reduction and customizability, it leaves a lot lacking in terms of consistent calibration to ensure long term reliable performance for users (Agrawal & Bhat, 2025).

A study looking at the life cycle of a 3D-printed Product and its environmental impact as opposed to traditional production methods. The researchers look at key factors to assess the life cycle of the product. This included material consumption, energy consumption during manufacturing and transportation requirements as well as end-of-life waste generation using cradle-to-grave life cycle assessment approach. The study found that environmental performance is heavily dependant on conditions prior and at production rather than the manufacturing method itself. The results showed that electricity consumption contributed to 70% of the total carbon footprint in some scenarios. In contrast to centralized manufacturing systems, these are more strongly influenced by the material production and larger scale processing stages (Cerdas et al., 2022).

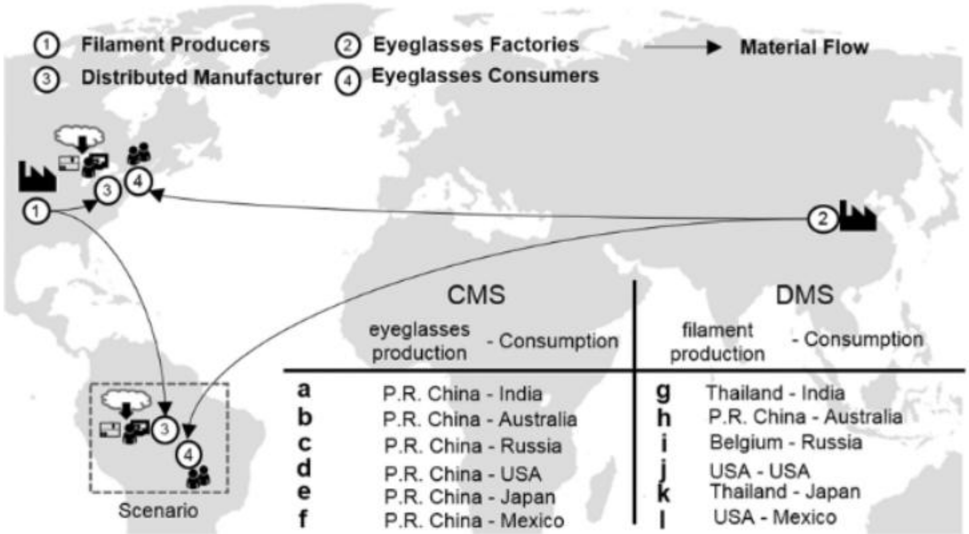


Figure 2.5: Description of the scenarios considered.

(Cerdas et al., 2022)

The study further Quantified that distributed manufacturing can strongly reduce transportation related emissions by allowing for local production. These benefits however can be nullified by high-energy demand during the printing process. The study found that overall sustainability performance is scenario dependant. 3D- printing being more environmentally friendly at lower-volumes for highly custom designs and localized production cases, while traditional manufacturing methods remain most efficient at large scales of high-volume production. These Scenarios are further Highlighted in Figure 2.5 with the findings aligning with Sustainable Development Goal 12 and 13(Responsible Consumption and Production/ Climate Action) by highlighting the importance of choice of material to efficiency, waste management and reduction and energy source optimisation all to aid in reducing carbon emissions (Cerdas et al., 2022).

A recent Study that focused on low-cost EMG-controlled Prosthetic hands that integrated the use of 3D-printed Structures and the use of machine learning based signal processing algorithms within them. These products demonstrated strong functional performance in the real world when it came to real time gesture classification. The system that was implemented achieved an accuracy level of approximately 92-95% using SVM-based models. These models were trained on several thousand EMG samples to enable them for reliable interpretation of basic hand gestures. Their performance evaluation showed an average response time of 80ms and a grasping success rate of 90% for loads up to 600g. The study also found that a low-cost example of prosthetics can have a cost of between USD 200 and USD 700 while more commercial products would normally exceed USD 10,000. These findings align with Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) and SDG 3 (Good Health and Well-being) by promoting the use of affordable assistive technology and improving user independence. However, with this being said, limitations still remain in EMG signal variability, electrode placement sensitivity, and long-term durability of the 3D-printed components. (Diab et al., 2024)

A review of 247 prosthetic studies that involved digital fabrication techniques and technologies, analysed the use of 3D printing, CAD/CAM Systems and the 3D scanning of upper and low limb prosthetics in their respective developments. The study looked at how fabrication methods, materials, clinical outcomes as well as manufacturing workflows were used across the said 247 studies. The researchers found that digital fabrication methods enabled for highly customized prosthetic devices that were tailor made to the individual user. 3D printing can reduce time to manufacture, the labour required to do so and the cost of production as compared to traditional manufacturing methods. They also identified 60 different materials shown in **Figure 2.6** that can be used for manufacturing prosthesis to demonstrate the rapid advancements in this field. Furthermore, it was found that current research and methods lack standardisation of testing procedures to identify a prosthetics validation in the industry. (Oldfrey et al., 2024)

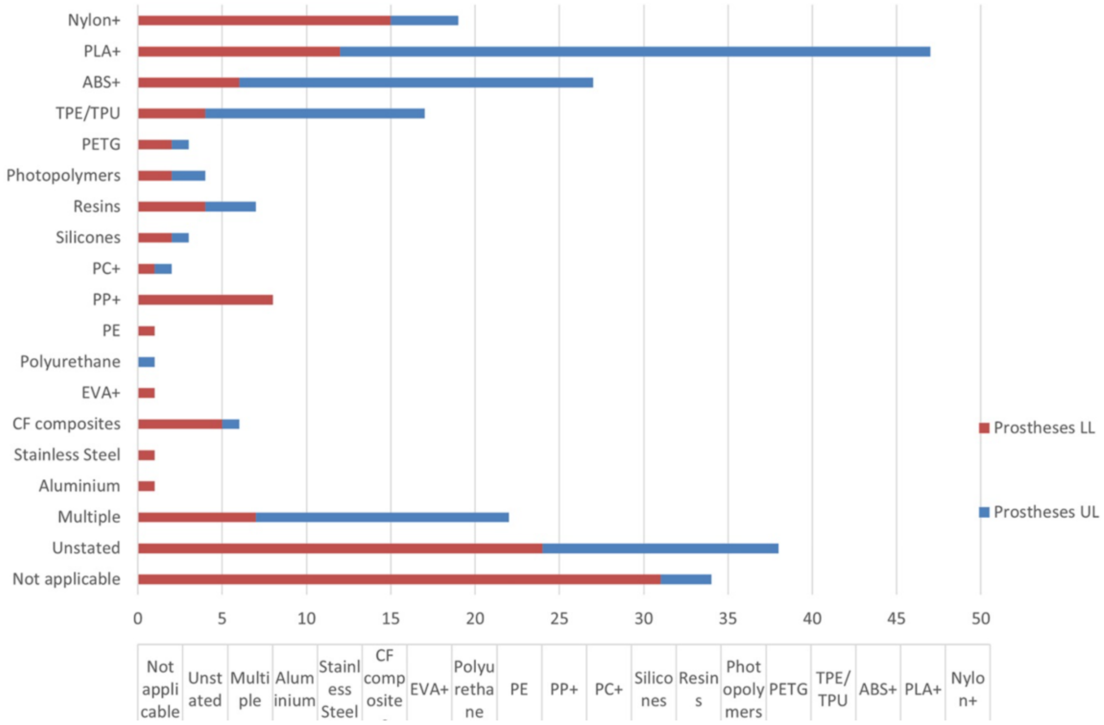


Figure 2.6

(Oldfrey et al., 2024)

The Improved access shown by the researchers to personalised prosthetic devices as well as the increase in the user mobility and quality of life aligns with the SDG 3 (Good Health and Well-Being) as we as SDG 10 (Reduced Inequalities). The reduction of material waste through additive manufacturing and improvements in resource efficiency shown in the study also align with the SDG 12 (Responsible Consumption and Production). Furthermore, the studies exemplification of the growth of digital manufacturing creates the demand for skilled workers in , engineering fields as well as biomedical fields. These professionals will need to work together to achieve their said goals. This adds to the SDG 8 (Decent Work and Economic Growth) as well as SDG 17 (Partnership for the Goals) (Oldfrey et al., 2024).

A study that Investigated the global ecosystems that were involved in the development of 3D-printed prosthetics conducted interviews between medical experts, researchers, engineers as well as prosthetic users from countries around the world. These interviews analysed and poked at the idea of how collaboration between those involved in the inception, development and use of prosthetics influenced their adoption and long-term sustainability. They learnt that the more stakeholders involved in its creation the better the prosthetics adoption would be and long-term user satisfaction. They found that follow up care for users to help with their experience and provide support significantly reduced device abandonment rates as well as boosted user engagement. Furthermore, they found that to build a sustainable prosthetic did not only depend on the technology used but also the social and organisational support systems around it (Savage et al., 2022).

Table 2.3: Summary of Literature Review

No.	Title	Year of Publication	Authors	Methodology	Outcomes	Advantages
1	Influence of filament fabrication parameter on tensile strength and filament size of 3D printing PLA-3D850	2023	S. Hamat, M.R. Ishak, S.M. Sapuan, N. Yidris, M.S. Hussin, M.S. Abd Manan	Experimental study on FDM 3D printing PLA filament. Varying printing parameters like layer height, print speed, nozzle temp	Tensile strength and filament diameter are significantly affected by print parameters	Shows how printing parameters impact mechanical performance, useful for designing durable prosthetic components
2	Electromyography Signals in Embedded Systems: A Review of Processing and Classification Techniques	2021	José Félix Castruita-López, Marcos Aviles, Diana C. Toledo-Pérez, Idalberto	Analysis of EMG signal processing techniques in embedded systems	Identified key methods for EMG acquisition, filtering and classification	Provides foundation for EMG-based prosthetic control, helping to design signal processing pipeline

			Macías-Socarrás, Juvenal Rodríguez-Reséndiz			
3	Experimental Evaluation of a Hybrid Sensory Feedback System for Haptic and Kinaesthetic Perception in Hand Prostheses	2023	Emre Sariyildiz, Fergus Hanss, Hao Zhou, Manish Sreenivasa, Lucy Armitage, Rahim Mutlu, Gursel Alici	Experimental study with hybrid haptic and kinaesthetic feedback system tested on human subjects	Hybrid feedback improved object handling, perception and manipulation accuracy among users	Demonstrates importance of sensory feedback in prosthetic dexterity
4	Impact of 3D printing technology for the	2021	Maxwell Salazar, Ricardo	Prototype development using CAD and 3D	3D printing enabled low-cost, customizable prosthetic arm with	Shows rapid prototyping and cost reduction using 3D printing for prosthetic arms

	construction of a prototype of low-cost robotic arm prostheses		Rosero, Oscar Zambrano, Paola Portero	printing and evaluation of PLA components	adequate mechanical performance	
5	Optimization of 3D printed parameters for socket prosthetic manufacturing using Taguchi method & RSM	2024	Lestari et al.	Taguchi design and Response Surface Methodology to optimize FDM printing parameters	Optimized parameters reduced printing time and socket weight while maintaining reliability	Provides framework to optimize 3D printing for low-cost prosthetic sockets, reducing material and time
6	Bio-signal based motion control system using deep learning models (EEG + EMG)	2021	Heba Aly, Sherin M. Youssef	Deep learning (CNN, LSTM, CNN-LSTM) on fused EEG and EMG signals for motion classification	Hybrid models improved motion classification accuracy by ~3.5% over traditional methods	Shows hybrid bio-signal control is effective for prosthetic motion prediction, reduces feature engineering

7	Multichannel haptic feedback unlocks prosthetic hand dexterity	2022	Moaed A. Abd, Joseph Ingicco, Douglas T. Hutchinson, Emmanuelle Tognoli, Erik D. Engeberg	Non-invasive multichannel haptic system tested on human subjects	Improved dexterity, reduced reliance on vision and offerred better grip and task completion	Demonstrates multichannel feedback significantly enhances prosthetic control
8	A review of technology, materials and R&D challenges of upper limb prosthesis for improved user suitability	2021	Robbie Brack, Emeka H. Amalu	Literature review on prosthetic technologies, materials, and R&D	Identified user dissatisfaction, limitations in materials, control and lack of feedback systems in commercial prosthetics	Highlights materials selection, sensory feedback, and actuator design as key areas for improved prosthetic adoption
9	The effects of vibrotactile feedback on task	2022	Sidney Nguyen,	Experimental study for 3D-printed prosthetic with	Vibrotactile feedback improved blindfolded	Demonstrates low-cost, non-invasive feedback enhances

	performance in a 3D-printed myoelectric prosthetic arm		Henrik Malmberg	touch sensor and vibrotactile feedback tested on a participant	task performance from users	performance and reduces visual dependency
10	The Role and Future Directions of 3D Printing in Custom Prosthetic Design	2024	Partha Protim Borthakur	Literature review on analysis of 3D printing in custom prosthetics	3D printing enables fast, low-cost and personalized prosthetic fabrication as well as identified challenges and future directions	Highlights rapid prototyping, personalization, material advancements, and future tech integration in prosthetics
11	Electromyography Monitoring Systems in Rehabilitation: A Review of Clinical Applications, Wearable Devices and Signal	2023	Al-Ayyad et al.	Analysis of sEMG monitoring, wearable devices and signal acquisition techniques	Reviewed EMG system designs, signal processing methods, wearable device implementation and clinical applications	Provides technical foundations for EMG acquisition, wearable system design, and integration into prosthetic control

	Acquisition Methodologies					
11	Electromyography Monitoring Systems in Rehabilitation: A Review of Clinical Applications, Wearable Devices and Signal Acquisition Methodologies	2023	Al Ayyad et al.	Review of sEMG acquisition systems, wearable devices, and clinical applications	Summarised EMG signal processing and wearable integration methods	Provides technical foundation for wearable EMG prosthetic control systems
12	3D-Printed EMG-Controlled Prosthetic Development (IIT Bombay Study)	2025	Agrawal & Bhat	Development of EMG-based prosthetic using additive manufacturing and microcontroller-based control	Demonstrated low-cost EMG prosthetic with calibration and signal consistency limitations	Supports SDG 3; highlights affordability and customisation but notes long-term reliability challenges

13	Life Cycle Assessment of 3D-Printed Products vs Traditional Manufacturing	2022	Cerdas et al.	Cradle-to-grave life cycle assessment comparing additive vs traditional manufacturing	Found energy consumption (up to 70%) as major environmental impact factor	Highlights sustainability trade-offs in distributed manufacturing and supports SDG 12 & 13
14	Low-Cost EMG-Controlled Prosthetic Hands with Machine Learning Integration	2024	Diab et al.	SVM-based machine learning applied to EMG gesture classification using 3D-printed prosthetic system	Achieved 92–95% accuracy, 80 ms response time, and 90% grasp success rate	Demonstrates affordability and high performance but highlights EMG variability and durability issues
15	Digital Fabrication in Prosthetics: A Review of 247 Studies	2024	Oldfrey et al.	Large-scale literature review of CAD/CAM, 3D printing and scanning technologies	Identified 60 materials and lack of standardised testing methods	Shows scalability and customisation benefits while highlighting validation gaps

16	Global Ecosystem Analysis of 3D-Printed Prosthetic Development	2022	Savage et al.	Multi-stakeholder interviews with engineers, clinicians, researchers, and users	Found that stakeholder collaboration improves adoption and reduces device abandonment	Emphasises importance of social and clinical support systems beyond technology

2.7 Summary

The combination of these findings from studies provide a practical foundation for the development of 3D- Printed EMG controlled prosthetics with the integration of sensory feedback. What has become clear is that modern prosthetics are no longer just about creating a function replacement for a lost limb but it is about restoring autonomy, dexterity and confidence to users. 3D printing has emerged to be a game-changing technology with enabling rapid and cost-effective solutions for users across all industries not just amputees. It allows designers to tailor the prosthetic precisely to the users anatomy. Equally vital is the role of EMG based control systems. By leveraging advanced signal processing techniques and deep learning models, prosthetic hands can achieve smooth and intuitive motion. However, even the most mechanically perfect prosthetic is limited without sensory feedback. Research consistently shows that integrating sensory feedback into prosthetics transforms users experience with an increase in object manipulation making the prosthetic feel alive and responsive.

Together these insights point towards a holistic approach to prosthetic design. One that balances mechanical soundness, intuitive EMG control and sensory feedback. The journals reviewed include 5 on 3D printing in prosthetics, 4 on EMG-based prosthetic control, and 3 on sensory feedback systems. The journals demonstrated the advancements in low-cost prosthetics using 3D printing. They demonstrated accurate and responsive EMG signal acquisition techniques and haptic feedback all being crucial to making a successful and user-centred prosthetic.

CHAPTER 3

CONCEPT DESIGN AND RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the concept design and research Methodology for developing the 3D-printed prosthetic arm. It will involve the use of numerous engineering principles to design, test and improve the arm. Mechanical design rules are used here to ensure the joints are able to move smoothly and freely are strong enough to support daily use. Material selection criteria and basic principles will help decide which filament is suitable by comparing a number of parameters from each type.

Kinematic principles will also be employed here to calculate how each joint should move to imitate a normal hands motion. Furthermore, control system design and principles will guide the selection of microcontrollers, motors and sensors. Finally, testing and validation will be conducted according to professional engineering standards along with a summary of key points and a conclusion of the chapter.

3.2 Investigation of tools and techniques

This section outlines the main tools and techniques used in this project. The development of the prosthetic arm will involve mechanical design in SolidWorks, embedded control systems with servos and sensors, programming in C++ and signal analysis. Choosing the right hardware and software is paramount to ensuring accurate motion of the arm and ensuring all systems communicate with each other efficiently. The selection, again, is based on engineering principles tailored to this project's objectives.

3.2.1 Prosthetic Arm Design

To determine the most suitable prosthetic arm to adapt for this project, several existing open-source designs were first reviewed and compared. The goal of this evaluation is to pinpoint a model that not only meets the project's requirements but still offers movement similar to that of a human hand. By then comparing models against the project's requirements and criteria, it Is possible to identify the most suitable model to adapt for the final prototype.

Table 3.1: Comparison of open-source Prosthetic Arm designs

Model	Pros	Cons
HACKberry Right Arm	Designed specifically for prosthetics Fully 3D-printable & open-source Good finger movement for daily tasks Internal volume for wiring & sensors Modular and easy to customize	Moderate grip force
Federiche hand (Federico prosthetic)	Research-backed, lightweight prosthetic design Biomechanically realistic human finger motion Comfortable prosthetic geometry	3D files not as openly available & less documented
InMoov Hand + forearm	Very popular open-source 3D-printed model Huge community + support Easy to print & assemble	Originally built for robotics not prosthetics

The selection of the Hackberry arm right hand is justified by its modularity, anthropomorphic design and its research backed reliability. The arm allows for the integration of additional sensors, which are integral for the systems feedback requirements. It is an open-source model which allows for easy customizability for this project's specific needs. Table 3.2 presents the arms specifications while figure 3.1 displays its mechanical design.

Component	Value
Total length from elbow to fingertip	260–280 mm
Forearm shell length	160 mm
Forearm outer diameter	75–85 mm
Forearm wall thickness	3–4 mm
Total mass of printed file	450–600 g
Total mass with electrical and mechanical components	800–1000 g

Table 3.2: HACKberry Arm Dimensions and Parameters

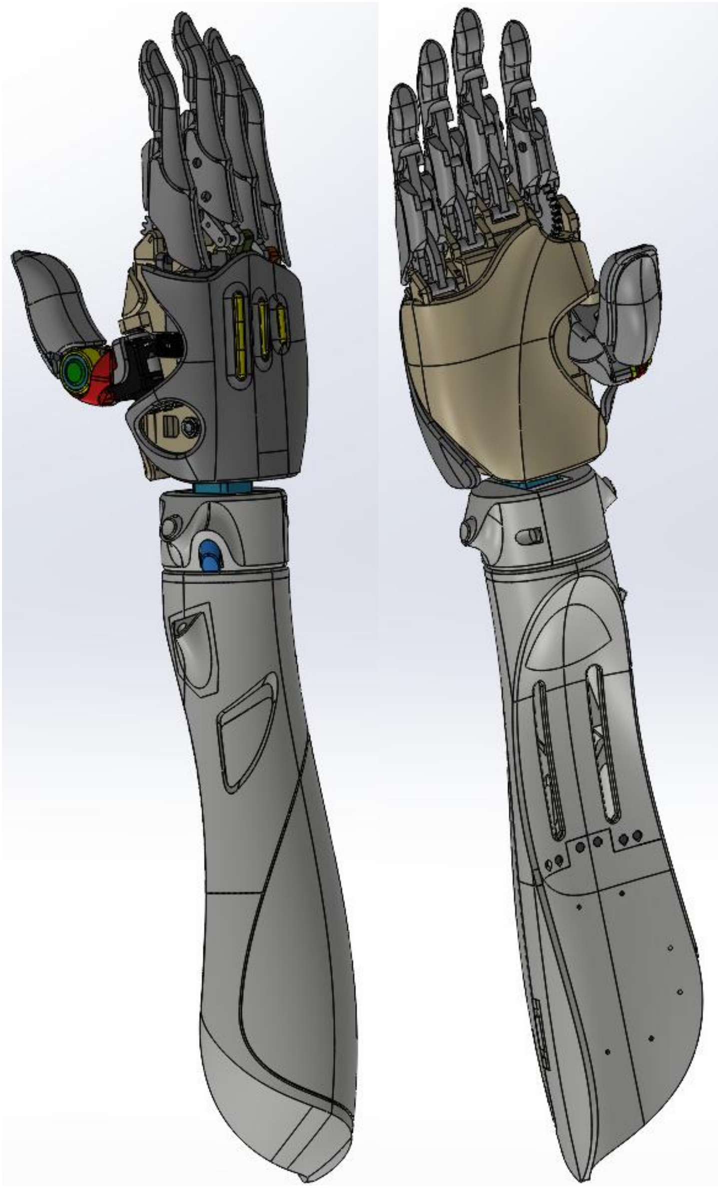


Figure 3.1: HACKberry Arm Mechanical Design

Joint	Degree of Freedom	Motion Range
MCP (base joint)	1DOF (flexion/extension)	0° – 90°
PIP (middle joint)	Passive (linked)	0° – 110°
DIP (tip joint)	Passive (linked)	0° – 80°
Total finger curl	Combined	~180° arc

Table 3.3: HACKberry Finger Degrees of Freedom

Table 3.3 Summarises the degrees of freedom and motion range of the HACKberry arms fingers. Its design consists of 1 active degree of freedom, specifically at the metacarpophalangeal joint. This allows for control from 0° to 90°. The remaining joints on the finger are passively actuated through a basic mechanical linkage. This combined configuration allows for a total finger curl of up to 180°. Table 3.4 and 3.5 highlight the arm thumb degrees of freedom and wrist degrees of freedom respectively. The arm persists by combining 2 degrees of freedom for the thumb while the wrist maintains motion similar to that of humans with a maximum of 90° and 1 degree of freedom.

Thumb Joint	DOF	Motion Range
Thumb base (opposition)	1 DOF	0° – 45°
Thumb flexion	1 DOF	0° – 90°
Thumb total DOF	2 DOF	—

Figure 3.4 HACKberry Thumb Degrees of Freedom

Motion Type	Degree of Freedom	Range
Wrist flexion	1 DOF	±30°
Wrist extension	1 DOF	±30°
Wrist rotation (optional module)	1 DOF	±90°

Figure 3.5: HACKberry Wrist Degree of Freedom

3.2.2 Design Software

To create or modify a highly geometrically complex 3D model, a decision must be made on which design software to use. The CAD Platform for this project is SolidWorks. It offers powerful parametric modelling and advanced assembly tools, ideal for this project's requirements.

In addition to this, SolidWorks offers a wide range of support for additive manufacturing workflows, particularly mesh generation for 3D Printing. It offers the ability to export files in the STL format which is key to ensuring that the components of the arm are printed reliably while maintaining as little mass as possible. An institutional license is also available for SolidWorks, which greatly reduces the costs while still enabling the use of all its features for simulation and validation. All this must be taken into consideration for the additional integration of sensors without the need for an excessive redesign of the arm. For these reasons, SolidWorks represents the most suitable option as a platform for the adaptation of the prosthetic arm design.

3.2.3 3D Printing and Materials

The selection of the main materials, the arm will be made of paramount to the arms' success. This is guided by two main factors of mechanical robustness and the materials thermal stability. However, this is not to disregard cost, printability and accessibility.

Evaluation of common engineering thermoplastics and 3D printing filaments highlighted PLA+ and PETG as the optimal combination. This is to strike a balance between heat resistance and the arms overall robustness.

PLA+ provides the arm with excellent structural stiffness and resistance to warping. This makes it ideal for structural elements such as finger linkages and the palms frame.

On the other hand, PETG provides temperature resistance with it having a higher glass transition temperature (Tg). This is the temperature at which a polymer transitions from being a hard, glass-like material to a softer and more rubber state. Below this state, the polymer chains in the material are relatively frozen and it behaves rigidly while above this the chains gain mobility and the material deforms and becomes more flexible. PETG also provides higher impact resistance as compared to other filaments and materials, this is key in components that experience repetitive mechanical stress. This can be hinge joints, motor mounts, tendons and miniature rib joints within the arm.

The combination of PLA+ and PETG allows the arm to maintain precise geometry where it is paramount for the arm precision while remaining resistant to heat deformation and mechanical stresses.

Material	Pros	Cons	GlassTransition Temperature (Tg)
PLA+	High print accuracy	Low heat Resistance	~55-60 °C
	Low wrap	Brittle under cyclic stress	
PETG	High heat resistance	Slightly Flexible	~80 °C
	High impact strength	Stringing during printing	
ABS	High Impact Strength	Prone to Warping	~105 °C
	Moderate heat resistance	Requires enclosure to print	
TPU	Elastic	Low stiffness	~30-50 °C
	Shock absorbing	Difficult to print precisely	

Table 3.4: 3D Filament Comparison

3.2.4 Microcontroller

The microcontroller will be the central processing unit for the arm. It will act as the brain of all the systems. It will acquire EMG signals, process sensor inputs, control servos and motors. Given the aim for the arm to be a standalone system not requiring an additional computing device such as a PC.

Options explored highlighted the Arduino mega as a possible contender. It bolsters 54 GPIO pins and requires simple programming to be and running. However, it leaves much to be desired in terms of speed of processing with it offering only a fraction of what the other options can do in terms of clock speed. It lacks the memory needed for high-frequency EMG signal filtering, PWM servo control and simultaneous sensory management capabilities. Furthermore, the lack of any wireless connectivity makes it fairly limited for expansion in terms of data telemetry and remote monitoring.

On the other hand, the raspberry pi Pico offers higher clock speeds and has support for python-based programming, but its higher power draw and non-deterministic timing make it limited for real time embedded control.

Microcontroller	Pros	Cons	Clock Speed
Arduino Mega	54 GPIO Pins	Limited Processing Power	16Mhz
	Relatively Easy to program	No Built-in wireless capability	
Raspberry Pi Pico	Higher processing power	Higher Power draw	133Mhz
ESP32	PWM Support for Servos	More complex to program	240Hz
	Low Power Draw		
	Wi-Fi and Bluetooth support		

Table 3.5: Microcontroller Comparison

With the options available evaluated, the ESP32 is selected as the best option for this project due to its ability to support standalone operation while still having real time control and signal processing. The esp32 is therefore capable of being used for EMG signal acquisition, motor control and sensor data processing all within a single embedded system. This reduces the need for additional microcontrollers to handle the systems’ functions thus reducing complexity and overall cost. The ESP32s low power consumption aids it here as reducing the need for large and often heavy batteries to power the arm. All in all, the ESP 32 proves to be the choice going forward as it encompasses efficient task scheduling, energy efficiency and scalability. This will ensure the project has reliable operation and efficiency.

3.2.5 Programming Language and Software Development Environment

The development of the arm requires a software environment that is capable of real-time signal acquisition and real-time actuator control. The selection of these is made with chosen components in mind as well as the projects objectives.

At the lowest level, a programming language is required to be the backbone of the entire system. This is how it thinks and communicates within itself. C++ will be employed here. It offers faster execution speeds, something critical to this project's success, low level hardware control and real time performance on command. As opposed to other options such as micro python and python itself, C++ offers direct embedded control over the system. This is crucial for real time EMG signal Processing.

To ensure the simplest pairing and workflow for this project, the Arduino IDE will be paired alongside C++. It provides a tailored development workflow, a vast library of potential add-ons and straightforward hardware integration. All this combined helps reduce development time in the project. Arduino IDE is also free and relatively user-friendly, however still allowing for advanced embedded systems capabilities.

3.3 Sensors and Actuators

This section outlines the sensory and actuation components used in the prosthetic arm. These selections were made with safety and real-time embedded operation being at the forefront. The selections were made to enable users of the arm to have intuitive control and environmental perception of their surroundings which is implemented through a closed loop feedback system to the user. Furthermore, all sensors were chosen based on their signal reliability, system integration capabilities and compatibility with the aforementioned ESP32 microcontroller.

3.3.1 Piezoelectric Transducer

To mimic one of the most unique human features, the sense of touch, piezoelectric transducers will be employed here. Piezoelectric discs create a potential difference when vibration or contact happens with textured surfaces.

These characteristics make them ideal for texture smoothness sensing through its voltage patterns. A different voltage will indicate a different level of surface smoothness, which can then be relayed back to the user to provide feedback of their environment. The proposed size for the transducer will be 20mm in diameter.

3.3.2 NTC Thermistor

Another aspect of human life that is key to survival is the ability to detect and feel temperature and changes around you. An NTC Thermometer is used in this here due to its high sensitivity to even the smallest temperature changes. It can be setup with the added help of a voltage divider to only detect specific temperature ranges. This analogue signal can then interfaced and integrated with the ESP32 ADC. It follows the trend of this project by being low-cost as well as compact and power-efficient enough to make suitable enough for a prosthetic and embedded system.

Its high sensitivity enables the monitoring of the surrounding environments temperature . In the case of motors and electronic components this can be pivotal in ensuring no overheating during prolonged operation and environmental damage. The calculations below better highlight its employment and how.

Firstly, The thermistor resistance R is calculated from the measured voltage using the voltage divider equation:

$$R = R_{fixed}\left(\frac{V_{out}}{V_{in} - V_{out}}\right)$$

The temperature is then calculated using the Steinhart–Hart equation :

$$\frac{1}{T} = A + B\ln{(R)} + C[\ln{(R)}]^3$$

Where $T$is the absolute temperature in Kelvin, R is the thermistor resistance and A, B and C are constants specific to a particular thermistor.

If the calculated temperature exceeds a predetermined threshold which is seen to be a threat to the user and the arm and its components the system will proceed to send a warning to the user to prevent any further damage. Software based calibration within C++ will be used to ensure accurate temperature estimations in spite of the thermistors non-linear characteristics.

3.3.3 Vibration Motor

The arm will need to provide feedback to the user to alert them to environmental factors. Haptic feedback through tactile vibration can be employed here to achieve this. This approach will help enhance the users perception of objects and further improve their dexterity with the arm.

When an object of surface is being grasped or felt, the system will trigger a surface dependent vibration. Smooth surfaces will trigger a long continuous vibration while rough and textured surfaces will trigger bursty and intermittent vibrations to mimic the irregularity of the surface.

These actuations will be determined in key part with the piezoelectric transducer. The ESP32 will be bridge that helps these two communicate. The ESP32 will use PWM to control the motor. Vibration motors are compact and have low power draw while still being fast-responding enough to be suitable for a wearable and prosthetic system.

3.3.4 Electrodes

Firstly, the approach to acquiring EMG signals is pivotal to this project's success. Muscle activity can be captured through various ways, in this case being a non-invasive approach with electrodes. Non-invasive surface electromyography (sEMG) measures electrical potential generated during muscle contraction. This method is preferred and chosen here over invasive techniques due to its safety, ease of placement and suitability for wearable prosthetic systems that are low-cost and more tailored for general user plug and play.

EMG Type	Description	Pros	Cons
Surface EMG (sEMG)	Non-invasive electrodes placed on skins surface	Safe and low cost	Sensitive to noise interference and sweat
Fine-wire EMG	Needle electrodes inserted into muscles	High signal specificity	Invasive and often painful
Implantable EMG	Permanently implanted electrodes	High signal stability	Surgical risks and high cost

Table 3.6: Comparison of EMG Signal Acquisition Methods

Three surface electrodes are used here: two of these will be active electrodes which will be specifically placed over target muscle groups. The final electrode will be used as a reference and ground which is placed on an electrically neutral area.

3.4 DATASET

A reliable benchmark is needed for true a understanding of typical EMG signal characteristics. Using an already established dataset allows for this without relying on limited test subjects and experimental measurements.

The dataset of choice here is Ninapro DB2 is a widely used, research-grade EMG database. It sole purpose was to support the study of arm movements for prosthetic and human-machine interface applications. It contains EMG recordings from multiple electrodes.

The Ninapro dataset is utilised here strictly during offline development. The dataset is used to analyse EMG characteristics and behaviours. Through this, appropriate signal processing techniques will be used to determine the feature extraction parameters. This offline analysis will then enable standalone prosthetic control without reliance on an external device for computing. The ESP32 will only execute a lightweight real-time signal processing algorithm.

Parameter	**Value**
Signal Type	Surface EMG (sEMG)
Subjects	40 intact subjects
EMG Channels	12
Sampling Frequency	2 kHz
Gesture Types	Finger, hand, wrist, elbow

Table 3.7: Ninapro DB2 contents

3.5 Proposed methodology

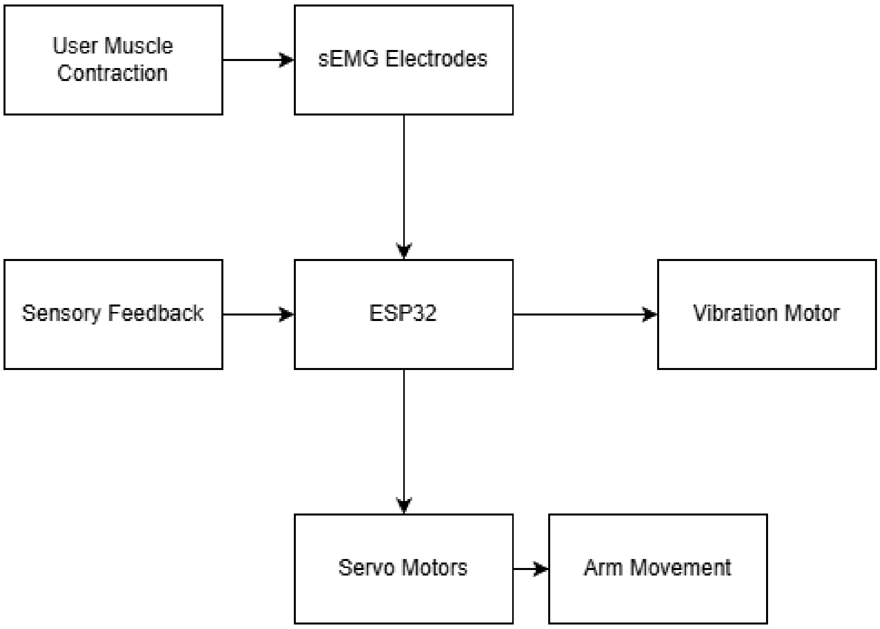


Figure 3.2: Embedded system block diagram

Figure 3.2 Shows the entirety of the systems basic block diagram. The system has 2 main types of sources of input. Data from the piezo electric and thermistor sensors and EMG signals from the electrodes. Both streams use the ESP32 to compute and make a decision for that data and both will engage the servo motors to enable arm movement based on that data. However the sensors will also engage a vibration motor to provide the user with information about temperature and texture.

3.5.1 EMG Signal Acquisition

The sEMG signals that are acquired from the electrodes are highly susceptible to noise. This noise is generated from power-line interference, electrode movement and skin impedance variations. Their amplitude also often ranges from 50μV to 5Mv while their frequencies range from 20Hz to 450Hz. Due to these challenges a strategic signal processing pipeline is necessary beforehand before the sEMG signals may be processed and reliably interpreted by the system.

When electrodes are placed on an individual's skin, each electrode tends to pick up unwanted electrical noise along with the real muscle signals. Instead of looking at each electrode's signal individually, the system will look at the signals between both electrodes as a whole and look for the differences between them. The system will then subtract one electrodes signal from the other.

$$Electrode\ 1 - Electrode\ 2$$

The noise appearing in both electrodes will then be cancelled out and nulled and what remains is the muscle signals which are different at each electrode. The third electrode will then serve as a stable baseline for the differential amplification which then further enhances the noise rejection. This produces a much cleaner and readable signal compared to what was initially acquired.

3.5.2 EMG Signal Processing

Once EMG signals have been amplified, filtered and the relevant muscle frequency range has been acquired, it is then sampled by the ESP32s 12-bit ADC. This converts the raw signal which is in a continuous-time analogue waveform into a discrete digital representation which is readable by the ESP32.

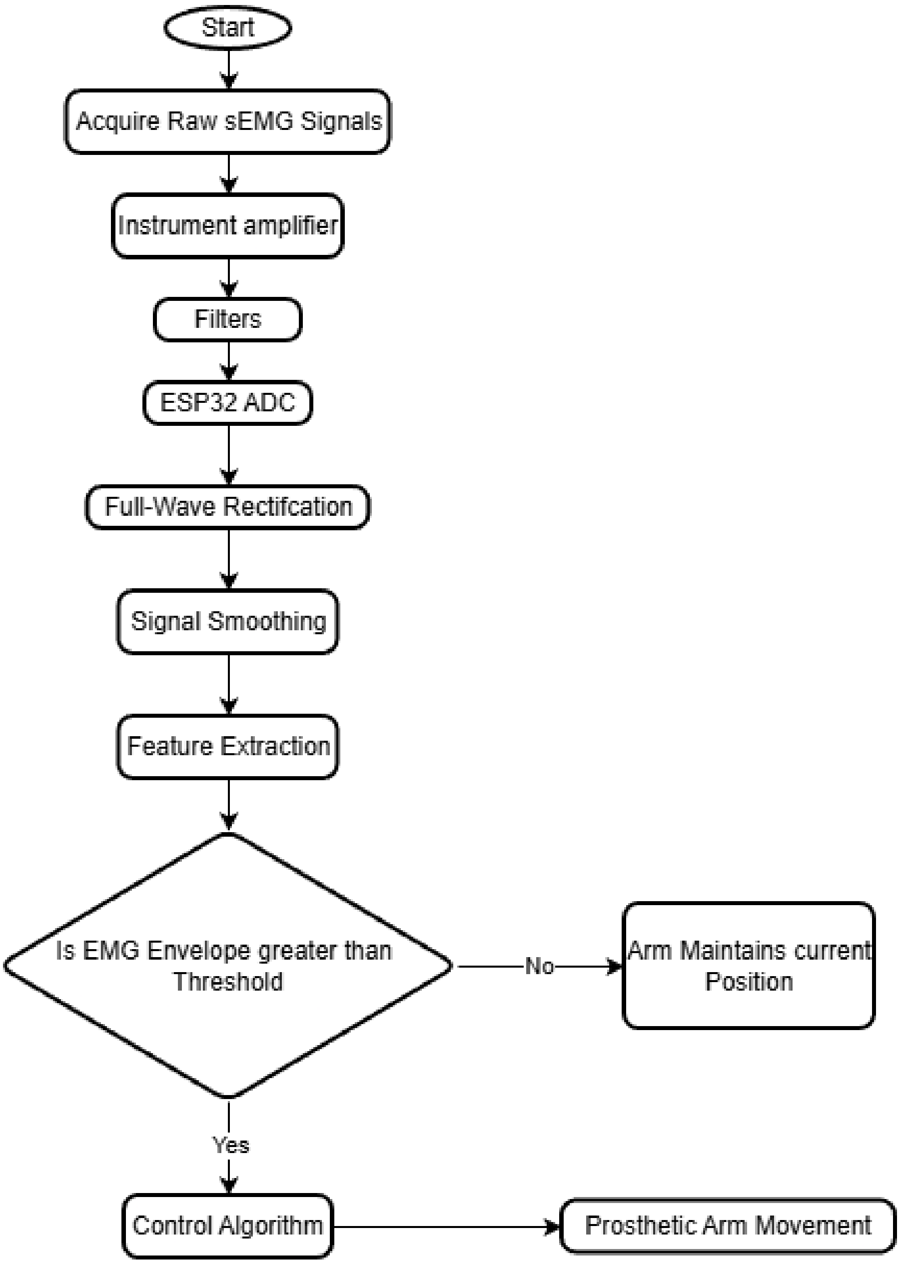


Figure 3.3: EMG Signal Processing Flow chart

3.5.3 Signal Rectification

However, these signals are not without their challenges. These now digitized sEMG signals are bipolar in nature. They tend to swing both ways and oscillate above and below the zero line, which is not indicative of the intensity of muscle contractions. With this kept in mind, to derive any meaning and useful information from these signals a process known as full wave rectification is required.

As the intensity of muscle signals is proportional to their magnitude and not their polarity, we can rectify these negative outputs into positive ones. Rectification converts bipolar signals into unipolar magnitude signals by altering none of the signal's information and instead flipping its polarity.

$$xrect[n] = | x[n] |$$

Next, once rectification is complete the signal that remains is still highly irregular and spiky due to the random firing of muscle fibres within the muscle. These random spikes in the signal remain even when a steady muscle contraction is present. If these irregularities were to be interpreted, they would cause unstable and jerky movement of the arm. Given that these irregularities are not the intention of the user and do not represent their intended muscle contraction strength, the need to address them is apparent.

3.5.4 Smoothing

Smoothing, also known as envelope extraction, acts as a way of averaging the signal over time. As opposed to reacting to every signal spike, whether the user's intentions or not, the system looks at the signal over a period of time. The system will look at what the current signal as a muscle contraction occurs and the signal prior to the contraction. Smoothing combines the values of the signal to produce a single smooth line, called the envelope that looks at the overall muscle activity.

To ensure smaller signal spikes have no effect on the overall signal a process known as exponential smoothing is used. Here each new smoothed value is obtained by using both a small portion of the current signal and a larger portion of the previous smoothed value. This will negate smaller irrelevant muscle activity and favour only useful real muscle contractions to cause the envelope to rise.

3.5.5 Threshold Detection

At this point in the signals pipeline, it has been rectified and smoothed. The signal will now begin to exhibit a singular smooth line that represents activities within the muscle. The line is high when a muscle is contracted and low when a muscle is relaxed. However, even when a muscle is relaxed the EMG signal is never exactly zero. As in smoothing, similar justification applies here. Background noise and minor involuntary muscle activity is still present and may cause the arm to move even when not intended by the user and thus making the system unreliable. Therefore, the system will be left clueless trying to decide whether the user is giving an input or not at all.

To address this a threshold on the signal is applied. Below this threshold the system will ignore the signal while above the threshold the signal will be interpreted as an intentional muscle activation.

This threshold is determined using the Ninapro Dataset. Here EMG signal from many users and specific gestures are analysed with the typical signal levels of muscle rest and contraction being analysed. A threshold that is both higher than the noise levels but lower than real muscle contraction levels is chosen. This allows the system to be robust and repeatable. Once this threshold is crossed the system marks this as a muscle active event. However, once the signal falls back below the threshold the system will interpret this as a muscle inactive event. This allows the system to detect and monitor a clear start and stop point for muscle actions.

3.5.6 Feature Extraction

Feature extraction is used to determine what the user’s input should be translated to in terms of servo and motor movement on the prosthetic arm. This means measuring properties about muscle contraction while it is occurring. These measurements are called features. As the previous processes have now cleaned up the signal, the system can now work with clean numbers that describe muscle action. The system does not attempt to extract a feature using the raw EMG signal. It instead detects which muscle group is active, measure how strong and long the period of activation on the muscle group is. With this information is maps measurements to a specific joints’ movement.

In terms of finger movements: open, close and how much grip force is necessary. The primary features for this will be the signals mean and peak amplitude as well as the RMS of the signal. The system will interpret these features of low amplitude as a weak contraction therefore giving slight finger movement. A strong contraction will be interpreted as full grip force while a long-sustained contraction will be perceived as a holding force. The Ninapro dataset highlights this by showing a strong correlation between EMG amplitude and finger flexion intensity.

For palm and wrist movement the system will look at the gradient of the slope of the signal. The rate of rise of this slop. Contraction duration and RMS are also used as key information for this movement. A fast rise in the slope of the signal indicates a quick wrist rotation while a slow rise in the slope will indicate a slow and smooth wrist motion. A short contraction will be perceived as a small rotation while a longer contraction will be perceived as a full rotation.

Next, elbow movement relies on longer and stronger signal strength. This can be illustrated by a brief contraction indicating that the elbow must be bent slightly while a longer contraction indicating that full elbow flexion is required here.

3.5.7 Control Algorithm

Processed EMG signals need to be translated into controlled prosthetic hand movements. A threshold-based control algorithm is implemented here. The control algorithm constantly monitors the EMG signals envelope, it then compares it to calibrated thresholds then finally generates a corresponding servo motor command. The algorithm will also incorporate information from the other sensors and enforce safety constraints.

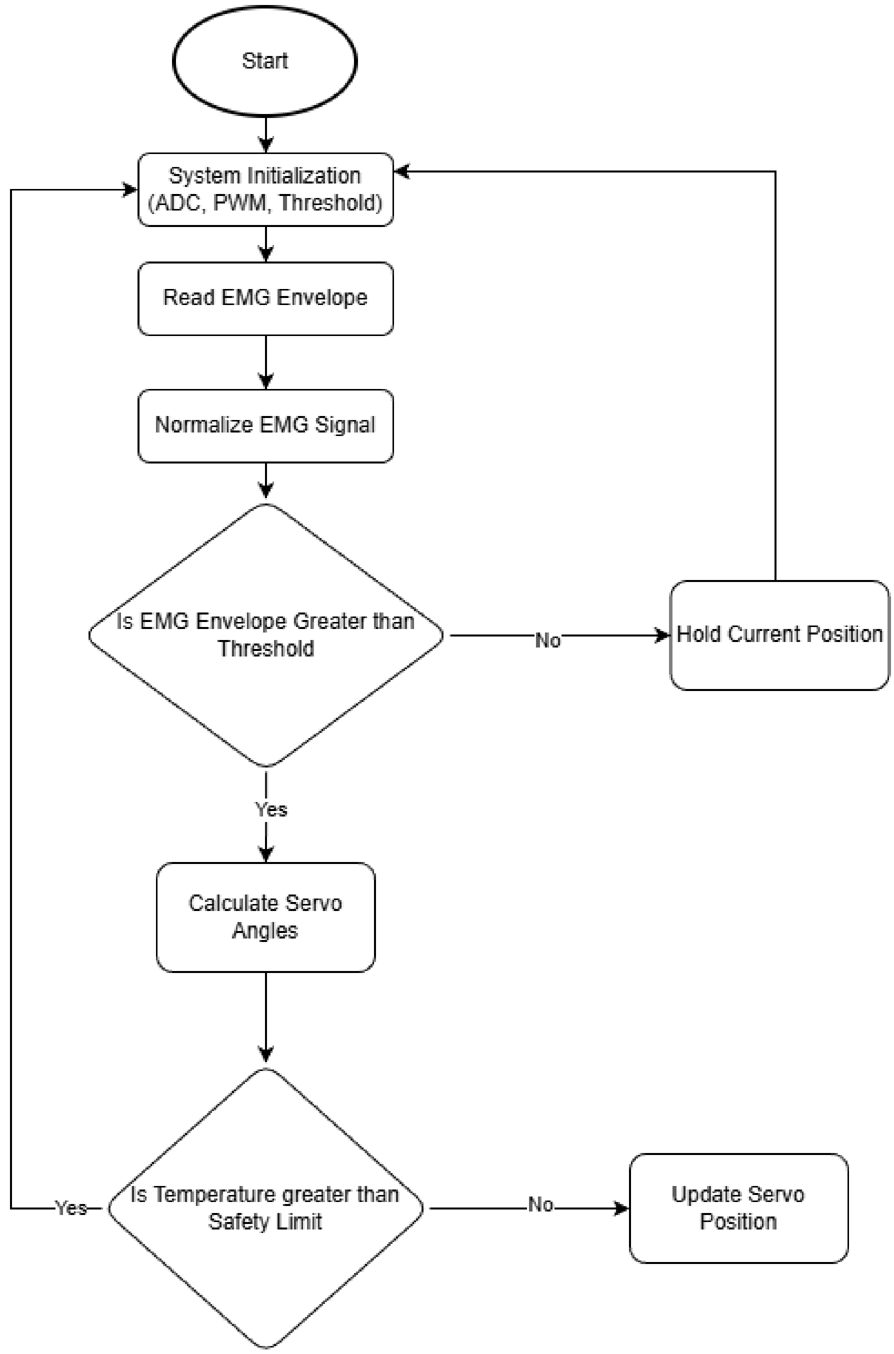


Figure 3.4: Control Algorithm Flowchart

Before this algorithm can be implemented, it is important to note that EMG signals vary significantly from user to user. The use of the NINApro dataset is brought here to calibrate the system and avoid false triggering. The output of this will be raw EMG samples for rest and contractions. The EMG signals within the dataset are averages for values at rest, maximum amplitude during contraction and a basic noise floor level. These parameters will define the operating range of the EMG signals.

A safe activation threshold is calculated using the dataset as well. This is carried out to prevent unintended movement caused by noise or small involuntary muscle activity.

$$EMGthreshold = EMGrest + k(EMGmax - EMGrest)$$

The system initialization phase will involve the ESP32 powering on the 12bit ADC and PWM modules. It will also move the servo motors to a safe initial open-hand position meanwhile calibration parameters are being loaded. This is done to ensure predictable and safe startup behaviour of the arm. Next the ESP32 will continuously read the processed EMG envelope from the processed signals at a sampling rate of 1000Hz to ensure real time response and information acquisition.

The values obtained earlier from the dataset now come into play. Normalization is now done to obtain consistent control regardless of absolute signal amplitude. The normal EMG Equation highlights this process. To satisfy this equation the range of normal EMG must be between 1 to 0

$$EMGnorm = \frac{EMGmax - EMGrest}{EMGcurrent - EMGrest}$$

The algorithm will then look at the normalized EMG values and compare them against the activation threshold. This is done to determine whether or not the user intends to move the arm. This logic can better be highlighted as:

If $EMG_{norm} < Threshold$→ No movement
If $EMG_{norm} \geq Threshold$→ Generate motion command

Next, the intensity of the EMG signals needs to be mapped to a specific and proportional servo rotation angle. This allows for variable grip strength rather than a simple full grip force or non mechanism. This is done through the use of equation []

$$\theta servo = \theta min + EMGnorm(\theta max - \theta min)$$

For the safety feedback loop a temperature check is carried out here. This is done to prevent the user from wielding hot objects that may damage the hand and its components. A simple temperature limit is set whereby if this temperature is met or exceeded the motors will stop and an alert will be issued to the user. if this temperature is not met the system will instead continue with its operation.

The next steps once the servo angle has been computed it will then need to be converted into a PWM signal to send to the servos to control the position of the arm. This logic is applied to the fingers, wrist and elbow.

$$PWM = 1ms + \frac{\theta servo}{180 \circ} \times 1ms$$

1ms here represents the minimum pulse width required by a standard servo to activate and reach a minimum angular position. It is added here to ensure that no matter the angle the servo will activate. For the fingers the servos will pull tensioned strings to produce the finger flexion and extension. However the wrist and elbow will rely fully on servo motor motion.

The entire control algorithm will be looped continuously at a rate of 100hz with 10ms intervals.

3.5.8 Piezoelectric Feedback System

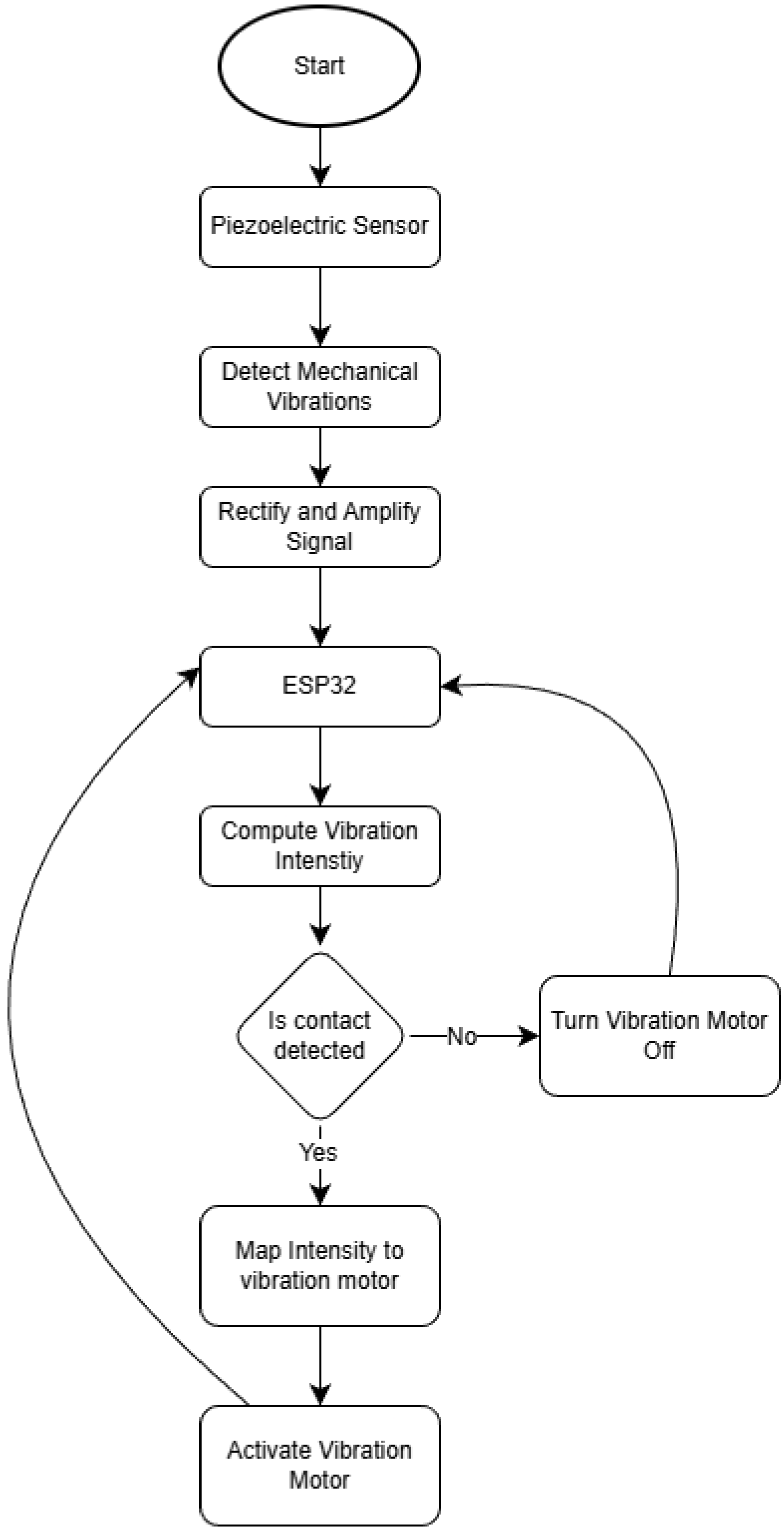


Figure 3.5: Piezoelectric Feedback System flowchart

Piezoelectric sensors are placed on the fingertips of the hand as the experience the highest contact vibrations and are what naturally users will use to feel objects. This is done as piezoelectric materials generate an electrical voltage when mechanically stressed.

The signal this sensor generates will vary based on surface roughness, sliding speed and contact force. Rough surfaces will produce a high frequency and amplitude vibrations while smooth surfaces will produce a low frequency and amplitude. Faster motion will also do the same with amplitude and frequency. However contact force will only increase the amplitude of the output signal voltage. However, piezoelectric raw output signals are not without their issues. The raw signals generated are AC voltage that often range between micro and millivolts and are prone to noise.

To extract any meaningful data from piezo sensors first an operational amplifier is introduced to boost the low-voltage signals to a more measurable range for the ADC to read. Next, rectification will be done to convert the bipolar AC signal to a unipolar one, similar to that of the raw sEMG signal. Similar processes to those done to the sEMG signals are then carried out, as an envelope and low-pass filter are introduced to extract a vibration intensity envelope and to create correlation with roughness perception respectively.

As the sensor will be sampling at a rate larger than 1kHz to capture high-frequency vibrations a Fast Fourier Transform shown in the equation can be applied to extract dominant vibration frequencies. Higher dominant frequencies will be determined as finer textures meanwhile lower frequencies will be indicated as coarse surfaces.

$$Vtexture = \frac{1}{N} \sum \mid Vpiezo, i \mid$$

Where Vpiezo,i is the instantaneous piezo voltage sample

Next, these signals will be mapped to the PWM control of the vibration motor. Low-intensity vibrations will indicate smooth texture while medium vibrations will indicate a moderate texture and lastly high-intensity vibration will indicate a rough texture surface. This PWM signal will be capped however to prevent user discomfort.

3.6 Concept design derived from fundamental engineering principles

The following Subsections portray the key fundamentals of the project. Its mechanical and electronic design concepts which are strictly derived from fundamental engineering principles. The design is supported by calculations for torque and power for the arms movement as well as conceptual circuit designs and 3D models.

3.6.1 Elbow Hinge and Motor Mount Conceptual Design

The elbow joint is an essential part of a functional humanoid arm. EXII did not design an elbow joint for the HACKberry arm. Thus, to achieve the projects objectives, an elbow joint compatible with the existing forearm must be created to facilitate this. The joint must be capable of closely mimicking a natural elbow joint thus it will require a single degree of free hinge mechanism.

To create an interfacing surface that will attach to the preexisting arm, first. measurements must be taken of the end of the forearm. **Figure 3.6** displays its general dimensions.

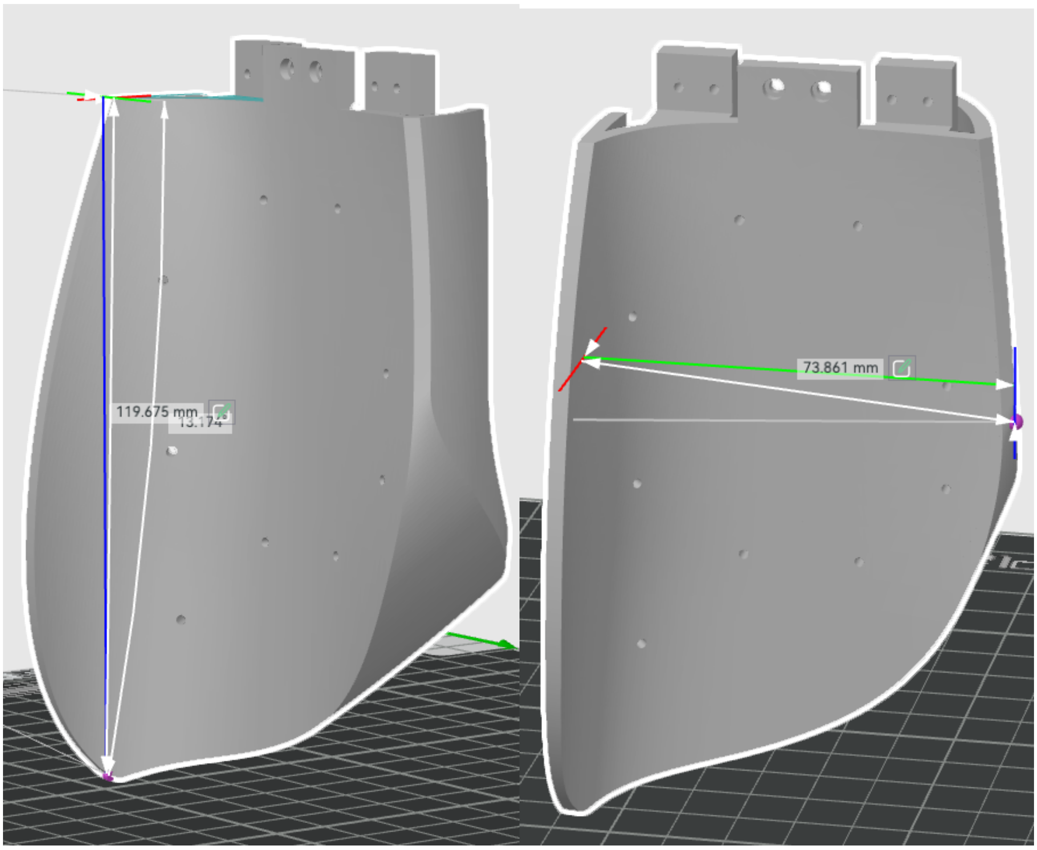


Figure 3.6: HACKBERRY Forearm Baseplate Measurements

The figure[] shows the forearm plate having measurements of 119.7mm x 73.9mm. a base plate to attach this plate unto must first be designed. This plate must be made without hindering the already existing arm from carrying out it duties or disrupting it in any structural way.

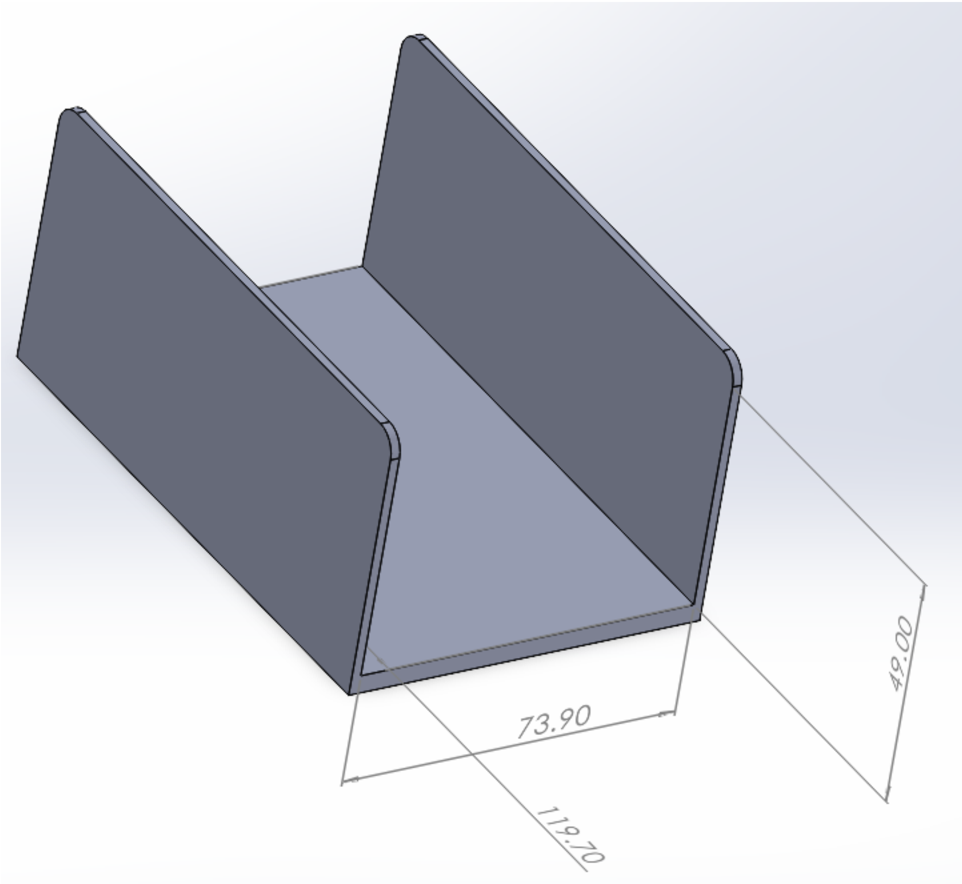


Figure 3.6: Forearm baseplate attachment

Figure 3.6 shows a baseplate to hold the forearm in place. The arms exact dimensions were used as the base and walls 2mm thick were added for support. Fillets on sharp corners were added for user safety as well.

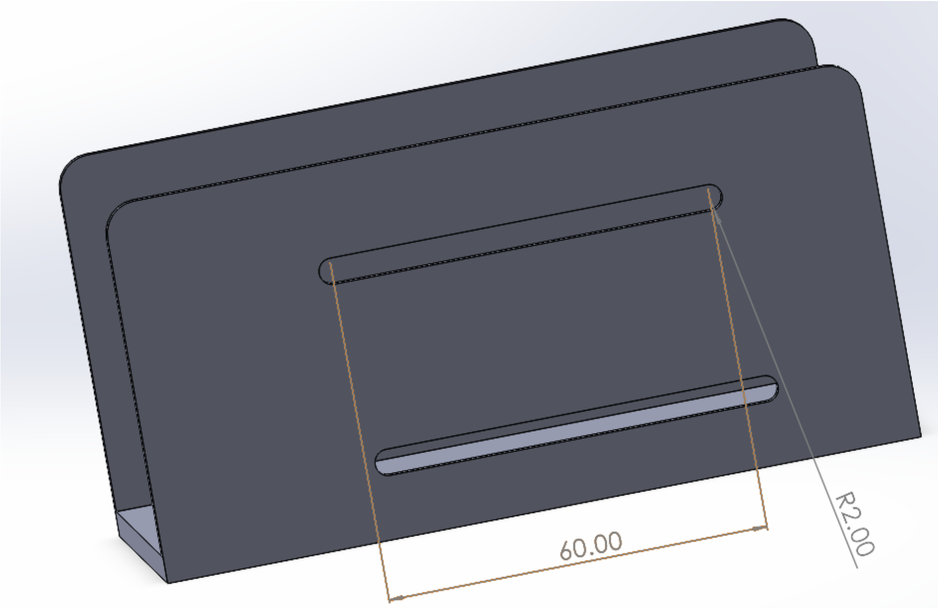


Figure 3.7 Baseplate Side Screw holes

Next, Figure[] shows the base plate attachment M4 screw holes to attach the forearm to the baseplate. The screw holes were made as sliders to allow for adjustment to the users preference.

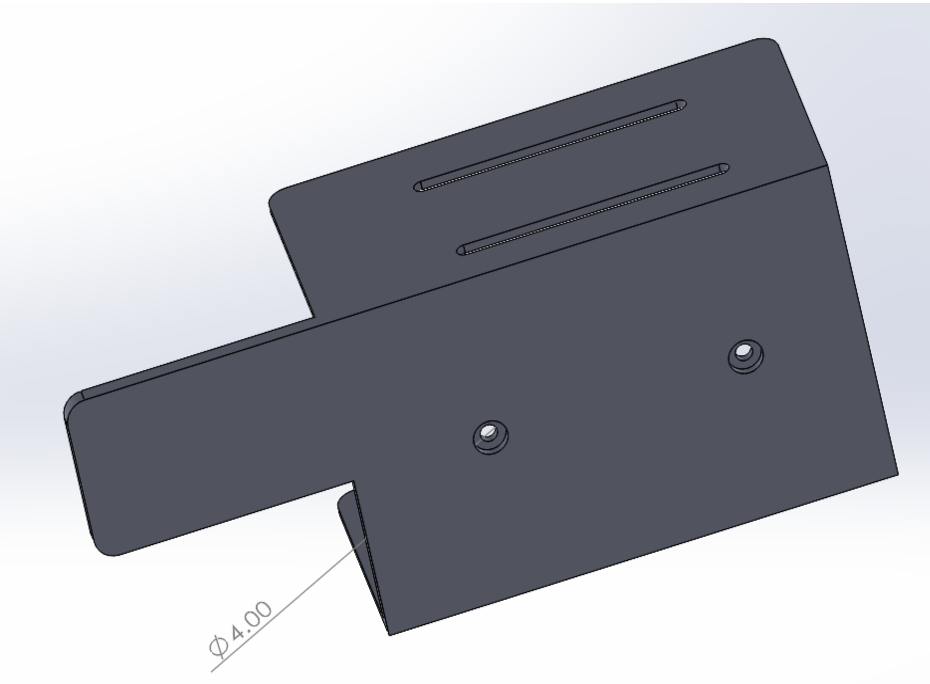


Figure 3.8 Baseplate Bottom Screw holes 1

Figure 3.8 Shows raised M4 Screw holes to hold the forearm in from the button on the plate. The figure also displays a 50mm extension of the plate, used later as the attachment to the elbow hinge.

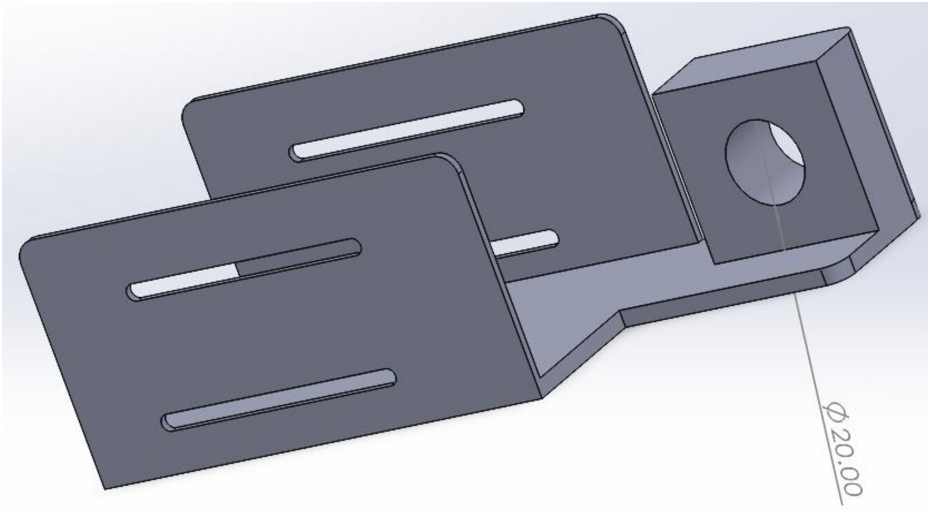


Figure 3.9 Baseplate Extension 1

Figure 3.9 Shows an extrude perpendicular to the base with a 20mm hole to allow for a bearing to be placed for smooth motion.

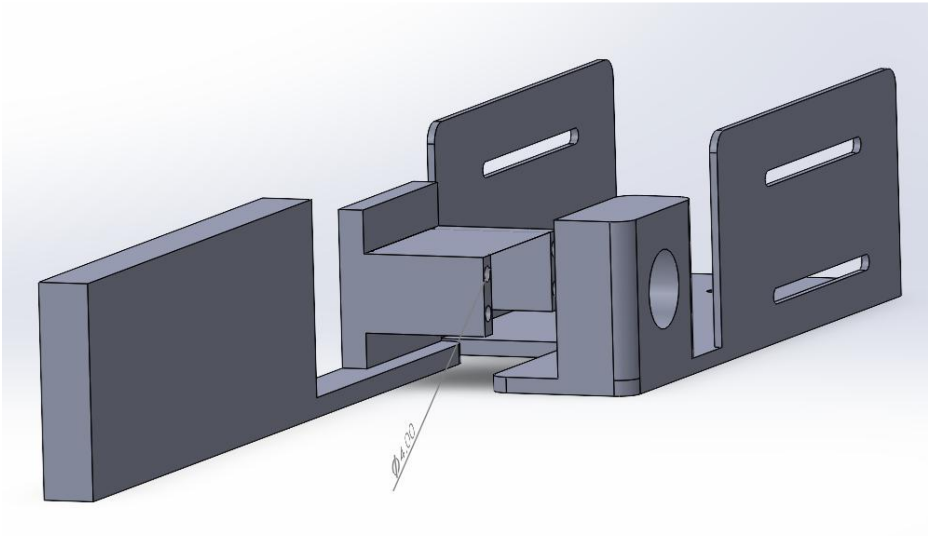


Figure 3.10: Baseplate elbow servo casing

Next, an extended rotating brace was added where the FT5330 M servo Motor will be added. Screw inserts for M2 screws were added, these were spaced 10mm apart and are added to screw in the servo motor to the frame. Here the motor will attach to a bearing and a shaft on the first base plate of the forearm.

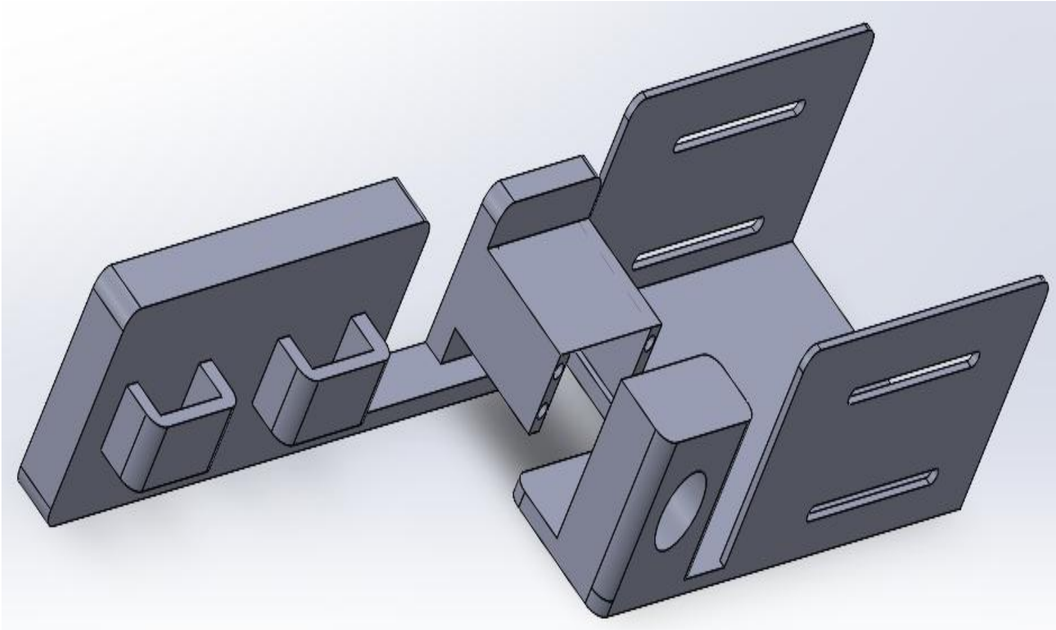


Figure 3.11: Baseplate Strap Holes

Figure 3.11 shows an addition of two slots on one side of the rotating brace. These slots are meant for hook-and-loop fastener straps. The straps will attach to the amputee's upper bicep and triceps muscles.

3.6.2 Servo Motor Torque Calculations

Given the employment of the hackberry arm as the basis for the arms design typically used servo motors for the arm will be used to streamline the process of motor selection. However, their characteristics and parameters will be studied to determine their viability for this project. This is to ensure computability and the ability to integrate with other components.

To enable finger actuation, a lightweight and compact servo is required. The HS-5645MG is used here to cover specifically that. It is a servo motor that is compact enough to fit within the arm with dimensions of 40.5mm × 20mm × 38mm. It provides torque up to a maximum of 1.17Nm.

In this case we can assume that the average weight of a finger segment on the hackberry arm to be 50g and the maximum object load in grip of a hand to be 500g. this 500g load can be assumed to be evenly distributed across all 5 fingers.

Given this information, we can then calculate the torque required.

Force $F = m \times g = 0.5\,\text{kg} \times 9.81\,\text{m/s}^2 = 4.905\,\text{N}$(distributed over 5 fingers ≈ 1N per finger)

Lever arm $r = 0.08$ m

Tfinger = 1N×0.08m = 0.08Nm

The torque required to move a finger on the arm is 0.08Nm and the servo here provides 1.17Nm of torque. The motor provides more than x10 the safety margin to ensure reliable finger actuation.

This servo motor can then be carried over to the palm and wrist joint of the arm. With the average palm mass of the hackberry arm being 500g and the objects mass being assumed to be 500g. the distance from the wrist pivot to the hand COM is set at 0.05m

F=(0.5 + 0.5)kg×9.81=9.81N

$\tau_{\text{palm}} = 9.81 \times 0.05 = 0.4905$ Nm

The torque required in this scenario is 0.4905Nm which is half of what the motor can provide, giving it a safety factor of at least 2.

As for the Elbow Joint, a much stronger servo motor is required here. Given the increase in mass and weight as it will have to move the entire forearm and wrist in its entirety. For this the FT5330M is selected for its higher torque capabilities at a maximum of 3.4Nm. It has a mass of 50 grams, dimensions of 40.5mm × 20mm × 38 mm. We can also assume the distance between the elbow to the centre of mass is 0.15m. Given the total mass of the forearm and wrist assembly is required to be moved here it can be assumed to be 1.5Kg we can then calculate the torque required.

F=1.5×9.81=14.715N

$$\tau_{\text{elbow}} = F \times r = 14.715 \times 0.15 = 2.2 \text{ Nm}$$

Given this, the motor provides a safety factor of 1.5 while still being capable of being compact enough to fit within an elbow mechanism.

3.6.3 Servo Motor Control and Sensor Circuit Design

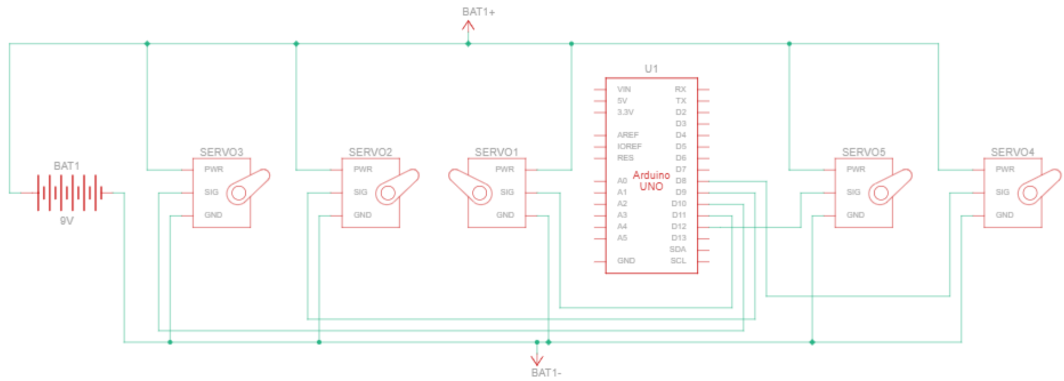


Figure 3.12: Finger Servo Motor Control Circuit Design

The servo motors used in this project all rely on the same standard for control, PWM. As the PWM signal is the only indicator for a servo motors position, external power can used to help supplement the power requirements of driving servos simultaneously.

Figure 3.12 represents the 5 finger circuit diagram. Each servo motors PWM signal input is connected directly to the microcontroller GPIO pins. However, external power here is provided by a 9V battery in this case.

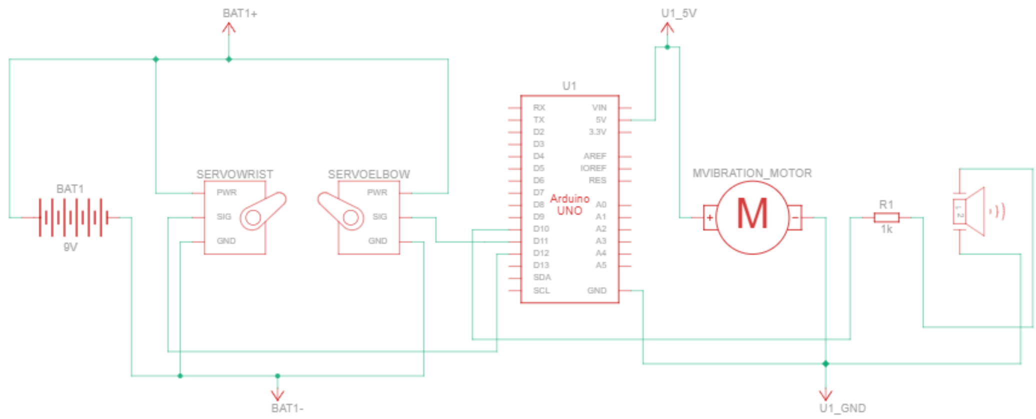


Figure 3.13: Piezoelectric Feedback Circuit Design

Figure 3.13 shows the combined circuit of the piezoelectric transducer as well as he wrist and elbow motors. The piezo electric transducer requires a bleeder resistor before its input maybe be connected to a microcontroller. this is done to prevent a buildup of current that could potentially damage the microcontroller and its adjacent components.

3.6.4 Power and Energy Calculations

Before the systems total power draw and energy consumption can be calculated, assumptions must be made upon the condition of the system and its behavioural characteristics. A basic assumption is that the arm will be operating for 16 hours a day. All components will need to be powered the entire time. This is taken as a worse case scenario baseline. The system contains 7 servos in total, each average a current draw of 300mA and operate at a voltage of 6V.

Pservo = V×I = 6×0.3 = 1.8 W

Accounting for the 7 servos gives us a total servo power can be calculated as:

$TotalP_{servos} = 7 \times 1.8 = 12.6$ W

Next, the EMG electrode sensors also account for power in the system, however minuet it may be, it will continuously be taking readings and will not turn of during the entire systems operations. The operating voltage of the sensors is 5V and they draw 10mA.

$P_{EMG} = 5 \times 0.01 = 0.05$ W

The NTC thermistor voltage divider circuit also draws a minuet amount of power but similar to the electrodes it will constantly be on. The voltage supplied to the circuit is 3.3V and the divider resistance total may be assumed to 20KΩ.

$$I = \frac{V}{R} = \frac{\mathbf{3.3}}{\mathbf{2000}} 0.165\ mA$$

Pthermistor = 3.3×0.000165 = 0.0005445 W

For the piezoelectric voltage divider circuit power is relatively straight forward to calculate. It has a supply voltage of 3.3V and a draw of 1mA

Ppiezo = 3.3×0.001 = 0.0033W

Finally, the vibration motor has an operating voltage of 3V and a current draw during full vibration of 80mA. However here we may assume the motor will only be active 30% of the time.

$P = 3 \times 0.08 = 0.24$ W

$P_{avg} = 0.24 \times 0.3 = 0.072$ W

As for the microcontroller, the ESP32 operates at a voltage of 3.3V and its average current draw is 240mA while Wi-Fi and Bluetooth are disabled.

$P_{ESP32} = 3.3 \times 0.24 = 0.792$ W

With all the components power draws being calculated, they may all be added to get the systems total worst case scenario power draw.

$$P_{total}=12.6+0.792+0.05+0.00054+0.0033+0.072$$

$$P_{total}=13.52W$$

The total power draw of 13.52W can then be used with the operating time to calculate the energy consumption over 16 hours.

$$E = P \times t = 13.52 \times 16$$

$$E = 216.32 \text{ Wh}$$

3.7 Professional engineering practices

The practice of professional engineering is not merely just technical design and knowledge. The practice of engineering extends further to include ethical, legal, social, cultural and health considerations that ensure safety, well-being and importantly fair treatment for all users. Engineers are required and expected to maintain professional and ethical standards. This is achieved through taking responsibility for societal impacts from their work, complying with regulatory requirements and maintaining the highest levels of safety and professionalism (BEM, 2023).

"A registered engineer shall at all times hold paramount the safety, health, and welfare of the public," according to BEM Code of Professional Conduct 1.0 (BEM, 2004). When designing assistive and prosthetic technologies that interact directly with the human body, this idea is very important. During the mechanical and electrical design phases of this prosthetic arm project, safety and health concerns are given first priority. The user won't be harmed during signal collecting thanks to the usage of non-invasive EMG sensors. In order to prevent excessive gripping force that could harm the user, mechanical safety is handled by regulating servo torque through software control. Additionally, the 3D-printed parts' rounded edges and suitable material choice lessen the possibility of cuts, pinching, or pain.

"The work should be certified satisfactory completion only if the engineer has control over the supervision of construction and installation of that work," according to BEM Code of Professional Conduct 1.2 (BEM, 2004). The project's structured development method reflects this idea. Prior to the complete system integration, each of the prosthetic arm's subsystems—the elbow joint, the forearm extension, the Hackberry-based hand mechanism and the EMG control system—are each independently developed, built and tested. The engineering concepts and calculations created in the concept design chapter are closely followed in the prototype's construction. Before any integration is done, functional testing is carried out for individual components to make sure each subsystem functions in accordance with its intended requirements.

Additionally, according to BEM Code of Conduct 2.1.4, "The registered professional engineer shall undertake assignments only if the engineer is qualified by education and experience in the specific field of the assignment to be involved" (BEM, 2004). This project, which combines aspects of mechanical design, electronics, control systems and signal processing, is within relevance to the field of mechatronics engineering. The prosthetic arm's design requires electrical knowledge and knowhow for EMG signal conditioning. Mechanical calculations for torque and stress analysis and embedded system programming with the ESP32 microcontroller. These assignments satisfy the criteria that the work be done within the engineer's area of expertise since they directly relate to the technical and academic background of a mechatronic engineering student.

Furthermore, "A professional engineer shall be objective and truthful in professional reports, statements, and testimony," according to BEM Code of Professional Conduct 3.1 (BEM, 2004). In the literature review and development of the prosthetic arm, reliable and current academic resources are used to uphold these guidelines. All external sources are correctly credited as well as open-source prosthetic manuals are referenced. Without overstating the capabilities of the system, design choices, calculations and constraints are given objectively where seen to be fit. Engineering analysis from fundamental principles, not personal preference, is used here to compare several design choices. This is shown throughout the project, such as in actuator and material selection. This guarantees academic integrity, honesty and openness in all project documentation.

In the context of this project, these principles guide every aspect. From its first inception to its development these principles ensure the device is safe and effective.

3.7.1 Safety

To design a prosthetic of any limb, safety must be at the forefront. The device must operate reliably without causing any harm to the user, whether physical or psychological. Thus, many considerations are taken into account before development. Mechanical safety considerations such as the inclusion of a rounded edges in the 3D-printed components, ensuring smooth motion of the fingers and the palm as well as appropriate torque limits on motors to prevent accidental pinching are taken into account. Proper insulation of electrically conductive components is also kept in mind here; this is particularly important as the EMG based control system is connected directly to the user's skin.

3.7.2 Health

The end user is at the forefront of this project. Their health is key to its success and the motive behind its inception. With that in mind, the prosthetic arm must be ergonomic, comfortable and cater to users of all sizes. The non-invasive approach to EMG signal acquisition Although it is not necessary here, comprehensive testing must also be conducted in a controlled environment before any user trials are done on assistive devices for users with disabilities (WHO, 2022).

3.7.3 Social

The ultimate goal of this project is to improve individuals with upper limb loss and amputees' quality of life. This goal needs to be achieved while still making those less financially privileged have access to it. The design further encourages social acceptance by being unobtrusive and user friendly. Stigma may still remain about users wielding a prosthetic, the design addresses this by being as close to a possible to human arm.

3.7.4 Legal

To ensure compliance with legal and regulatory requirements, this project respects intellectual property rights by properly citing and modifying open-source designs. This extends further with the use of opensource Datasets with the individuals privacy and personal information being redacted. Furthermore, the design process aligns strictly with the Board of Engineers Malaysia (BEM) Code of Professional Conduct regarding professional accountability and risk management (BEM, 2023). Any liability issues that may arise are mitigated and handled through thorough documentation of design choices and procedures.

3.7.5 Sustainability and Environmental Impact

The conscious decision to remain sustainable was carried out throughout this project's development. Materials choices were greatly influenced by this such as the 3D printer filaments used. PLA and PETG are selected for their recyclability and minimal environmental impact as opposed to traditional plastic manufacturing methods. Components within the system remain modular to reduce waste and energy efficiency is highlighted as a key point of contention for this project. This aligns with the BEM guidelines on responsible engineering practices and WHO recommendations to consider environmental impact for assistive technology design and development.

3.8 Project management, Finance and Entrepreneurship 1

3.8.1 Project management

Project management ensures projects are executed efficiently, systematically and within a given budget and time frame. This project is divided into two phases. Phase 1 consists of literature review and Methodology as its main pillars thus an effective project management strategy is paramount to its success and smooth progression between project phases.

Phase 1 consisted of weeks 1 to 3 of the project's inception with the FYP topic selection being made here. Furthermore, preliminary project research was done before to ensure basic understanding of a topic's needs and requirements. Upon selection and approval, the next steps taken were the creation of the project specification form where the projects aim and objectives were clearly laid out to ensure clear deliverables. Phase 2 comprised of the next three weeks in the project. An emphasis was done here on literature review and development of the methodology of the project. These sections are carried out to aid in execution of the project later in part 2. During this phase extensive research was carried out on existing prosthetic designs, 3d printing techniques and EMG based signal acquisition methods. In parallel, a methodological approach was conducted to define a design approach to the arm. Phase 3 continues here with the development of the elbow joint concept design. Here careful consideration was required for electronic component placement, user comfortability and overall structural integrity. This phase translated the theoretical research carried out prior into practical design decisions and choices. Phase 4 involved the projects overall review, its refine and documentation. It also comprised of final evaluations and meetings with supervisors for any corrections and guidance where needed.

Figure 3.14 portrays the Gantt chart for these phases. It demonstrates effective time management, task prioritization and the added need for a buffer during any project.

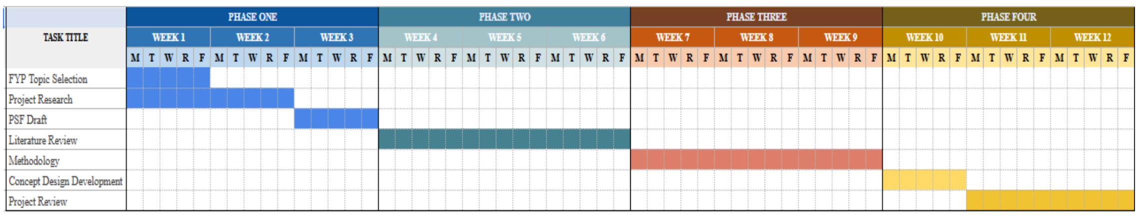


Figure 3.14: Project Gantt Chart 1

3.8.2 Finance

Throughout this project, an emphasis has been placed on creating a low-cost and accessible prosthetic using readily available components. Table 3.8 highlights this by showing the total component costs of the prototype being a fraction of commercially available prosthetics.

Component	**Quantity**	**Purpose**	**Estimated Unit Cost (RM)**	**Total Cost (RM)**
ESP32 Microcontroller	1	Main control unit and EMG processing	35	35
Surface EMG Electrodes	1 set	Non-invasive muscle signal acquisition	40	40
Servo Motors	7	Finger, thumb, and wrist actuation	25	175
PLA Filament (1 kg)	1	Structural and non-structural parts	80	80
PETG Filament (2 kg)	2	Load-bearing components	90	180
Piezo Sensor	1	Force/contact detection	5	5
NTC Thermistor	1	Temperature monitoring and safety	3	3
Vibration Motor	1	Vibrotactile sensory feedback	8	8
Miscellaneous (wires, connectors, fasteners)	—	Assembly and integration	20	20

Total Estimated Cost				**546 RM**

3.8.3 Entrepreneurship

There has been a notable advancement of additive manufacturing and embedded biomedical systems. These advancements have made it possible to create a low-cost and customisable prosthetic arm with sensory feedback. The open-source designs, 3D printing and use of non-invasive EMG control systems are a low-cost alternative to standard commercial prosthetics that are dependent on proprietary hardware, complicated calibration and high-cost manufacturing. The proposed prosthetic design and methodology reduce the barrier of entry into the adoption of assistive technology by removing costly fabrication techniques and closed-source control systems and designs.

Traditional prostheses are also limited by high prices and a lack of customization for users. These challenges effect amputees, especially in low and middle income areas. Furthermore, the approach applied to this system uses PLA and PETG for 3D printing, an ESP32 microcontroller, feedback from electronic sensors and actuation through servo motors to create a functional prosthetic device at a fraction of the price. High affordability, ease of customization and modularity of this design make it appealing to hospitals, rehabilitation facilities, non-governmental organizations, and angel investors who want to provide an individualized solution in the world of prosthetics.

This project is a step towards the future where assistive robotic technologies are readily available. Moreover, the gap between the high-tech capabilities of the prosthetics and their affordability in the real world will be bridged. The suggested prosthetic system integrates digital design, embedded systems control and sensory feedback to promote inclusive innovation and emphasize the high entrepreneurial potential it presents to global healthcare markets.

3.9 Summary

In this chapter, an investigation on 3D-printed prosthetic hand systems, EMG based control methods and sensory feedback techniques was performed.

The tools and techniques required to implement the project were outlined and discussed with selections being made using engineering principles and project requirements. Proposed methods and algorithms were illustrated with flowcharts were appropriate and the prosthetic arms additional shoulder mechanism concept design was conceived. EMG signal conditioning methods were discussed and elaborated on with clear digital signal processing techniques being implemented. Supporting circuits and designs were also added in this section to give a bird’s eye view of circuits. Furthermore, project management, financial considerations were also discussed as well as the measures put in place to ensure that the project complies with professional engineering practices. Lastly, considerations were also taken into account to ensure that the project complies with the Board of engineers Malaysia code of professional conduct.

CHAPTER 4
FINAL DESIGN AND SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter consists of the final design of the prosthetic arm as well as the implementation of the EMG and sensory feedback system. Outlined in this chapter is the systems hardware integration, software architecture and key modifications done to the initial concept design. Following the initial design phase many improvements to the system were implemented to address challenges, these include EMG signal noise reduction, Hackberry arm design modifications and control system reworks. The final implementation of the system uses the ESP32 Microcontroller, single channel EMG control, tactile sensing through the use of piezoelectric sensors and thermistors. These were done to ensure the implementation of the heat reflex and response mechanism for safety of the user and the arm as well as haptic feedback delivery for texture detection.

This chapter shows the design evaluation for each component beginning with justification for important modifications and redesign decisions. This is followed by an updated system methodology and control framework. The hardware design and system circuit schematics provide a birds-eye-view of sensor placement, actuator integration and the control systems architecture. The software implementation section shows how the EMG signals are processed and a calibration and control algorithm are carried out. Lastly, this chapter conveys the operational principles of the prosthetic arm, these are supported by the hardware and software integrations and their outcomes.

4.2 System Implementation

4.2.1 Justification for Changes Done

Several design and system modifications were introduced during the development of the arm to help improve functionality, save costs and ensure reliable control of the prosthetic arm system. Each modification was made with rationale and detailed consideration for other possible solutions, the following paragraphs address this.

Transition to a Single-Channel EMG system

The initial method for EMG signal acquisition from the user was to use a multi-channel EMG Setup. This is considered a more complex EMG acquisition approach involving multiple signal inputs. This would require the ESP32 to handle multiple computational tasks and decrease its reliability and responsiveness. Thus, the move to a single-channel EMG system would lighten the load on the microcontroller with lower memory utilization and improve real-time performance. Furthermore, having a single acquisition channel will substantially reduce the systems likelihood of cross-talk between adjacent muscles where electrical activity unintentionally alters measurements which makes the signal easier to read. This move also aids in reducing the end user’s setup time and complexity. Placing multiple electrodes each time in the correct positions can be cumbersome to end users and increase the variability between placements.

Removal of Dataset-Based Control and Implementation of User Calibration and training for EMG Control

The use of surface EMG electrodes presents substantial variability between end users due to their physiological differences. Physiological differences such as skin impedance, muscle mass distribution and electrode placement position greatly alter the end output signal and measurement strength. The use of a dataset driven approach to calibrate the signal introduces a lack of attention to these physiological differences especially when real-world users deviate from the training conditions performed in the dataset.

An individual based user calibration system gives each specific user a unique activation threshold and parameters for control to improve the arms end user compatibility. The addition of a user training method enables adaption to each user unique muscle activation patterns without requiring the large computational need for a Dataset, further reducing the processing strain on the ESP32. This further removes the need for an offline processing infrastructure and introduces a plug and play solution for users. A calibration-based system also provides a way for adaptability to long-term physiological changes for users such as long-term muscle fatigue or an altered electrode placement. This ensure that throughout a user's physiological change the prosthetic remains usable.

Active Wrist Mechanism Redesign

The original Hackberry EXII arm design consisted of a passive wrist articulation design. This limits the end users object manipulation capabilities. Introducing an active wrist design allows for more range of motion for the user, similar to that of a natural hand. This will increase the degrees of freedom of the arm and improve its functional dexterity. The addition of an active wrist will lower strain on the user's shoulder and elbow as these joints will then compensate for the lack of any wrist actuation. This greater range of motion of the arm will aid in object interaction as it increases its behaviour to mimic upper limb kinematics.

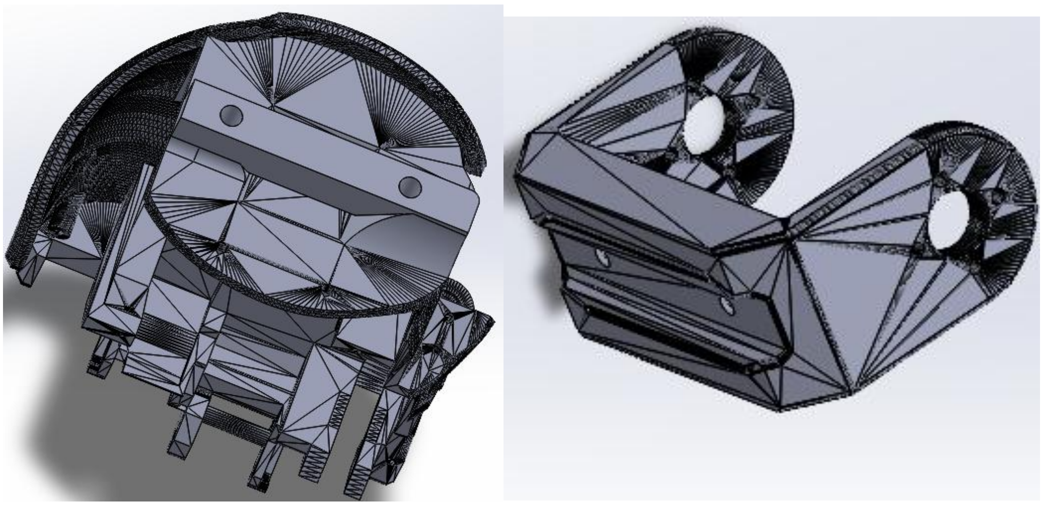


Figure 4.1: Active Palm wrist and Fork Redesign

Figure 4.1 shows the redesigned palm casing end with holes at the to fit M3 threaded inserts. The inserts are screwed in and attached to a form which will be mounted to the servo for movement in 180 degrees.

Integrated Elbow Extension Redesign

The initial decision to use an external attachment mechanism to introduce a secondary structural interface point. This point would see increased stress concentration and a higher likelihood of failure. The use of an integrated elbow joint structure would improve load distribution throughout the prosthetics frame. Furthermore, this would reduce the overall component count while improving long term mechanical reliability as there is now less point of possible failure.

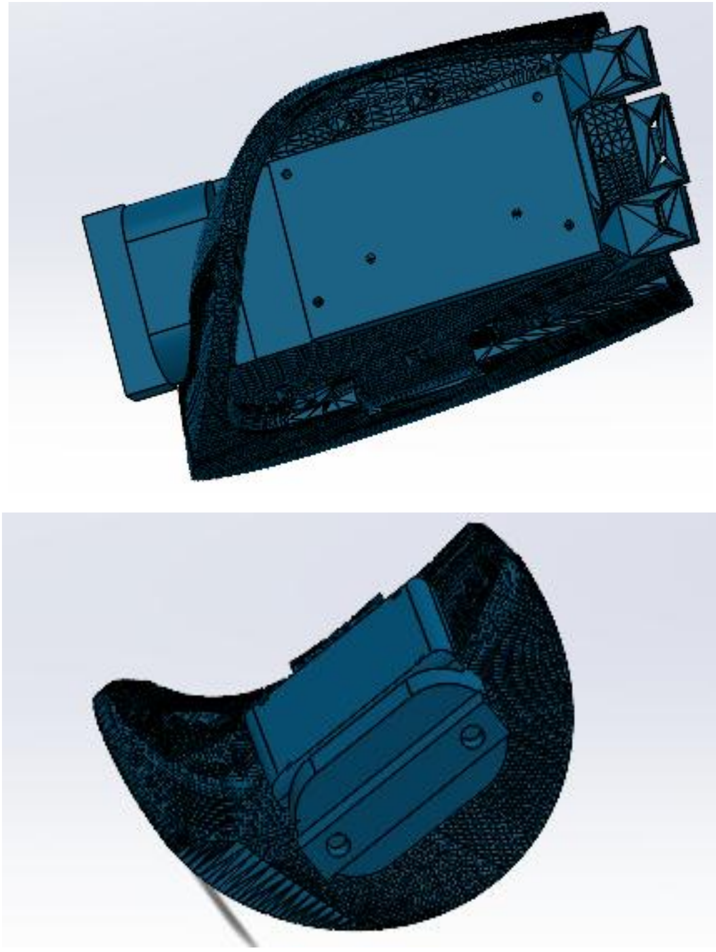


Figure 4.2: Elbow Joint Redesign

Figure 4.2 Shows the Elbow joint attachment redesign CAD model. The design features an added end with slots to fit threaded M3 screw inserts. The slots attach to a fork of similar size to the one used for the wrist and servo housing. Spots for the buck converters were added on either side to easily mount them to the arms chassis as well.

4.2.2 Updated System Architecture and Methodology

In **Figure 4.3** the ESP32 serves as the central brain of the prosthetic system. It handles real time EMG signal acquisition, signal conditioning, feature extraction and control of the actuators as well as getting data from sensors. A single channel surface EMG system is used to capture the users muscle intentions. Through 2 active electrodes and 1 reference electrode the ESP32 ADC processes the motion of them through calibrated thresholds and mapping.

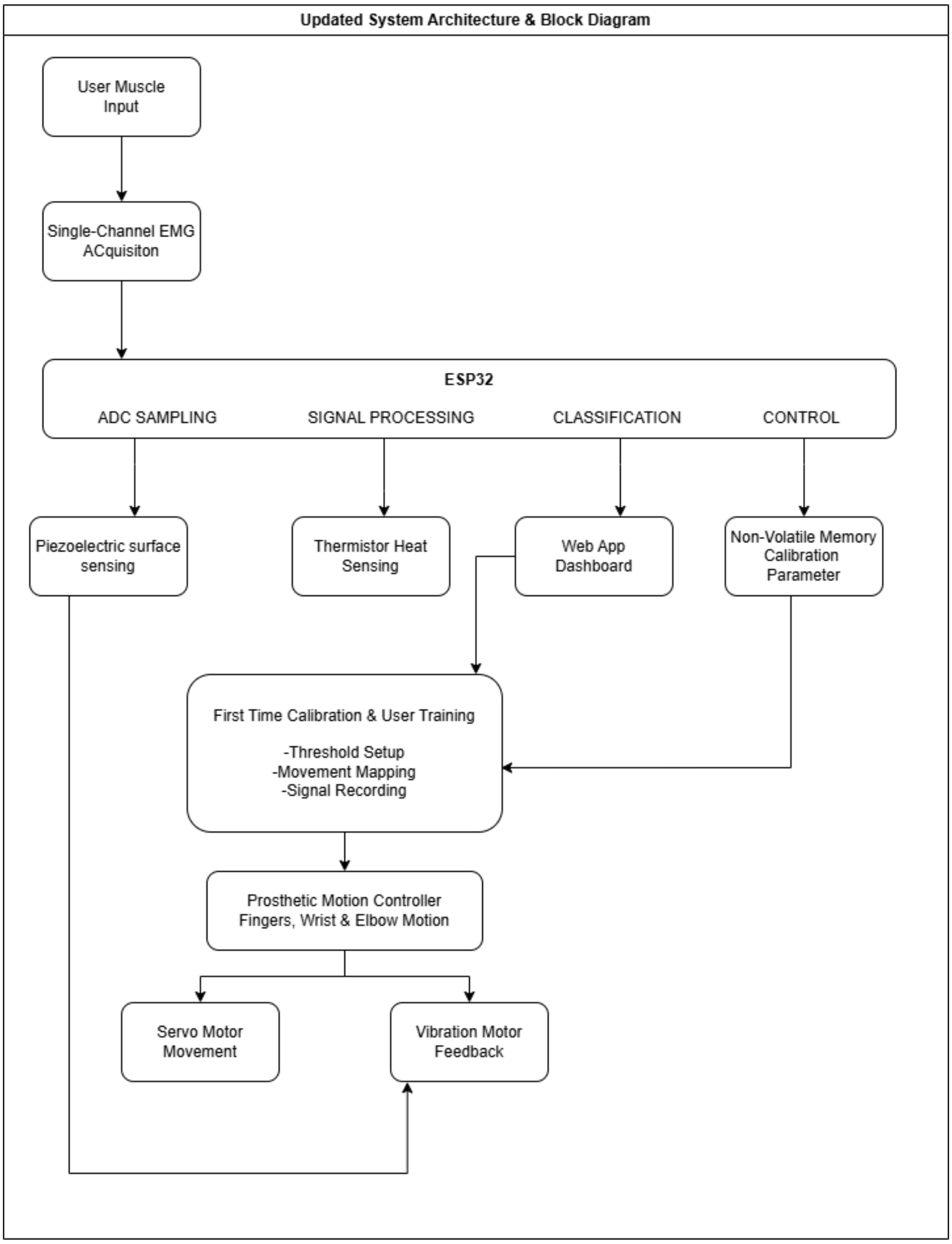


Figure 4.3 Updated System Architecture and Methodology

To increase user-friendliness a Wi-Fi based dashboard is used during the initial calibration phase of the arm. It is here where ethe user records specific muscle activation patterns for intended prosthetic motion. These patterns and values are then stored directly onto the ESP32s non-volatile memory for later use even after the system has been fully shut down. During normal use of the arm these stored values are then continuously matched to the real time signal being produced at any given time by the users' muscles. Once these are decoded the system then decides which gesture to produce based on that initial calibration and mapping. For the feedback system, piezoelectric sensors are integrated for surface contact sensing and is then paired with a vibration motor to give user a sense of feel of objects. This is paired with a thermistor to provide safety for the user and the arm. Just like a normal hand, involuntary movement is done here when the thermistor detects rapid temperature increase to thresholds unsafe for humans and the arms plastics and electronics. It reacts to this temperature by pulling arm from the heat source and warning the user as such.

The overall system integrates all these sensors and actuators to create an intuitive and low latency control system. The system adapts to the users specific EMG signals and provides feedback to the users surroundings to allow for a more fluid and natural experience.

4.2.3 Prosthetic system and Hardware Design

The prosthetic System utilized the HACKBERRY EXII arms initial design with modifications made to the wrist joint to make it active. The same treatment was applied to allow for an elbow joint to be added for better movement and functionality. **Figure 4.4** Shows the completed system.

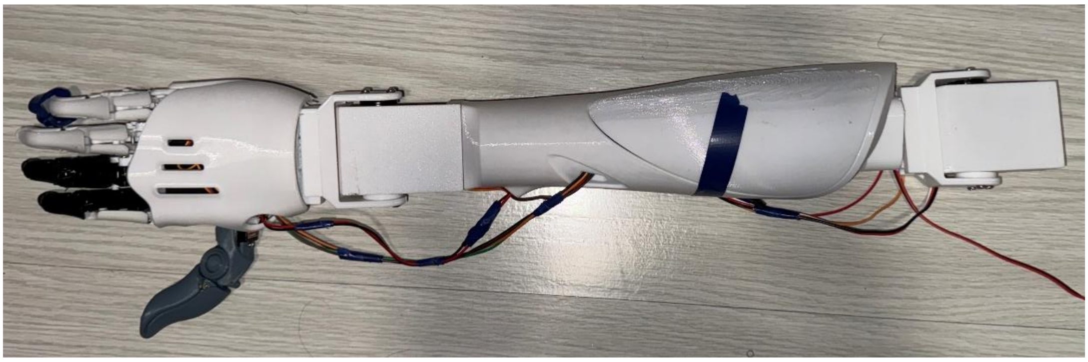


Figure 4.4: Completed Prosthetic System

Cable channels present on the original design were used to route cables cleanly around the arm. A housing for the ESP32 was created and attached to the bottom of the forearm as show in **Figure 4.5**

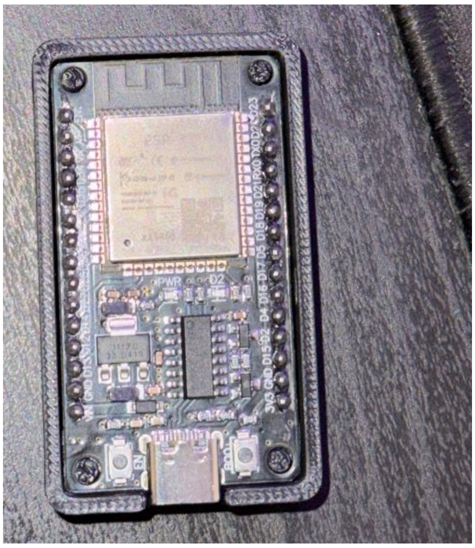


Figure 4.5: ESP32 Housing

Buck convertor mounts were added to each end of the forearm, the first buck convertor is set to 6V to provide steady voltage for the servo motors for finger actuation and wrist movement while the second buck Convertor was set to 8.4V to provide power to the elbow servo as it draws more voltage as shown in **Figure 4.6**.

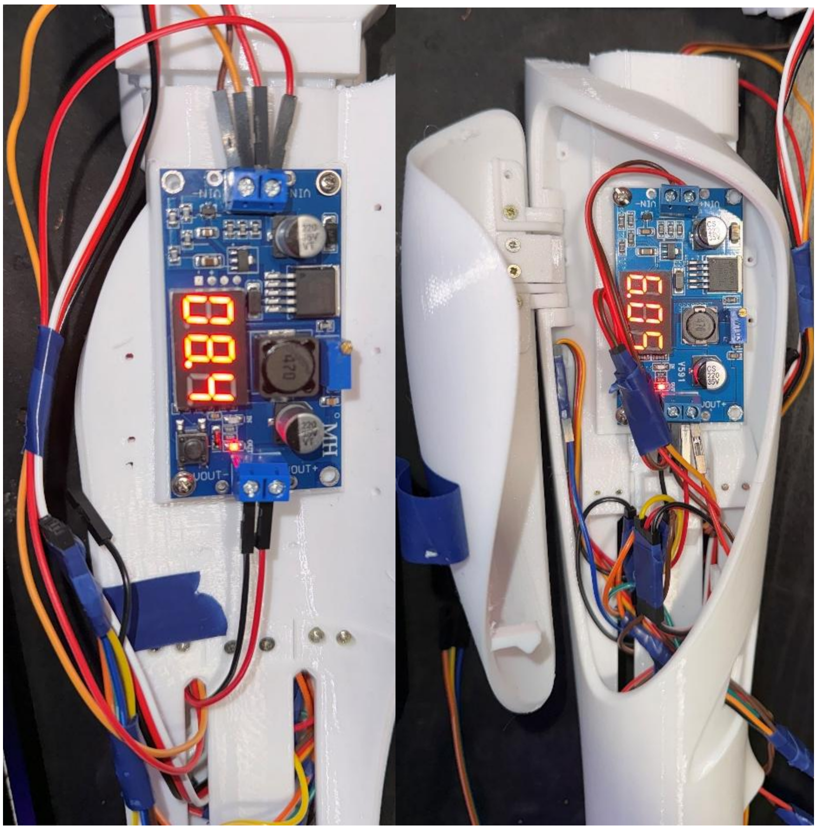


Figure 4.6: Step-down Buck Converters Mounting

Figure 4.7 shows the servo motors for finger actuation that are placed and housed within the palm itself. The index finger receives the biggest servo, the HD3001HB for increased grip while the thumb uses the EMAX ES08MD servo, this is the same type of servo motor used by the remaining fingers as they are all attached to each other these are all connected to the same buck converter as they all require a steady 6V.

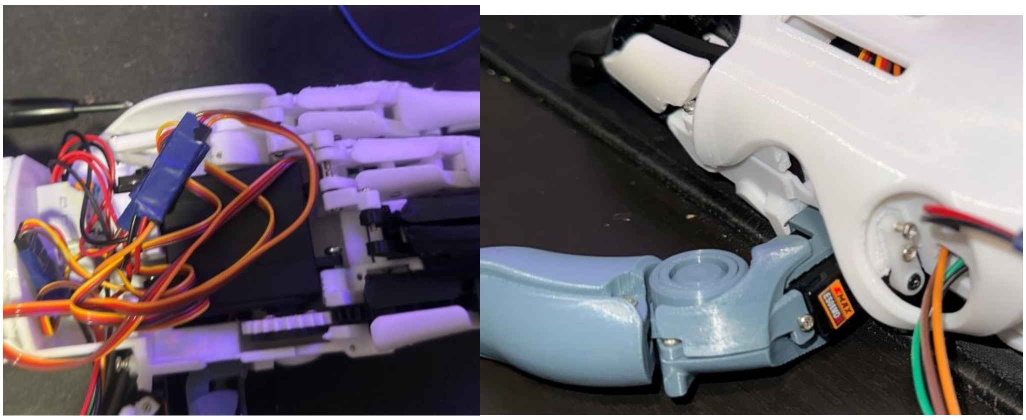


Figure 4.7: Palm Servos

The wrist servo is mounted and attached to a form for rotation of the palm. It sits in a custom designed housing which is attached to the front of the forearm as shown in **Figure 4.8**. the servo motor used is the MG996R and it is connected to the 6V buck Convertor. It is mounted to the housing via a set of M3 screws and threaded inserts which are heat pressed into the palm

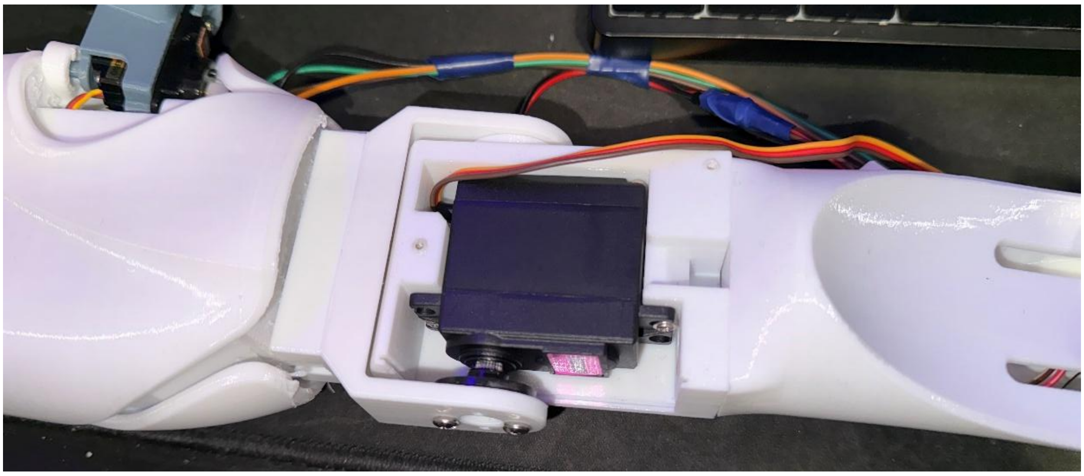


Figure 4.8: Wrist Servo in housing

The elbow servo is then placed in a housing similar to the wrist servo with a modified end of the forearm being attached to it with a fork for added rotation. The servo used here is the TD-7020MG and is connected to the 8.4V Buck converter. The Elbow housing is then connected to the main strap block which features a 12V battery, haptic motor and EMG sensor housing.

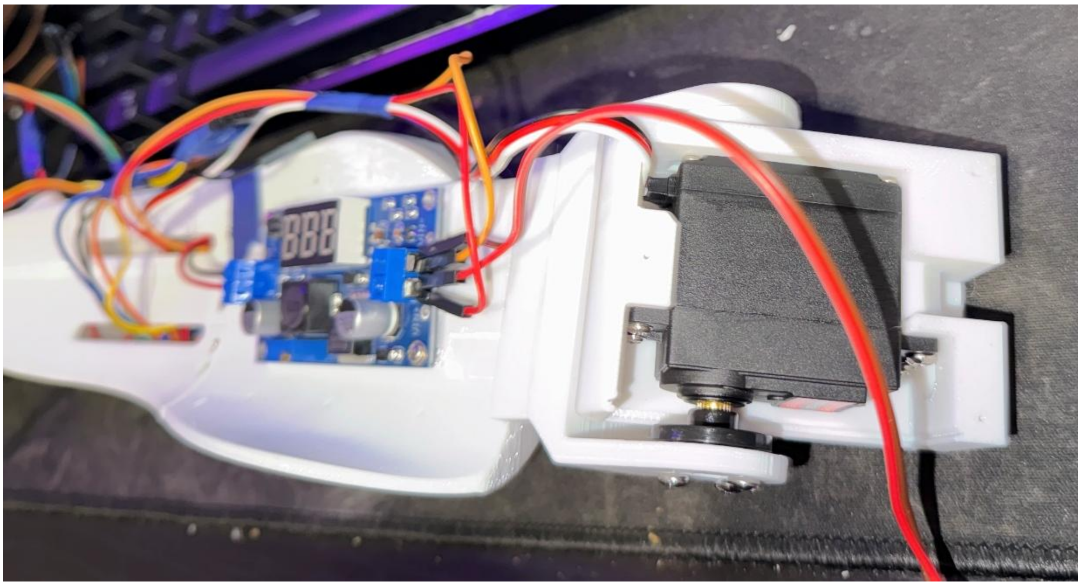


Figure 4.9: Active Elbow Joint Housing

The surface EMG electrodes are them positioned on the arm on the user to ensure consistent muscle signal acquisition as shown in **Figure 4.10**

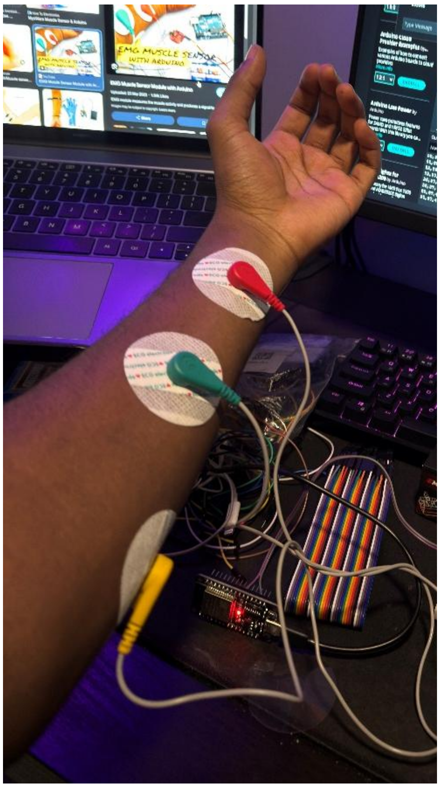


Figure 4.10 EMG Electrodes

Next the piezoelectric sensors are placed on the base of the palm and the fingers to be the most natural place humans use to feel texture. These are then connected directly to the ESP32.

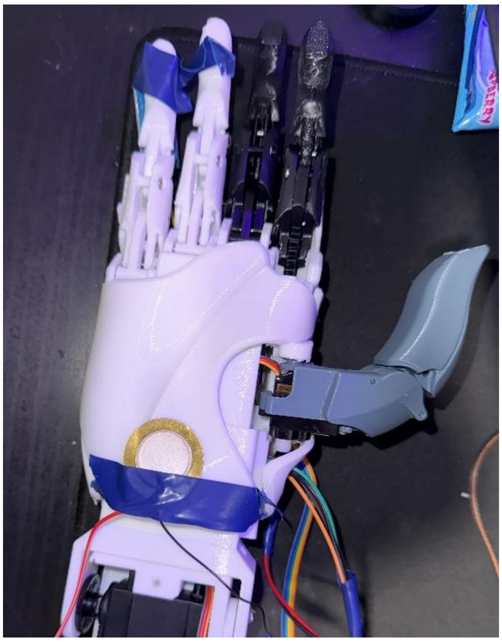


Figure 4.11 : Piezoelectric Sensor

The thermistor is placed on the fingers towards the edge to ensure rapid response from temperature changes shown in **Figure 4.12**

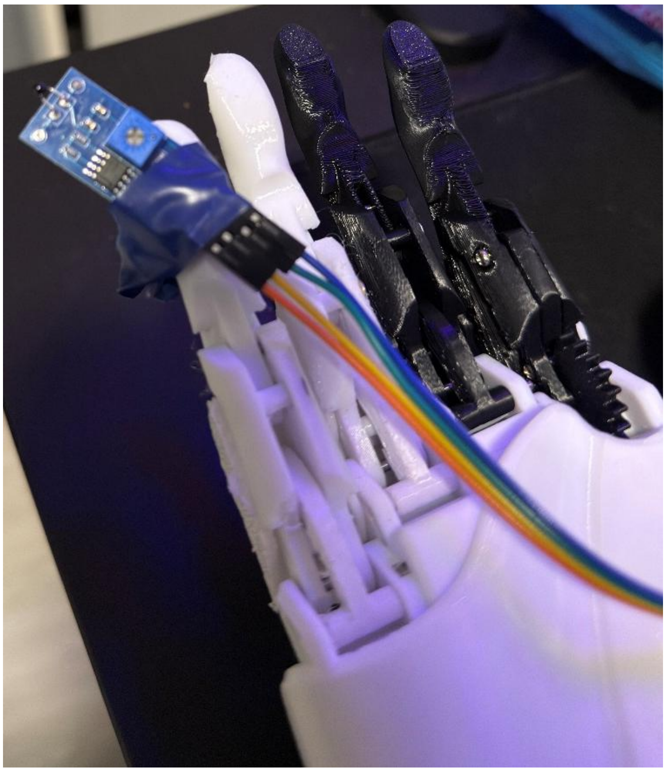


Figure 4.12: Thermistor Unit

The Vibration motor is then placed on the mount that attaches to the user to be as close as possible for haptic feedback as shown in **Figure 4.13**

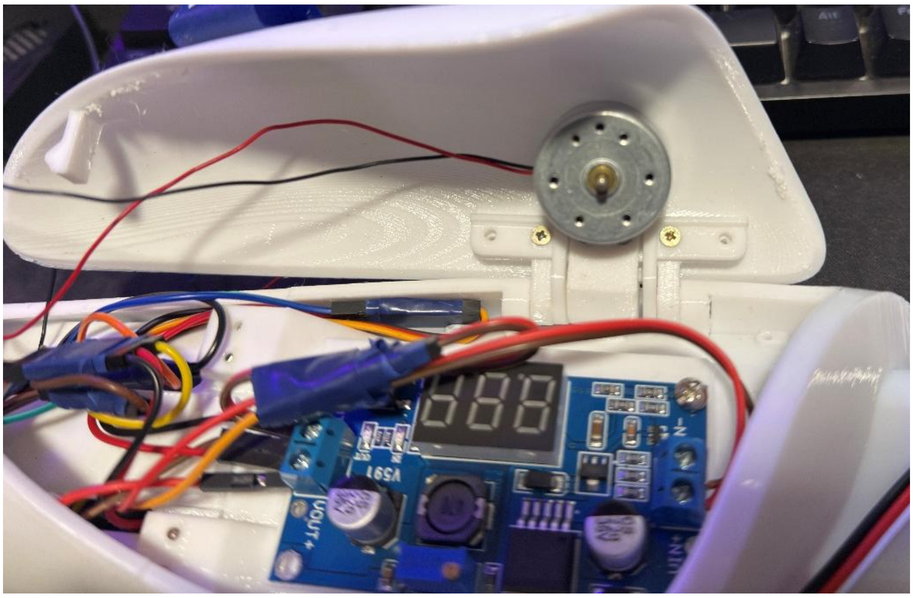


Figure 4.13: Haptic Vibration Motor

4.2.4 Overall System Circuit design

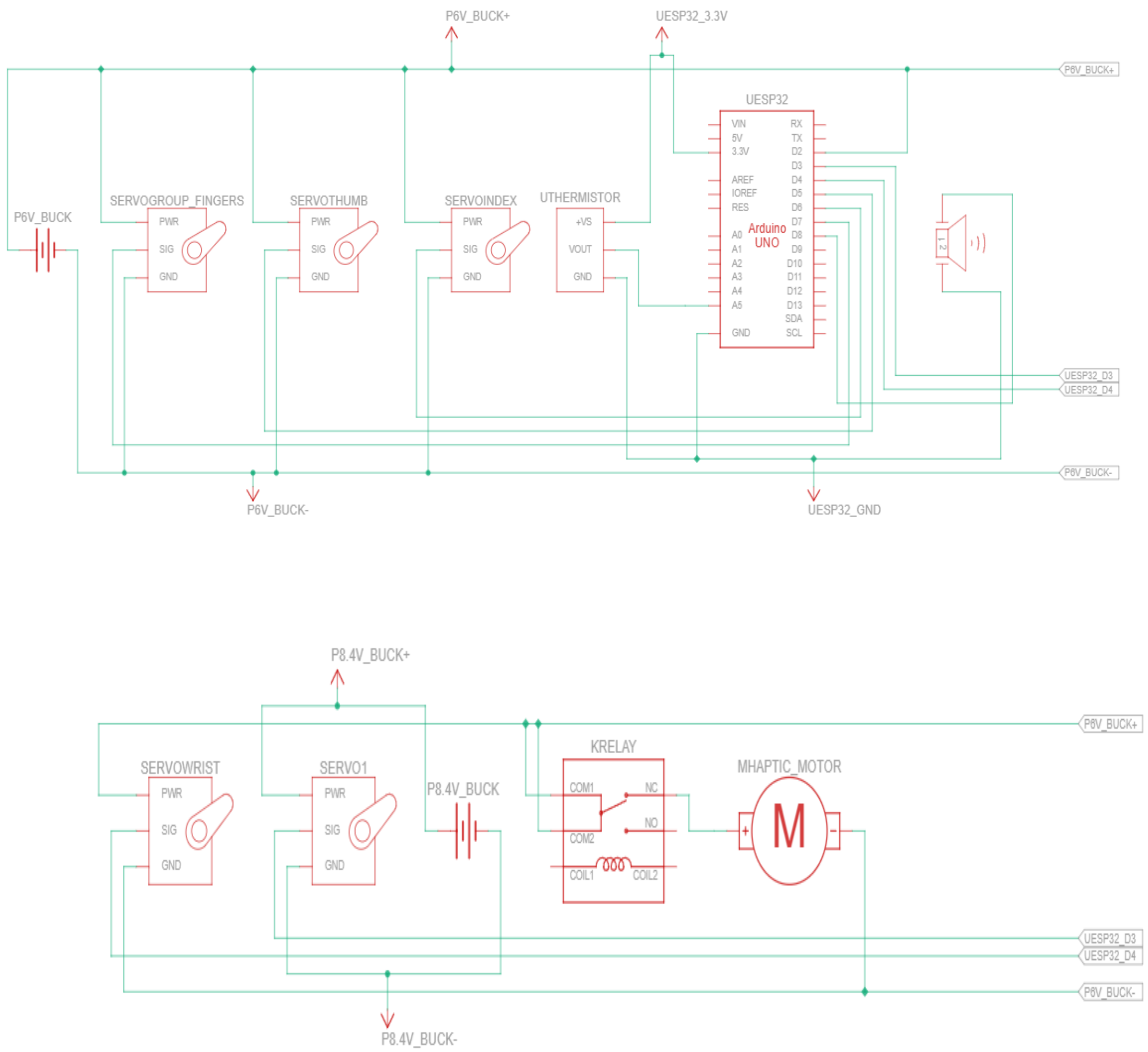


Figure 4.14: Updated System Circuit

The overall circuit shown in **Figure 4.14** is powered by a 12V battery to power servos and a set of two 9V batteries to power the EMG sensors. The circuit shows the overall integrating power distribution of the prosthetic system as a whole. The ESP32 (here displayed as an Arduino Uno for simulation) integrates sensing and processing of inputs for control and feedback. High current actuators are separated from low-noise sensing circuits to reduce interference. Furthermore, multiple power distribution rails are added to improve circuit efficiency and actuator performance.

Microcontroller Unit

The system is controlled and managed by the ESP32 Microcontroller and devboard. This acts as the central brain of the system. As such it is paramount to have a steady supply of power to it not only for safety but for control of the prosthetic arm. The ESP32 Provides a 34 general purpose input and output (GPIO) pins. A key feature of this is its ADC for analogue signal input, this is used to input the signal from the EMG sensor. PWM outputs are used for servo motor control and the built in Wi-Fi is used for user configuration and setup with the web-based dashboard.

Power Source

Power for the entire system is provided by a 12V lithium-ion battery. Its voltage is too large for each component in the system thus the use of buck convertors to step down the voltage to useable levels is done here to generate different voltage rails. A 6V buck convertor is used to create a rail that power the three palm servos for finger actuation, this same power rail is then used to power the wrist servo. A second buck convertor is then used to power the elbow servo, this being the servo that carries most of the system weight, it requires a steady 8.4V from a separate buck convertor for high torque movement. Furthermore, a regulated 5V supply powers the ESP32 and lower power electronics such as the haptic feedback motor and the thermistor. Common ground connections are implemented across all components to maintain signal integrity.

Inputs Components – Sensors

Inputs in this system is driven mainly the single channel surface EMG sensor. This sensors output is connected directly to the ESP32s ADC input for real-time signal processing. The piezoelectric sensors are also connected to another ADC input of the ESP32. The thermistor sensor here is also connected here to the 5V source for power as well as a voltage divider circuit and its ADC is also connected to the ESP32.

Output Components – Actuators and Feedback System

Here for the output actuators, the 3 palm servos are used here and are firstly connected to the ESP32 via PWM Signal Pins. The same is done to the wrist and elbow servo motors and they all share a common ground with the ESP32 for signal integrity. The vibration motor is first connected to a relay for power and is only actuated and activated by the ESP32s high output signal to turn it on and off.

4.2.5 Software and Pseudocode

Calibration and Training Algorithm

```
START

INITIALISE ESP32
INITIALISE ADC, WIFI, MEMORY MODULE

CHECK calibration data IN non volatile memory

IF calibration data NOT FOUND THEN

    START calibration mode
    CONNECT TO web dashboard

    READ resting emg baseline

    FOR each movement class (finger, wrist, elbow)

        DISPLAY instruction to user

        SET sample buffer = empty

        FOR i = 1 TO N samples

            READ raw emg
            FILTER raw emg
            STORE filtered emg INTO sample buffer

        END FOR

        COMPUTE mean value = AVERAGE(sample buffer)
        COMPUTE peak value = MAX(sample buffer)
        COMPUTE threshold = mean value + offset factor

        STORE movement class parameters
        (mean value, peak value, threshold)

    END FOR

    SAVE calibration data TO non volatile memory

END IF

END
```

Figure 4.15 Calibration & Training Algorithm Pseudocode

Figure 4.15 shows the systems calibration and training algorithm. To initialize the system first checks for existing calibration data in the ESP32s memory at startup. If no data is found here the system, then enters into calibration mode via the Wi-Fi based dashboard. The user performs the instructions as they appear on screen for each movement classification. Multiple samples for each movement are collected in this case. The raw signal is always filtered to remove any noise and artifacts. Activation thresholds are then computed using an adaptive offset that changes based on the user's calibration profile. The final mapped values are then stored into the ESP32s non volatile memory for later use.

EMG Signal Processing and Classification Algorithm

```
START

LOAD calibration data FROM memory

WHILE system is active

    READ raw emg signal

    APPLY filter for noise reduction

    COMPUTE activation level

    IF activation level > finger threshold THEN
        movement command = FINGER FLEXION

    ELSE IF activation level > wrist threshold THEN
        movement command = WRIST MOVEMENT

    ELSE IF activation level > elbow threshold THEN
        movement command = ELBOW MOVEMENT

    ELSE
        movement command = NO ACTION
    END IF

END WHILE
```

Figure 4.16 EMG Signal Processing Pseudocode

Figure 4.16 shows the ESP32 continuously reading the raw EMG signals from its ADC input where the EMG signal is filtered to reduce noise. An activation threshold is computed here from the processed de-noised signal. This threshold is compared against a stored calibration threshold value. The movements of the arm are classified based on these into finger, wrist and elbow movements. If the threshold is not exceeded the system will remain at rest and remain idle.

Sensory Feedback Algorithm

```
START

WHILE system is active

    READ piezo sensors
    READ thermistor value

    COMPUTE contact intensity FROM piezo sensors
    COMPUTE temperature level FROM thermistor

    IF contact intensity > threshold THEN
        ACTIVATE vibration motor proportional response
    END IF

    IF temperature level > safety limit THEN
        ACTIVATE vibration motor warning pattern
    END IF

END WHILE
```

Figure 4.17: Sensory Feedback Pseudocode

For the sensory feedback system as shown in **Figure 4.17** shows the piezoelectric sensor detect surface roughness through different vibrations caused by them and the contact intensity. The thermistor is also shown here to monitor temperature during object interaction. The readings from these sensors are then matched to the vibration motor proportionally with the piezoelectric and a warning vibration from the thermistors to provide a close-loop haptic feedback system for the user.

Involuntary Withdrawal Response Algorithm

```
START

WHILE system is active

    READ thermistor value

    IF temperature level > safety threshold THEN

        OVERRIDE user commands

        ACTIVATE warning vibration

        MOVE elbow servo to safe position

    ELSE

        ENABLE normal EMG control

    END IF

END WHILE
```

Figure 4.18: Involuntary Movement Pseudocode

Figure 4.18 shows the Involuntary withdrawal Response Algorithm. Here the system with monitor temperatures from the thermistor which are compared to prestored safety threshold. If the temperature here exceeds the threshold the system identifies this as a potential thermal and heat hazard. An involuntary response is then triggered without requiring the users input where the elbow servo lightly moves and is actuated to retract away from the source of heat. At the same time, the vibration motor activates to issue a warning to the user. Normal operation of the arm is overridden during this response to prevent further damage to the arms components until the temperature readings go below dangerous levels.

4.2.6 Working Principle

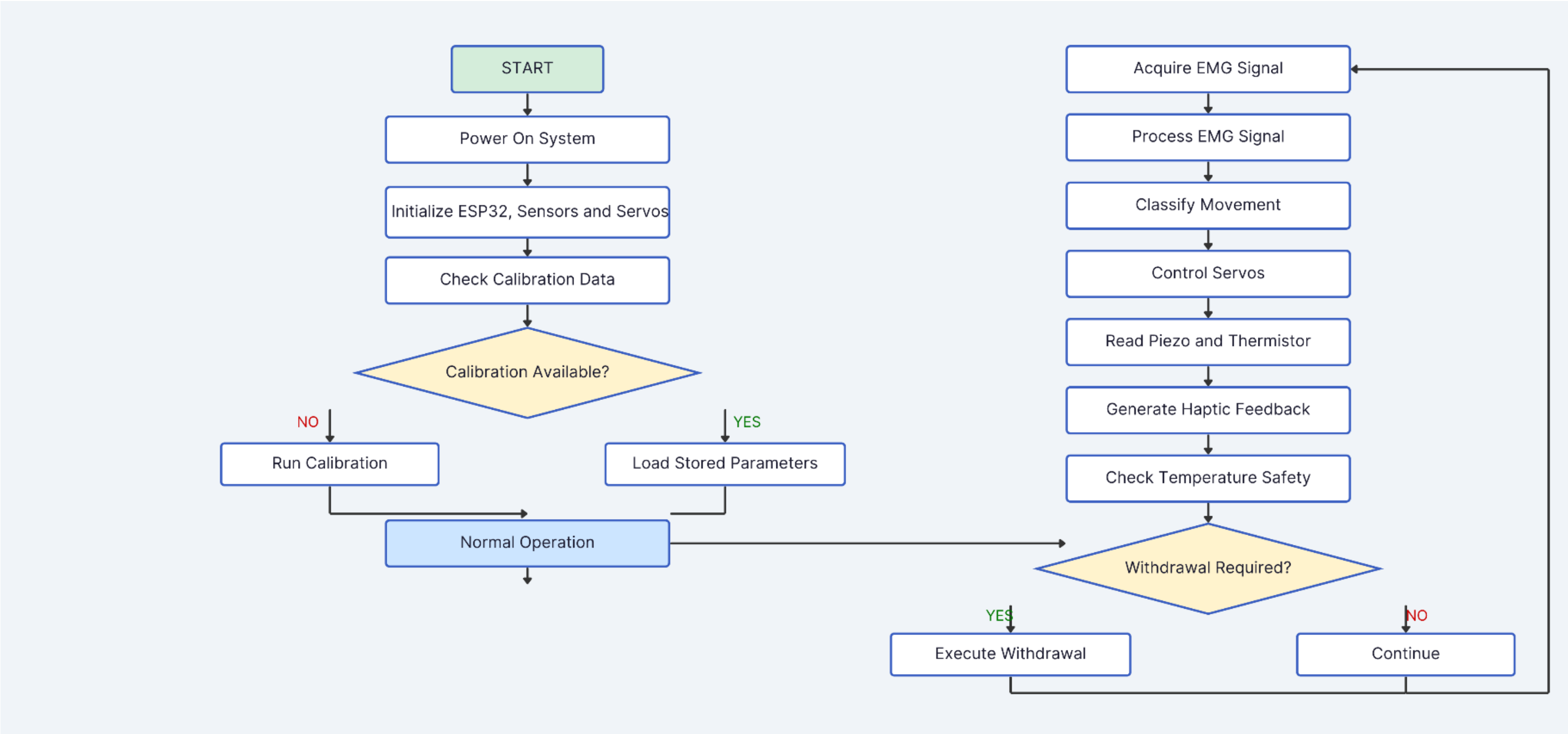


Figure 4.19: Overall System Working Principle

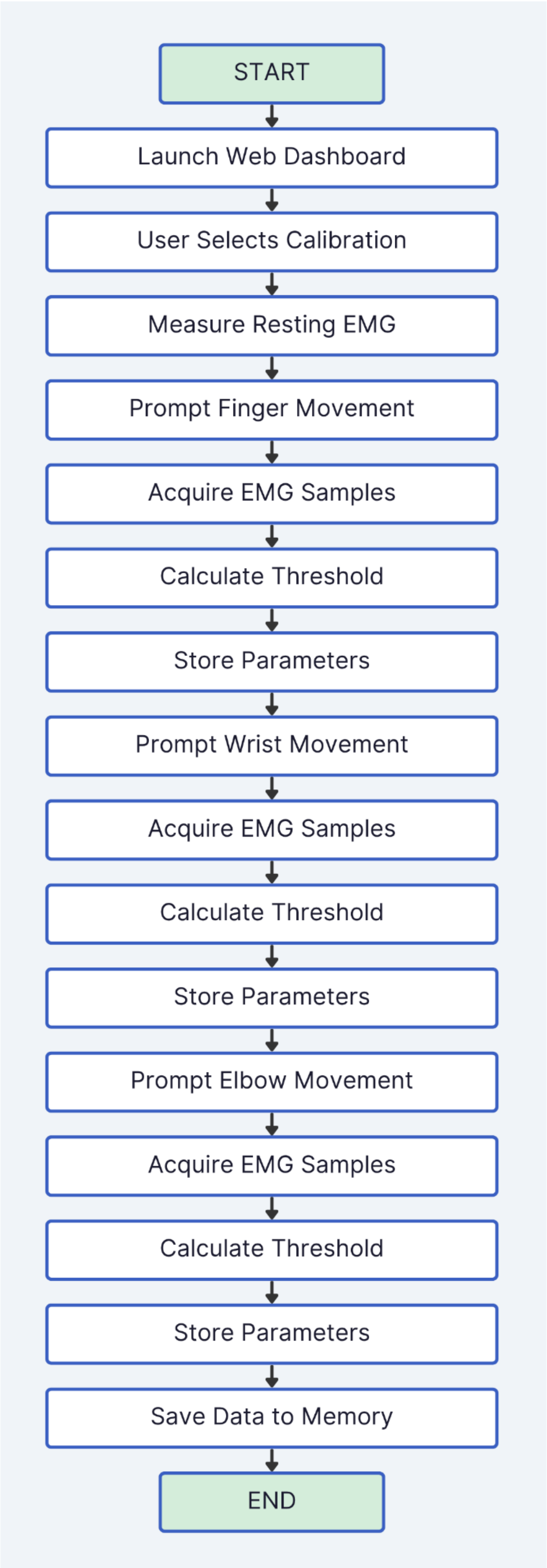


Figure 4.20: Calibration and Training Flowchart

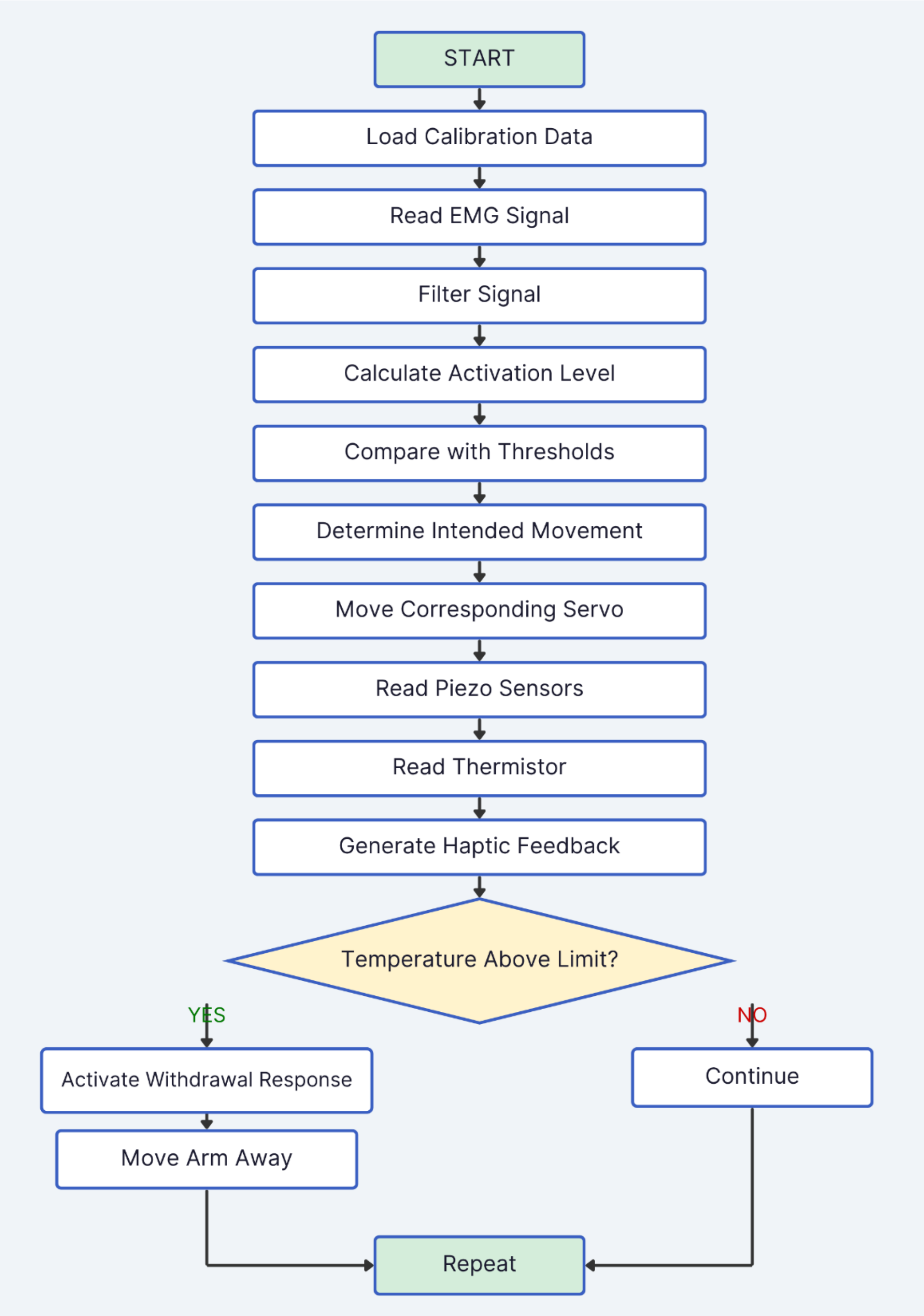


Figure 4.21 : Normal Operation and Safety Flowchart

4.3 Hardware and software results

4.3.1 Calibration WEBAPP

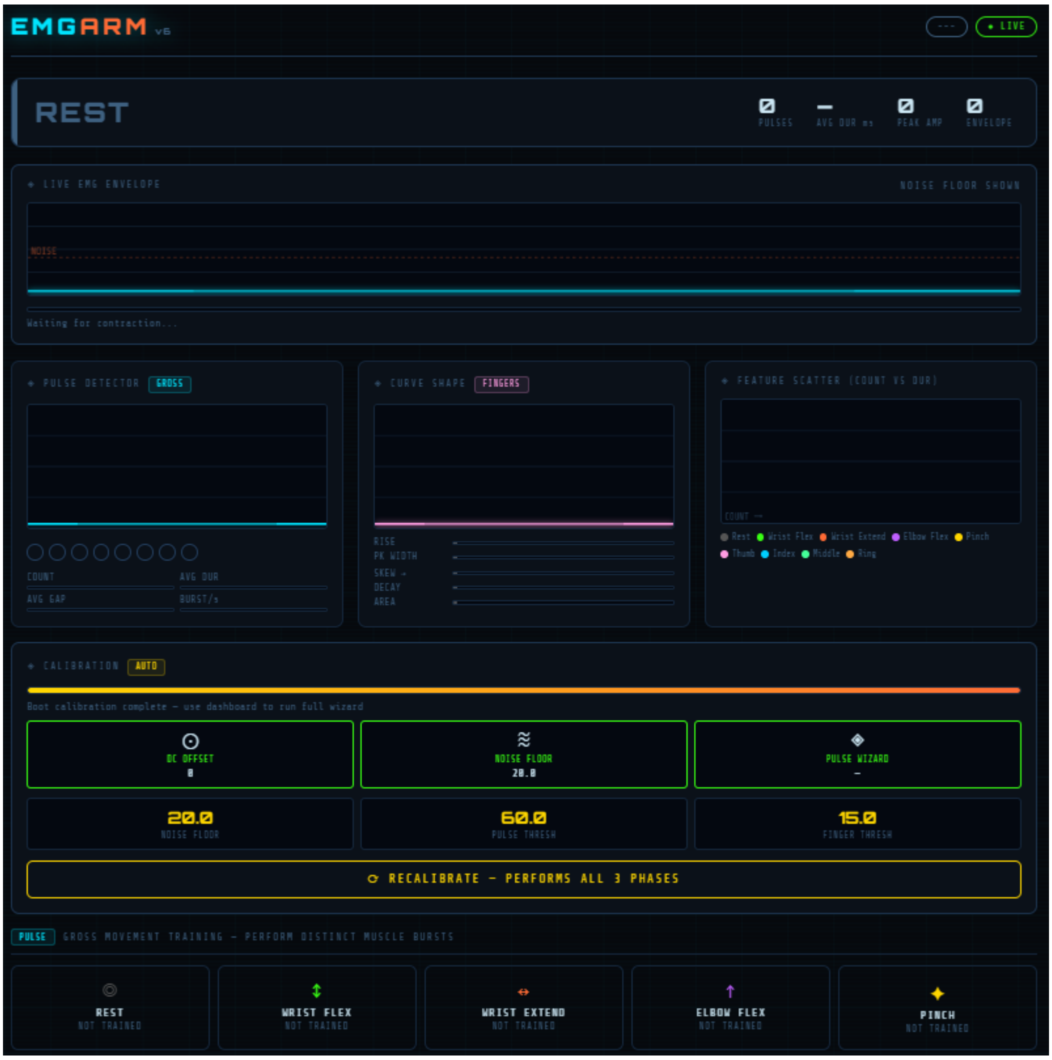


Figure 4.22

Figure 4.22 shows the ESP32 hosted webapp that can allow the user to calibrate the EMG levels to their specific physiology. The Dashboard guides the user through the calibration process. The real-time EMG values are displayed during muscle contractions, and the system automatically detects to assist the user when generating consistent activation levels. The system records a baseline, peak and most importantly, threshold values for each movement. These values are then stored in the ESP32s non-volatile memory for later use and confirms this to the user when the calibration is complete.

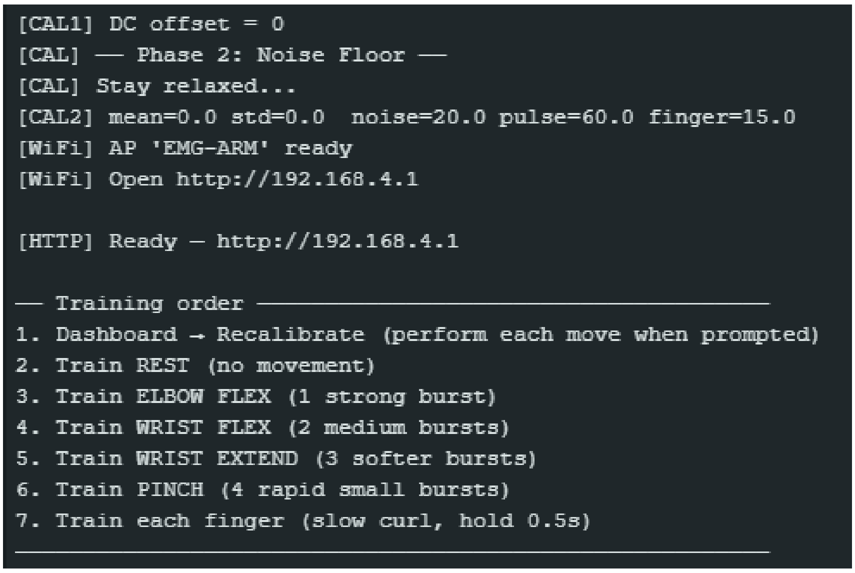


Figure 4.23: Serial Monitor Calibration Steps

Figure 4.23 shows the steps required to calibrate the arm and train it on specific movements as shown in the ESP32s serial monitor. This is done as a first setup process and is done through the serial monitor in the event that the ESP32 is unable to relay the Webapp dashboard.

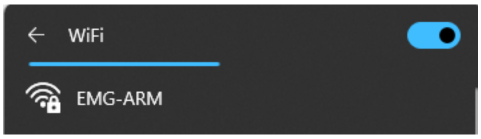


Figure 4.24: EMG-ARM Wi-Fi Network

The user simply connects to the WIFI network as shown in Figure 4.24 created by the ESP32 and proceeds to the provided IP address from the serial monitor.

4.3.2 Movement Recognition Output

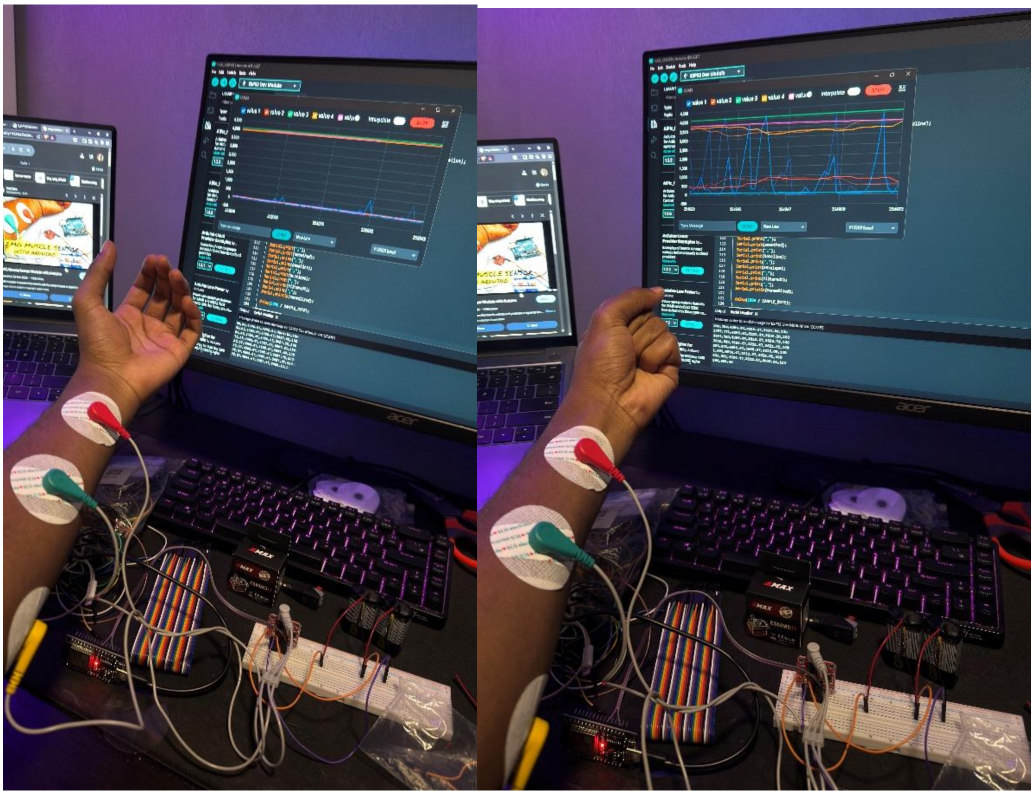
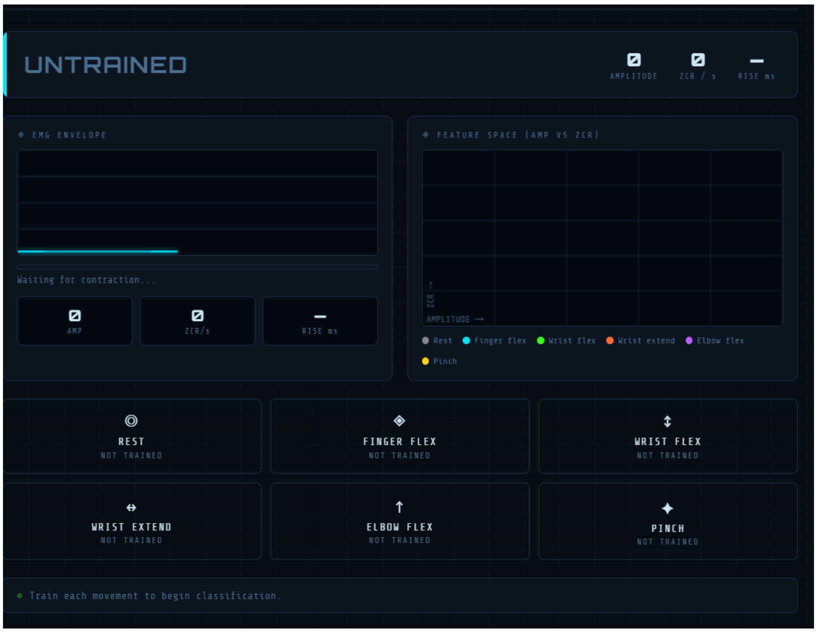


Figure 4.25: Initial EMG testing with Forearm

Figure 4.25 shows the system being initially tested with real-time signal graphs displayed in the background. The surface EMG electrodes are attached to the arm to get initial readings. The two images show rest (on the left) and contraction (on the right) of the palm. Clear amplitude differences are visible with the image where the arm is at rest having a lower amplitude and spikes in the graph but noise and artifacts are still present as this is done prior to calibration. The image on the right with contraction portrays amplitude spikes much higher and this confirms stable signal acquisition under real world conditions.




```
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
[EMG] amp=0.0 zcr=0.0 rise=0.0 → UNTRAINED  [IDLE 0%]
```

Figure 4.26: Untrained EMG Output

The system performs real-time feature extraction on the incoming EMG signal, including, in this case as shown in **Figure 4.26,** the amplitude, zero crossing rate and the signal rise characteristics. The signal looks at how long the signal rise is for and determines whether or not it is not noise based on the duration of time. If it is too short it determine it as noise. In the figure, the current output is labelled as untrained as it is prior to any calibration and mapping to be done by the user. This demonstrates the use of the safety mechanism of the classification algorithm, where only trained and threshold mapped signals are allowed to trigger movement. This stage is done as the preprocessing layer before movement classification into finger, wrist and elbow movement is done

Figure 4.27: Brief Muscle contraction

Figure 4.27 Shows how the system reacts when a contraction is too brief. It simply disregards it and deems it to be involuntary and or noise. This prevents any unintentional movement due to noise while the figure 4.28 shows the complete opposite and how the system reacts when untrained but recognises when an intentional contraction occurs.

Figure 4.28 Muscle Contraction but Untrained System Reaction


```
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
[EMG] amp=64.0 zcr=187.5 rise=0.0 → REST  [IDLE 0%]
```

Figure 4.29: At Rest Graph after calibration and training

Figure 4.29 shows the systems reaction and plotting of amplitudes for the user to be at rest and for it not to react. This is done after calibration, and this rest value is what is used as the baseline for noise rejection and further contraction.


```
[EMG] amp=891.8 zcr=0.0 rise=288.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=890.8 zcr=0.0 rise=288.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=893.1 zcr=0.0 rise=288.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=888.5 zcr=0.0 rise=288.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=1349.4 zcr=15.6 rise=288.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=1182.9 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=965.7 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=858.4 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=867.1 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=847.7 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=858.5 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
[EMG] amp=825.6 zcr=31.2 rise=6.0 → FINGER_FLEX  [CONFIRMED 100%]
```

Figure 4.30

Figure 4.30 Shows the system reacting to mapping to finger flexing once the user has mapped it to their desired contraction level. The system recognizes this and gives it an accuracy score based on how close it is to the recorded value stored earlier. If the value exceeds it assumes it 100%, only values below it will go below 100%. The steps are then repeated for wrist flex, extension, elbow flex as well as the pinch movement.

4.3.3 Servo Actuation

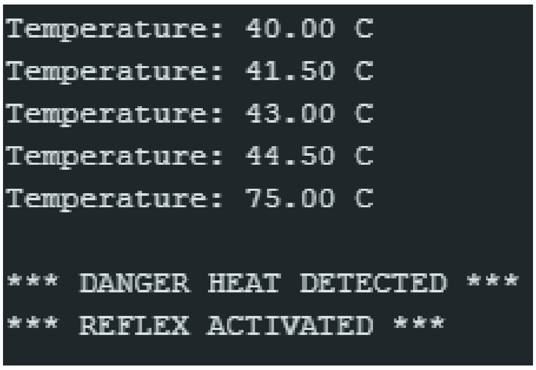
The servo system here is used to control the prosthetics movement and provide its key functionality. Three servos embedded within the palm of the arm operate in coordination to achieve grasping and release actions. The wrist servo provides controlled rotational movement to improve object interaction while the elbow servo enables arm movement as a whole to allow for reaching and lifting.

Figure 4.31

In **Figure 4.31** the palm servos are seen repeatedly doing a grasping and release motion and their angles are confirmed on the serial monitor with their positional output as the ESP32 receives feedback from them. The Palm and wrist servo are next given an action to flex as well as shown in **Figure 4.32**.

Figure 4.32: Wrist Flexion and Extension

4.3.4 Involuntary Heat Safety Reflex System



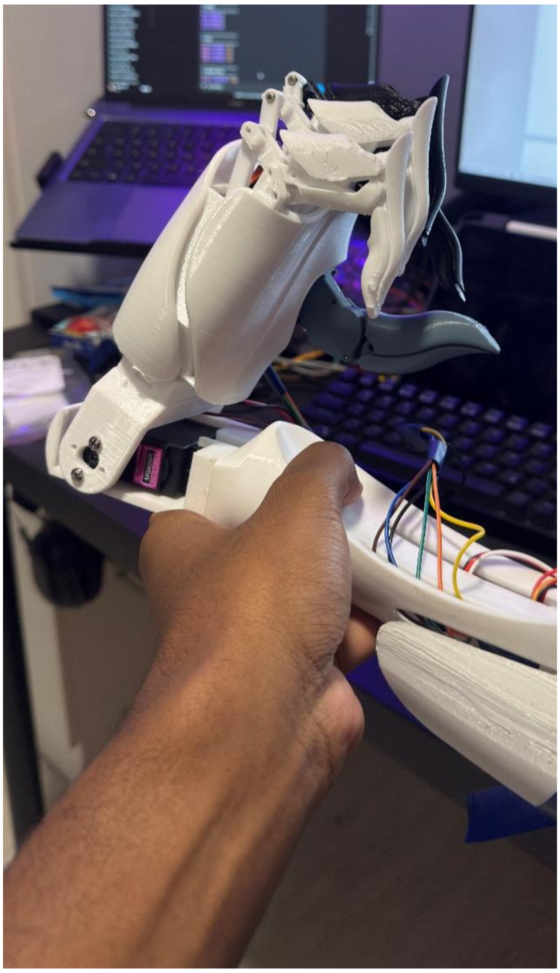


Figure 4.33: Involuntary Safety Reflex System

The prosthetic system implements an automatic withdrawal response as shown in Figure 4.33 when an unsafe temperature level Is detected. The thermistor continues to monitor levels of temperature until they exceed or reach the predefined limit, in this case being 75 degrees. This overrides the systems EMG functionality and control. The wrist servo automatically rotates and the fingers crumple away from the heat source. Meanwhile the vibration motor vibrates rapidly to give the user feedback and a warning. Normal operation of the arm only resumes once safe temperature conditions are met

4.3.5 Haptic Feedback Activation System

```
Piezo: 1200  Surface: NORMAL
Piezo: 3000  Surface: ROUGH
Piezo: 200  Surface: SMOOTH
Piezo: 1200  Surface: NORMAL
Piezo: 3000  Surface: ROUGH
Piezo: 200  Surface: SMOOTH
Piezo: 1200  Surface: NORMAL
Piezo: 3000  Surface: ROUGH
Piezo: 200  Surface: SMOOTH
Piezo: 1200  Surface: NORMAL
Piezo: 3000  Surface: ROUGH
```

Figure 4.34 Piezoelectric Feedback

The haptic feedback system uses the piezoelectric sensor to detect surface smoothness and roughness. Contact intensity here is directly proportional to the output being given to the haptic motor for it vibrate. Rougher surfaces will show from staggered vibrations while smooth ones will do smooth slower ones. This improves user awareness of object contact and grip strength. The feedback loop confirms sensory interaction between the environment and the user.

4.4 Summary

To sum up this chapter as a whole, it presented the final design and implementation of the EMG controlled prosthetic arm by highlighting the integration of mechanical, electrical and software subsystems into one cohesive device. The hardware architecture was successfully developed and integrated into the software for standalone EMG control with calibration for specific user control. The mechanical design incorporated finger, wrist and elbow actuation mechanisms while still maintaining structural stability. The ESP32 was implemented to develop the electronic architecture with dedicated power distribution rails through the use of multiple buck converters. The software system enabled real-time EMG signal acquisition and calibration. This allowed for specific user feature extraction to classify movements and actuator control. A web based calibration interface provided the user a method to personalize training and storage of their specific movement parameters. Hardware validation integrated the use of the piezoelectric sensors and thermistor as well as the vibration motor for safety, environmental feedback and general dexterity improvements for the user. Overall, the developed system addresses the objectives of achieving a personalized prosthetic with EMG control and sensory feedback while still being a standalone system and compact enough to run on embedded architecture.

CHAPTER 5

DISCUSSION – PROJECT FINDINGS & TESTING

5.1 Testing the Proposed Design

The testing phase of the prosthetic arm system was structured to evaluate and analyse the performance of its respective systems. Each test is meant to assess a specific component of said system to ensure both mechanical elements as well as control elements are validated independently of each other before systems overall evaluation. These have been divided into 2 categories, namely hardware testing and software testing. Hardware testing for the system will focus on evaluate the physical comments which include the servo actuation performance, mechanical movement and circuit power capabilities. Software testing will focus on the control systems performance which includes the EMG signal Processing, its response time and latency as well as the accuracy of commands given by the user to motion by the arm.

5.1.1 Test 1 – EMG Level vs Thumb-index Distance

This test evaluates how muscle activation from the user and the strength of signal provided from that affects the prosthetic finger movement. This measures the relationship between input signal intensity, here categorised as low, medium and high and the physical output which in this case will be the distance between the thumb and index finger in millimetres. This test will also identify non-linearity or saturation effects on the system especially higher EMG levels where the servo limits may affect its performance. This is done to validate the systems performance in terms of mapping to be stable, predictable and suitable for prosthetic applications.

Manipulated Variable: EMG signal level

Responding Variable: Thumb-index distance (mm)

Table 5.1:Results of Thumb-Index Accuracy Test

Trial No.	**EMG Level**	**Distance (mm)**	**Accuracy Level (%)**
1	Low	42	95.0
2	Low	40	100.0
3	Low	39	97.5
4	Low	41	97.5
5	Low	38	95.0
6	Medium	28	92.5
7	Medium	26	97.5
8	Medium	25	100.0
9	Medium	27	95.0
10	Medium	24	97.5
11	Medium	23	95.0
12	Medium	22	92.5
13	Medium	21	90.0
14	High	15	75.0
15	High	14	77.5
16	High	13	80.0
17	High	12	82.5
18	High	11	85.0
19	High	10	87.5
20	High	9	90.0
21	High	10	87.5
22	High	8	92.5
23	High	9	90.0
24	High	7	95.0
25	High	6	97.5
26	High	6	97.5
27	High	5	100.0
28	High	6	97.5
29	High	5	100.0
30	High	4	97.5

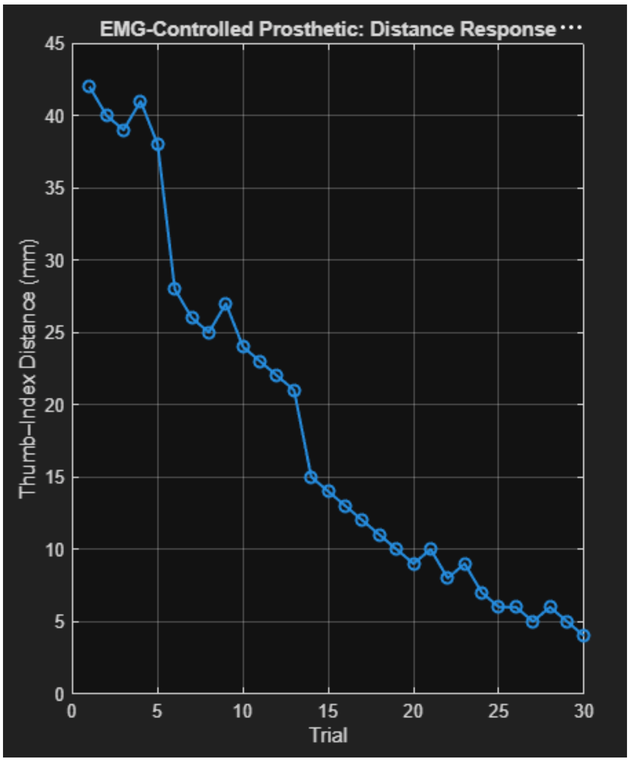


Figure 5.2: Index - Thumb distance Graph

Based on the results plotted, the distance between the Thumb and index finger increases with EMG strength proportionally. The LOW EMG average accuracy results showed a 96% rate; the medium level showed a 95% accuracy rate while the highest EMG level showed an accuracy rate lower at 94% due to mechanical saturation of the fingers as they neared full closure. This also occurs due to non-linear closing behaviours are full pitch from the servos as well as minor EMG signal fluctuations during very strong contractions. The overall trend of these results shows that there is a strong monotonic relationship between EMG intensity and the fingers distance closure.

The prosthetic system achieves high positional accuracy above 90% across all EMG states while proving that EMG intensity can be mapped to specific steps in finger motion and controlled actuation. The minor deviations shown here are high activation levels represents mechanical limitations of the system rather than the control systems lack of control.

5.1.2 Test 2 – Servo internal system latency

For this test, the internal systems latency is tested within the ESP32. Using the serial monitors timestamp feature, the EMG signal detection to servo execution latency can be monitored. This test is repeated 5 times to check consistency and reliability. Each trial represents a repeated measurement under the same conditions.

```
01:59:43.415 -> 45194,EMG_DETECTED
01:59:43.505 -> 45244,SERVO_MOVED
01:59:43.639 -> 45417,EMG_DETECTED
01:59:43.723 -> 45467,SERVO_MOVED
01:59:43.723 -> 45476,EMG_DETECTED
01:59:43.770 -> 45527,SERVO_MOVED
```

Figure 5.3: Serial Monitor internal system latency for Servo Movement

Table 5.2: Results of Servo Timestamp for Latency

Trial	EMG Detection Time (ms)	Servo Movement Time (ms)	Latency (ms)
1	45194	45214	20
2	45419	45467	48
3	45467	45527	60
4	45591	45649	58
5	45780	45848	68

The results in table 5.2 indicate the systems minimum latency across the 5 tests to be 20ms while the maximum latency is 68ms. Across these tests the average internal latency was 50.8ms. This demonstrates the systems real-time response time.

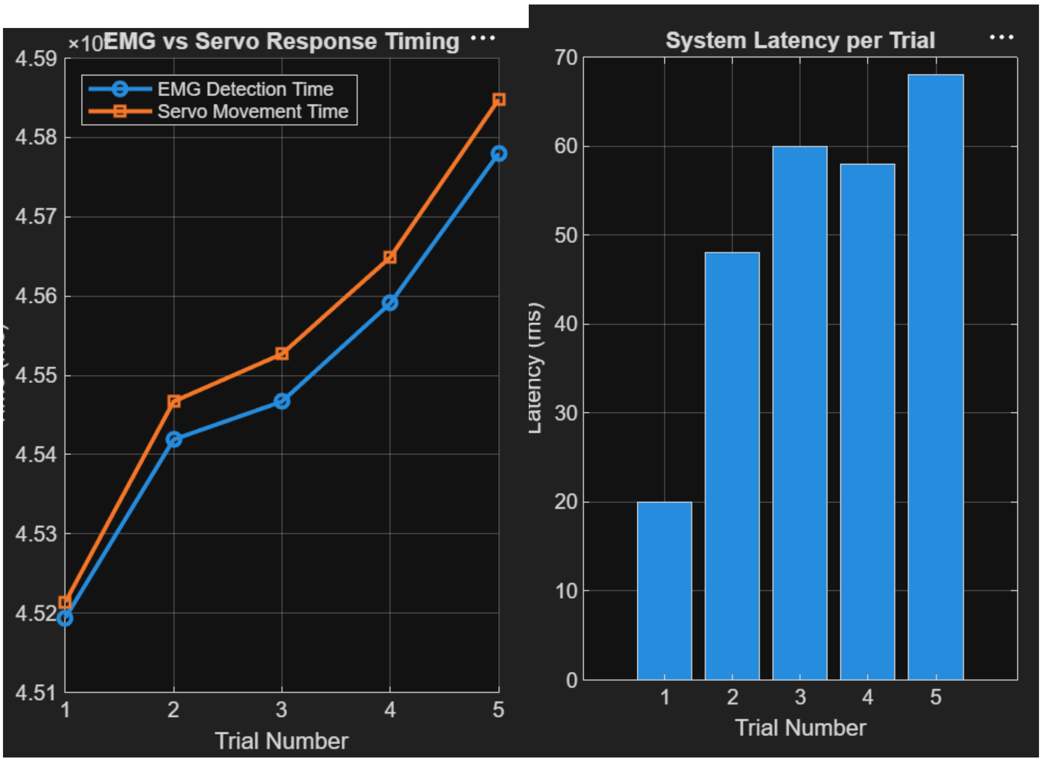


Figure 5.4: System internal latency Graphs

The results shown in the graphs in Figure 5.4 suggest a trend in increasing latency as the servo signals get sent to the actuators. The ESP32 processes variability and has to wait for the servos to mechanically settle before sending the next command. This is indicated by the first trial being the fastest as it did not need to wait for the previous command to end. This indicates that the system responds within 20ms to 70ms as real time capable. This is acceptable in a prosthetic setting as humans cannot distinguish latency below 100ms. This latency is acceptable but suitable only to a limit of 100ms where the response is now noticeable and the prosthetic starts to act as a perceived tool and not another limb.

5.1.3 Test 3 – Power Consumption During Prosthetic Arm operation

Here the power consumption test is carried out to evaluate the systems power consumption and requirements. The objective is to identify the power demand of individual subsystems and determine the maximum power consumption on the complete system during its full and partial operation. A 12V battery is used as the primary power source for the prosthetic arm. The current drawn by the components is measured as the system is running and performing various modes as shown in table 5.3. during each test only the required components for specific movements are activated. For simplicity, the ESP32, corresponding sensors are summed up as well. The total power consumption is measured using P =VI where P is the power consumption in Watts, V is the supply voltage at 12V and I is the total measure current in Amperes. For consistency each operation mode was done for a maximum of 10 seconds. The manipulated variable here is the operation mode of the arm while the responding variable is the total current consumption and power consumption. The controlled variables are the arms components and the time to do each movement.

The prosthetic arm shows its lowest power consumption at idle when no actuators are running and only the ESP32 and its sensor systems are active with being only 3W. Power consumption however increased drastically to 22.20W when the system engaged the hand close movement as 3 servos and the vibration motor were activated in this case. The trend from these results indicates that the increase in actuator use substantially increases total system power draw. Maximum system power draw was seen when every actuator was in full operation with the sensory subsystem at 49.80W. the indicates that the power distribution architecture of the system working to support all conditions. The measured power requirements remain within the capabilities of the selected 12V battery system to demonstrate its standalone capability.

Operation	Index Servo (mA)	Thumb Servo (mA)	Group Finger Servo(mA)	Wrist Servo (mA)	Elbow Servo (mA)	Vibration Motor (mA)	ESP32 & Sensors (mA)	Total Current (mA)	Power (W)	Energy (Wh)
Idle	0	0	0	0	0	0	250	250	3.00	0.0083
Hand Close	500	500	500	0	0	100	250	1850	22.20	0.0617
Hand Open	500	500	500	0	0	0	250	1750	21.00	0.0583
Hand Close +Wrist	0	0	0	800	0	0	250	1050	12.60	0.0350
Elbow Flexion	0	0	0	0	1500	0	250	1750	21.00	0.0583
Elbow Extension	0	0	0	0	1500	0	250	1750	21.00	0.0583
Heat Reflex Response	0	0	0	0	1500	100	250	1850	22.20	0.0617
Full System Operation	500	500	500	800	1500	100	250	4150	49.80	0.1383

5.1.4 Test 4 – Grip Repeatability

This test evaluates how consistently the arm can reproduce the same grip position when given the same command. This is done to measure the system overall consistency with producing grip positions with the results shown in Table 5.4.

Table 5.4: Grip Repeatability Results

Trial	Commanded Grip	Thumb Angle (°)	Index Angle (°)	Fingertip Gap (mm)	Repeatability Error (mm)
1	Full Grip	170	168	2.0	2.0
2	Full Grip	171	169	1.6	1.6
3	Full Grip	169	167	2.4	2.4
4	Full Grip	170	170	1.8	1.8
5	Full Grip	172	169	1.5	1.5

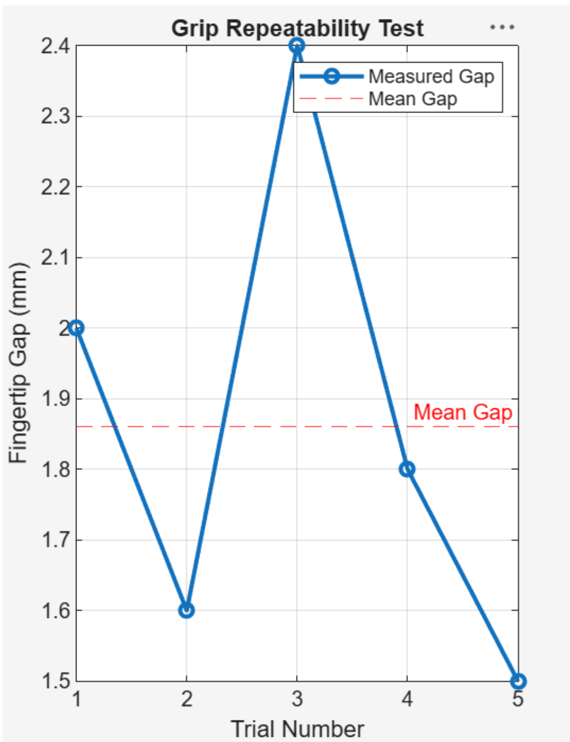


Figure 5.5 Grip Repeatability Graph

A full grip was executed 5 times and the resulting fingertip gap between the thumb and index finger were measured for each attempt. The fingertip gap ranged between 1.5 mm and 2.4mm with an average deviation of 1.86mm. this small variation shows that the servo driven fingers mechanism maintains a high level of positional consistency across repeated operations. Overall, the system demonstrates good repeatability, indicating that the prosthetic can reliably reproduce movements and in this case consistent grip positions. Minor discrepancies observed here are likely due to the tolerances from the servo manufacturers and their respect calibration settings. These differences can also be attributed to mechanical backlash in the finger joints and slight variations in PWM response from the ESP32.

5.1.5 Test 5 – Piezoelectric Surface Texture Sensing

To evaluate the ability of the piezoelectric sensor, this test is carried out in order to distinguish between different surface textures. The sensor works by analysing vibration patterns generated during sliding contact and thus produces a voltage based on the strength of these vibrations. Here the system uses vibration intensity and signal variability as indicators for surface roughness and or smoothness. The testing setup is as follows:

Surface type and conditions:

- Smooth (glass / plastic sheet)
- Medium rough (wood / matte plastic)
- Rough (sandpaper / fabric)
- Sliding speed (cm/s): 1 – 10 cm/s
- Applied normal force (N): 1 – 5 N

Dependant Variables:

- piezo signal RMS voltage (V RMS)
- Peak-to-peak voltage (Vpp)
- Signal variance (noise intensity)
- Vibration frequency content

Table 5.5 Piezoelectric Response Results

Surface Type	**RMS Voltage (V)**	**Vpp (V)**	**Signal Variance**	**Texture Index (0–10)**
glass	0.12	0.35	Low	1
Plastic	0.25	0.68	Low-Mid	3
Wood	0.48	1.20	Medium	6
Fabric	0.62	1.75	High	8
Sandpaper	0.85	2.40	Very high	10

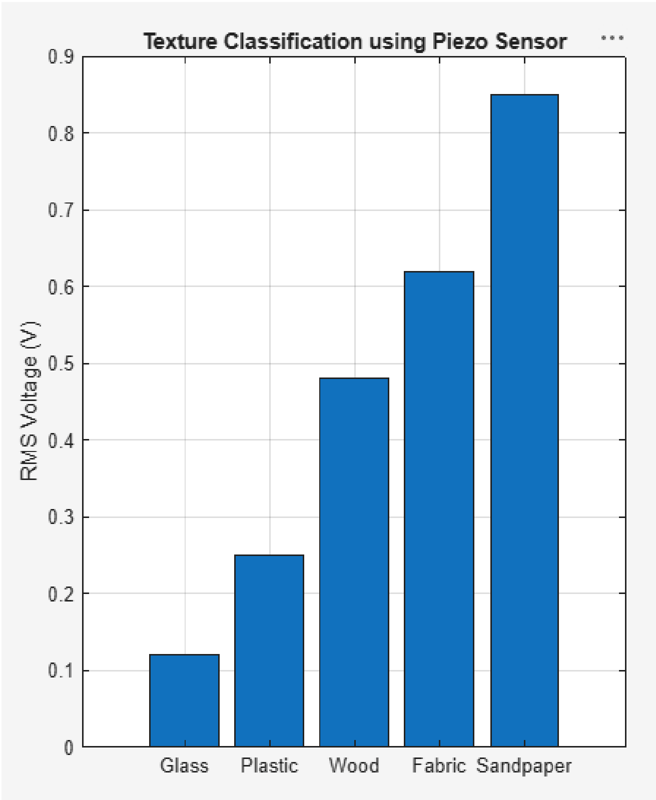


Figure 5.6: Piezo electric Response Histogram

The results shown in **Table 5.5** and **Figure 5.6** Indicate that as a surface gets rougher, the interaction between the piezoelectric and that material become irregular. The irregularity creates stronger vibrations which the piezoelectric then uses this data and converts it into higher voltage fluctuations. The system interprets this as roughness and determines the surface texture based on that.

As the results show, the smoothest surface being glass as it has almost no friction at micro level, thus the sensor slides across it without “catching” on any irregular rough patches. Its RMS value was the lowest at 0.12V and a low peak to peak of 0.35V. Fabric produced an RMS value of 0.62 as the sensor interacts with the fibres where it is re-engaging constantly causing higher variance and being interpreted as rougher than glass. Sandpaper produced the Highest RMS value of 0.85V and 2.4V peak to peak as the roughest surface. The results indicate that the piezoelectric can successfully distinguish between surface textures with it mapping their RMS produced value to a surface type.

5.2 Discrepancy between Theoretical and experimental Results

Throughout testing theoretical models made during the design phase were predicated to be a set ideal assumption. This included stationary EMG signals as well as them being free of any noise. These also included assuming the behaviours of rigid-body kinematics with the prosthetic joints and their respective actuators response characteristics. Experimental results revealed that these often deviated from the ideal predictions.

5.2.1 EMG signal Amplitude Variability

Theoretical EMG models assumed that the consistent muscle contractions from users would produce a stable EMG signal amplitude throughout testing. However experimental results showed discrepancies with this with trial values differing from expected values due to human physiology differences and muscle contraction intensity. These values are also skewed due to electrode placement variability. Basic skin conditions also played a role in these discrepancies as contact resistance increases and introduces more noise into the EMG signal. User fatigue also introduced more errors and weakened the EMG signal. These factors occasionally affected gesture detection accuracy and increased the variability in the servo response time and force.

5.2.2 Servo Motor Response Latency

For servo motor response, theoretical models and calculations assumed that their response would be immediate after an EMG signal detection. However, upon testing it was revealed that these showed additional delays caused by the ESP32 processing and PWM Signal generation. Weight from prosthetic joints and tendon driven mechanisms of the arm also increased mechanical load on the servo motors as well as friction within tendon routing caused minor delays during finger movement and joint operation. The variation in tendon stiffness and tension also played a part in this but nevertheless measured servo latency remained below acceptable response times for upper-limb prosthetic applications.

5.2.3 Piezoelectric Surface Texture Classification

For the Piezoelectric sensory feedback system, theoretical assumptions predicted that each surface would generate a unique vibration pattern in order to distinguish between them. However experimental testing showed some overlap between vibrations patterns and signals produced by surfaces with similar textures. It was found that sensor mounting positions and hand vibrations introduced additional noise into the system. Some texture samples also produced lower classification accuracy than predicted by simulation models. Overall classification still remained useable and successfully differentiated between smooth and rough surfaces.

5.3 Possible Sources of Error and Troubleshooting

5.3.1 EMG signal acquisition Subsystem

Errors caused by electrical noise from surrounding electronic devices was present throughout the testing of this system. It was found that something as trivial as a laptop charger not being grounded properly would introduce noise into the system when first testing. Inconsistent electrode placement on the muscles of users would create variabilities in signal strength to the EMG. Poor skin contact as well as muscle fatigue and motion artifacts would also dilute and create an unreadable signal for control.

To combat this a baseline high-pass filter was applied as well as a notch filter to remove any power-line interference. A low pass filter was then added to create a signal envelope and to remove any high frequency electromagnetic noise. A calibration system was then added to determine the cut off frequencies for these values were the system learnt from the user's physiology itself.

5.3.2 Mechanical Actuation

During operation of the arm servo motors would often create backlash causing positioning inaccuracies. This would be compounded by friction inconsistencies from joints and tendon guides within the arm itself and most notably the fingers. Variations in servo torque under load conditions also attributed to errors in this regard. Furthermore, misalignment od 3D-printed components also played a role in this. To aid this tendon tension with springs would constantly be monitored and adjusted regularly, servo calibration would be done before positioning and structures would be reprinted with reinforced materials to prevent structural bending and load.

5.3.3 Piezoelectric and temperature Sensing system

Unintentional vibrations generated from servo motor movement and actuator control would create inconsistencies in piezoelectric readings. Discrepancies in contact pressure from surface to surface would also create errors during texture sensing. Furthermore, the temperature sensing and response system would often lag during sudden temperature changes. To combat these sensors were mounted using more vibrationally damped materials such as foam padded double sided tape. During testing it was also clear that consistent pressure during surface detection was required for predictable values.

5.4 Comparison with Previous Research

The developed prosthetic arm has numerous improvements compared to the existing open-source prosthetic systems. The original Hackberry arm is primarily designed for simple hand actuation, whereas the proposed design incorporates active wrist articulation, integrated elbow movement and environmental sensing capabilities. Furthermore, the developed arm is different from many low-cost prosthetics that only use mechanical movement for actuation, by including an EMG-based control system, haptic feedback as well as thermal sensing to provide a more interactive experience for the user. The use of a calibration system also increases the adaptability by allowing individual users to configure movement mappings based on their own muscle activation patterns. These additional features contribute to increased functionality while preserving the affordability and accessibility advantages associated with 3D-printed prosthetic devices.

5.5 Sustainability Development and Environmental

The development of a low-cost EMG-controlled prosthetic arm aligns with the United Nations 2030 agenda for sustainable development by specifically addressing SDG3 (Good Health and Well-being), SDG 9 (industry, innovation and infrastructure) and SDG 10 (Reduced Inequalities).

5.5.1 SDG3 - Good Health and Well Being

This is addressed the through the systems capacity and intention to restore upper-limb function to amputees at the fraction of the cost of conventional solutions. Most prosthetics retails for between USD5000 AND USD 70,000 which is out of reach for most individuals. The WHO estimates that fewer than 10% of people requiring devices in low-income countries have access to them at all which is primarily due to the steep costs. However, this project has a bill of materials below USD150 with the proposed design delivering multi-grasp EMG control, active wrist and elbow movement as well as haptic and sensory feedback. These are capabilities absent in many commercially available prosthetics. This directly supports SDG3 on the universal access to such a device and making it more feasible to produce which further contributes to the holistic well-being outcomes.

5.5.2 SDG 9 — Industry, Innovation and Infrastructure

This is advanced through the project's reliance of 3D-Printing which enables for geographically distributed, on demand fabrication of patient specific components. This eliminates the need for centralised clinical infrastructure of specialised tooling. This is particularly useful in post-conflict zones or disaster affected areas where prosthetic demand rises and supply chains for them are normally disrupted. The system employs the use of commercially available microcontroller and sensor platforms which can easily be swapped out for what is available in that region and easily adapted for what components are available. It supports incremental upgrades and easy repair which an emphasis on resource efficiency. This project also contributes to the indigenous biomedical engineering capacity in Malaysia, reducing the need and dependency on imported prosthetic devices.

5.5.3 SDG 10 — Reduced Inequalities and SDG 12 – Responsible Consumption and Production

SDG 10 is addressed by removing the primary barrier required to obtain a prosthetic. With it being cost and geographic remoteness as well as dependency on clinically infrastructure. The onboard calibration routine enables users to independently configure the device without repeated prosthetist visits. This is critical for users in underdeveloped regions with limited access to advanced healthcare.

From an environmental perspective, the use of polylactic acid (PLA) as a main structural material is done due to its bio-derived properties. It is an industrially compostable thermoplastic. This reduces the reliance on petroleum derived polymers and plastics for its construction. FDM fabrication achieves material utilization efficiencies above 90% which significantly outperforms subtractive manufacturing methods. The arms parts are modular with a repairable design to further extends its lifespan to reduce waste which is consistent with SDG

12. It is however acknowledged that the electronic components constitute as e-waste at their end of life and future iterations should include a design for disassembly principle to address this limitation.

5.6 Moral Professionalism and Ethical Consideration

Throughout the Project Phase 2, moral and ethical considerations were made to be essential in its development and implementation. These are done to ensure both safety and efficiency of the project. These values are ensured to be upheld across various disciplines which include, safety, health, cultural and legal perspectives

5.6.1 Health and Safety

User health is paramount for any ethical obligations for biomedical development of assistive devices. The duty to do no harm is always kept in mind here ass the prosthetic arm interfaces directly with the residual limb of amputees. Any software or mechanical failure as well as electrical fault carry the potential risk of physical injury to a vulnerable population. Several design choices were made with this in mind. Servo actuator torque and angles were capped in software to prevent joint over extension to beyond safe levels. Voltage levels for sensors were kept at safe human interaction levels below 5V and the Heat reflex system was implemented to protect not only the user but the arm itself. Beyond physical Safety the arm carries positive health implications as well. The restoration of active limb prehension allows for the user to reengage in daily life activities while clinical evidence shows that this improved self-sufficiency enhances quality of life.

5.6.2 Cultural

For a prosthetic with an anthropomorphic design this was done to deliberately mimic natural finger morphology. It is done to show that cosmetically naturally looking prosthetics designs help improve users body image and social acceptance. This reduces the social visibility of the disability of amputation and facilitates reintegration into occupational and social roles. The 3D printed form factor also allows for skin-tone matched filaments to be used to further support cultural sensitive personalisation.

5.6.3 Social

The use of a prosthetic arm carries significant social implications. This is rooted in the principles of justice for equitable distribution of benefits to be given to people regardless of economic status or geographic location. The lower cost of the prosthetic compared to more conventional solutions is deliberate to ensure its outreach to users. The open-source design philosophy of it encompasses freely distributable fabrication files and firmware to further operationalise this social commitment by enabling replication and adaptation by engineers, clinicians and stakeholders without the barrier of finances and intellectual property.

5.6.4 Legal

The development and deployment of a biomedical device in Malaysia is subject to a defined legal and regulatory framework that imposes obligations on developers, manufacturers, and clinical deployers. Under the Medical Device Act 2012 (Act 737) administered by the Malaysian Medical Device Authority (MDA), any device intended for use in diagnosis, prevention, monitoring and or treatment of a medical condition must be registered prior to commercial supply or clinical use. The prosthetic arm falls within this scope. However, as a research prototype, the system is exempt from registration during the academic development phase. However, any transition toward clinical deployment or commercialisation would require the full MDA registration. Furthermore, the system employs data collection and is thus subject to data protection laws. As such all data collected as localised to that specific device. Data is not shared or sent to any other device and remains fully localised and central to the user.

5.7 Project Management, Finance and Entrepreneurship

5.7.1 Project Management

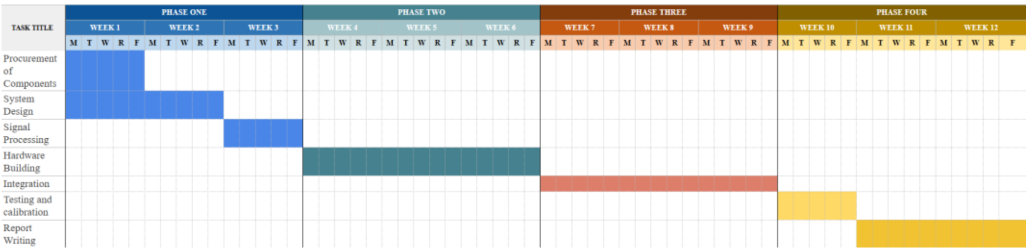


Figure 5.6: Phase 2 Gantt Chart

Based on the Gantt chart shown in **Figure 5.6**, the project is divided into four phases. Here the following are carried out: component procurement and system design, signal processing and hardware development, system integration and testing as well as report writing. The timeline is done across the 12 weeks to ensure a structured development process from the planning to the system final evaluation. Phase 1 covers procurement of all the components and the system design. During this, all required parts were sourced and bought then the overall system structure was designed. This included planning the EMG-based control system and defining how the prosthetic arm will function. Phase 2 focused on the signal processing and the hardware building part of the project. In this phase, EMG signal processing was developed which included filtering and preparation of control signals. At the same time, the actual physical prosthetic arm was built using servo motors and mechanical components which were also acquired during the procurement part. Phase 3 involves the integration, where the hardware and software are combined into one working system. This stage is done ensures that EMG signals correctly control the movement of the prosthetic arm.

Phase 4 focused on testing, calibration, and report writing. The system is tested to measure performance such as response time and movement accuracy. Adjustments are made to improve stability and reliability, while the final report is completed during the last weeks. Overall, the project follows a step-by-step development approach, where each phase depends on the completion of the previous one. This ensures smooth progress from design to a fully functional system.

5.7.2 Finance

Throughout this project, there has been a target that has been placed on creating a low-cost and accessible prosthetic using readily available components.

Component	**Quantity**	**Purpose**	**Estimated Unit Cost (RM)**	**Total Cost (RM)**
ESP32 Microcontroller	1	Main control unit and EMG processing	35	35
Surface EMG Electrodes	1 set	Non-invasive muscle signal acquisition	40	40
Servo Motors	7	Finger, thumb, and wrist actuation	25	175
PLA Filament (1 kg)	1	Structural and non-structural parts	80	80
PETG Filament (2 kg)	2	Load-bearing components	90	180
Piezo Sensor	1	Force/contact detection	5	5
NTC Thermistor	1	Temperature monitoring and safety	3	3
Vibration Motor	1	Vibrotactile sensory feedback	8	8
Miscellaneous (wires, connectors, fasteners)	—	Assembly and integration	20	20
Total Estimated Cost				**546 RM**

The Finance table highlights this by showing the total component costs of the prototype being a fraction of commercially available prosthetics. The cost breakdown of the prosthetic arm system shows that the total estimated development cost is RM546. This is covering all electronic, mechanical and supporting components required for full system functionality. The highest cost contribution comes from the servo motors, which account for RM175 in total due to the use of 7 units required for finger and wrist actuation. This is followed by PETG filament at RM180, which is used for load-bearing and structural components of the arm. While still reflecting the need for a stronger and more durable material in mechanically stressed parts of the design. PLA filament contributes a lower cost of RM80 as it is primarily used for non-structural or lightweight components.

On the electronic side of the project , the ESP32 microcontroller at RM35 and surface EMG electrodes at RM40 form the core of the control and signal acquisition system. This is done to enabling muscle-based input processing for prosthetic actuation. Additional sensing and feedback components, including the piezo sensor, NTC thermistor, and vibration motor, collectively contribute a small portion of the cost at just RM16, but are essential for implementing tactile sensing, safety monitoring and sensory feedback features. Miscellaneous hardware used such as wiring, connectors and fasteners added RM20 to support full assembly and integration. Overall, the cost distribution reflects a system that prioritises functional actuation and structural reliability, while still maintaining relatively low-cost sensing and control implementation which is suitable for a prototype-level prosthetic design.

5.7.3 Entrepreneurship

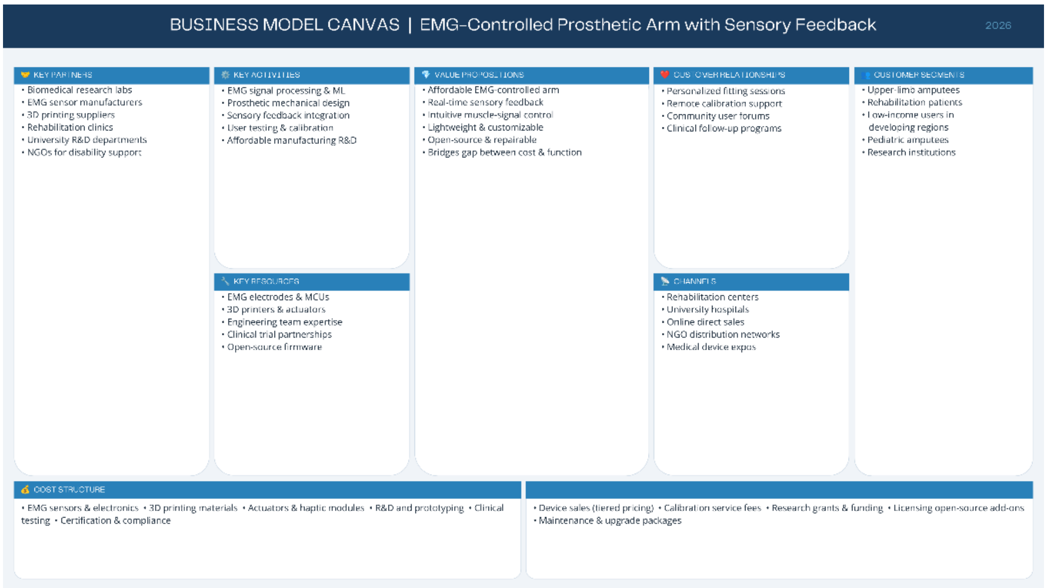


Figure 5.7 Business Canvas Model

The prosthetic arm business model canvas provides a clear pathway from technical prototype to a viable assistive arm with emphasis on affordability, accessibility and functional performance. The primary value proposition is a low-cost EMG controlled prosthetic arm with integrated sensory feedback, which will improve the quality of life for users who may not have access to expensive commercial prosthetics. Key customer segments for this are Amputees, Rehab Centres and Healthcare Providers. Key activities are Further Refinement of Product, User Testing and System Optimisation. Possible revenue streams for this device can include direct device sales, partnerships with hospitals and potentially working with NGOs or government healthcare programs. This can especially be done in developing regions where affordability is a concern.

5.8 Contribution in This Project

This projects creation involved the development, implementation and the evaluation of an EMG-controlled prosthetic arm. The Key contributions made in this project included the modification and adaptation of the Hackberry prosthetic arm in order to incorporate active wrist and elbow motion. The development of a personalised EMG calibration system for users was done as well as the integration of environmental sensing capabilities through piezoelectric and temperature sensors for involuntary movement control. In addition, an embedded control system was developed using the ESP32 to enable for real-time processing, actuation, and feedback without external computing required. The project also contributed to the exploration of low-cost prosthetic technologies capable of improving user interaction, functionality and accessibility.

CHAPTER 6
CONCLUSION & RECOMMENDATIONS

6.1 Conclusion relating to the Research objectives

The research objectives set out at the beginning of this project were addressed and successfully fulfilled. The development of a fully standalone ESP32 based prosthetic arm was made. The system integrated EMG signal acquisition, real time processing and multiple joint movement through the use of servo motors. The system also introduced and achieved sensory feedback to demonstrate the viability of a self-contained and low-cost prosthetic arm while still providing the user with useful sensory feedback. During development seven independent controllable degrees of freedom were achieved through the control of the fingers, wrist and elbow movement. The system also featured a personalised EMG calibration and training system which was delivered to the user through a web-based interface. The onboard calibration system uses the users specific contraction thresholds and stores them on the ESP32 to improve movement decoding accuracy and achieved an accuracy level of 90% and above. The haptic feedback system was implemented by the use of a piezoelectric sensory system to allow the user to distinguish between surface texture without the need for an external source to tell them apart. This was discreetly done through the use of a vibration motor. The heat sensing and reflex system was also implemented with the use of a thermistor and the activation of a heat protection system that reacts to heat at dangerous levels and protects the arm and user by reacting the same an involuntary movement would for a normal arm would. Experimental validation confirmed the systems reliable operation within the defined functional acceptance criteria. All performance metrics were verified to meet or approach their designed and expected targets demonstrating that the integrated system achieves its integrated system achieves its intended functionality.

6.2 Limitations

While the system did meet its defined objectives, several technical and operational limitations were found and constrains its full capability.

The use of a single channel EMG acquisition system restricts the ability simultaneous movement commands. With only 1 differential electrode channel the classifier is limited in the number of orthogonal movements intentions the user can do and it can resolve only 1 at a time. Furthermore, user physiological variability introduces inconsistent classification performance. Differences in users muscle anatomy and skin impedance mean that classifier performance is done during single user testing and may not generalise without recalibration which limits the systems plug and play capability.

The piezoelectric sensing system on provides single point vibration measurements, these can sometimes be insufficient to distinguish between surfaces and texture grades. Furthermore, the operational duration of the arm is restricted by the power budget allowed from servo actuation subsystem. The servo actuators themselves do not fully replicate the force-velocity and do not fully mimic biological muscle movement. They have fixed gear ratios and rigid position control and cannot produce variable levels of stiffness resulting in motion profiles that remain mechanically distinguishable from biological limb movements.

Validation of the arm was conducted under a control environment with a limited user cohort for testing. The lack of real-world trials across a diverse set of prosthetic users and amputees means that factors such as long-term mechanical wear and classifier drift over extended periods of time could be analysed and studied.

6.3 Recommendations and Suggestions for Further Research

To improve the current systems performance, the expansion to a multi-channel EMG setup is necessary. The addition here to distinguish between the multiple single channels would be furthered by the use of machine learning classifications. Incorporating this with the use of linear discriminant analysis or neutral networks would allow for resolving a rich set of simultaneous movement intentions from the user. Furthermore, the integration of resistive and capacitive fore sensing contact surfaces. This would allow for a close-loop grip force control system and would prevent object slippage and crushing and further improving the systems utility for manipulation of fragile and deformable objects. Moreover, exploration into lightweight actuator technologies including shape memory allows or pneumatic artificial muscles would ait in this project's further development. The actuators offer higher force to weight ratios and have a biological inspired characteristics compared to servo motors and represent a promise case for reducing limb mass while improving motion naturalness. To further improve the system performance, real world trials with clinical representatives would be necessary. Extending testing across a large and diverse group of amputees would aid in any prosthetics development let alone this one.

6.4 Summary

Chapter 6 concludes the project by evaluating the extent to which the defined research objectives have been achieved. It examines the limitations of the system and issues encountered during development and testing while still proposing a direction for improvements and future research. All primary objectives were successfully fulfilled with an EMG control accuracy exceeding 90% full control of all 5 fingers, wrist and elbow joints was also achieved. Furthermore, the systems sensory feedback system was implemented, and its limitations as a whole were addressed to further improve its performance in further research. The systems performance can be categorised as predictable and safe while the improvements suggested simply further its functionality to get as close as possible to a real biological arm.

REFERENCES

BOARD OF ENGINEERS MALAYSIA. (2016) *Code of Professional Conduct.* [Online] Available at: http://bem.org.my/code-of-professional-conduct

IEM. (2020) *Professional Conduct and Discipline*. [Online] Available at: https://www.myiem.org.my/content/professional_conduct_and_discipline-68.aspx

HAMAT, S., ISHAK, M.R., SAPUAN, S.M., YIDRIS, N., HUSSIN, M.S. AND ABD MANAN, M.S. (2023) Influence of filament fabrication parameter on tensile strength and filament size of 3D printing PLA-3D850. ScienceDirect.

CASTRUITA-LÓPEZ, J.F., AVILES, M., TOLEDO-PÉREZ, D.C., MACÍAS-SOCARRÁS, I. AND RODRÍGUEZ-RESÉNDIZ, J. (2021) Electromyography Signals in Embedded Systems: A Review of Processing and Classification Techniques. ScienceDirect.

SARIYILDIZ, E., HANSS, F., ZHOU, H., SREENIVASA, M., ARMITAGE, L., MUTLU, R. AND ALICI, G. (2023) Experimental Evaluation of a Hybrid Sensory Feedback System for Haptic and Kinaesthetic Perception in Hand Prostheses. ScienceDirect.

SALAZAR, M., ROSERO, R., ZAMBRANO, O. AND PORTERO, P. (2021) Impact of 3D printing technology for the construction of a prototype of low-cost robotic arm prostheses. ScienceDirect.

LESTARI, R. ET AL. (2024) Optimization of 3D printed parameters for socket prosthetic manufacturing using Taguchi method & RSM. ScienceDirect.

ALY, H. AND YOUSSEF, S.M. (2021) Bio-signal based motion control system using deep learning models (EEG + EMG). ScienceDirect.

ABD, M.A., INGICCO, J., HUTCHINSON, D.T., TOGNOLI, E. AND ENGEBERG, E.D. (2022) Multichannel haptic feedback unlocks prosthetic hand dexterity. SpringerLink.

BRACK, R. AND AMALU, E.H. (2021) A review of technology, materials and R&D challenges of upper limb prosthesis for improved user suitability. SpringerLink.

NGUYEN, S. AND MALMBERG, H. (2022) The effects of vibrotactile feedback on task performance in a 3D-printed myoelectric prosthetic arm. ScienceDirect.

BORTHAKUR, P.P. (2024) The Role and Future Directions of 3D Printing in Custom Prosthetic Design. MDPI.

AL AYYAD, R., AL KHALIFA, K., ALFARIS, R. AND ALHAIDARI, F. (2023) Electromyography Monitoring Systems in Rehabilitation: A Review of Clinical Applications, Wearable Devices and Signal Acquisition Methodologies. MDPI.

WHO. (2022) World report on assistive technology. Geneva: World Health Organization.

Mission-Arm. (n.d.). GitHub - mission-arm/HACKberry. GitHub. https://github.com/mission-arm/HACKberry

AGRAWAL, R. & BHAT, S. (2025) Development of a 3D-Printed EMG-Based Prosthetic System Using Additive Manufacturing and Embedded Microcontroller Control. IIT Bombay.

CERDAS, F., THODE, C., LAMPE, M., WIESNER, S. & HERRMANN, C. (2022) Life Cycle Assessment of 3D-Printed Products in Comparison to Conventional Manufacturing Methods. Journal of Cleaner Production.

DIAB, M., MOHAMMED, A. & JIANG, Y. (2024) Development of a Low-Cost Prosthetic Hand Using Electromyography and Machine Learning. arXiv.

OLDFREY, B., KUMAR, S., ALI, R., PATEL, M. & JONES, D. (2024) Digital Fabrication in Prosthetics: A Review of 247 Studies on CAD/CAM and 3D Printing Technologies. International Journal of Medical Robotics and Computer Assisted Surgery.

SAVAGE, S., MILLER, J., HARRIS, P. & NGUYEN, T. (2022) Global Ecosystem Analysis of 3D-Printed Prosthetic Development and Stakeholder Collaboration. Journal of Rehabilitation Engineering.

APPENDIX A

STUDENT’S ONLINE FYP SELECTION

Project Title Proposal

Project Title: EMG Controlled Prosthetic Arm for Intuitive Movement and Feedback		**Status**	APPROVED

Proposed By	ISAMELDIN AHMED HUSSEIN AHMED
Description	This project improves prosthetics by using muscle signals (EMG) to give users more natural and easy control of an artificial arm Aim: To build a low-cost, 3D printed prosthetic arm that uses muscle signals to move in a simple and intuitive way while giving the user useful information such as temperature of surfaces, texture and temperature. Objectives: To collect muscle signals with sensors, filter and process the signals as well as turn them into movement commands and finally test the arm for accuracy and usability and document findings from the study of EMG signals Targeted Users: People who have upper-limb loss who need a practical and affordable prosthetic arm for daily activities to help improve their quality of life
SDG	SDG3
Keywords	Intelligent Algorithms; Medical System; User-friendly interface; Good Health and Well-being
Preferred Supervisor(s)	IR. TS. DR. YVETTE SHAAN-LI SUSIAPAN, KRISHNA A/L RAVINCHANDRA, ASSOC. PROF. IR. DR. SIVA KUMAR SIVANESAN, TS. SURESH A/L GOBEE, IR. EUR. ING. TS. DR. LAU CHEE YONG
Assigned Supervisor	Shaan
Assigned Second Marker	sathish

Remarks

Name	Remarks	Date
IR. TS. DR. REENA SRI A/P SELVARAJAN	Good.	Oct 3, 2025, 2:15:09 AM
IR. TS. DR. REENA SRI A/P SELVARAJAN	Good.	Oct 3, 2025, 2:15:11 AM

[FYPPGBank] Reassignment of Supervisor and Second Marker Summarize this email

N no.reply@apu.edu.my
Dear Student, Please be informed that your title has been reassigned to new Supervisor or Second Marker by Project Manager. Kindy proceed to the system... Thu 5/21/2026 10:04 AM

I Ir. Ts. Dr. Reena Sri Selvarajan
Reply Reply all Forward
To: Assoc. Prof. Ts. Dr. Sathish Kumar Selva Perumal
Thu 5/21/2026 10:09 AM
Cc: ISAMELDIN AHMED HUSSEIN AHMED; Dr. Teow Hsien Loong

Dear @Assoc. Prof. Ts. Dr. Sathish Kumar Selva Perumal and @Dr. Teow Hsien Loong,

The following student is Dr. Yvette's student.

The second marker has been reassigned as the main supervisor (Dr. Sathish), and a new second marker has been assigned (Dr. Teow).

@ISAMELDIN AHMED HUSSEIN AHMED pls refer to the above lecturers for your FYP progress.

All the best!

Best Regards,

Ir. Ts. Dr. Reena Sri Selvarajan, ACPE, LSSBB, MIEM
Senior Lecturer,
FYP Manager (School of Engineering, APU)
Affiliate (Young Scientists Network of Academy of Sciences Malaysia, YSN-ASM)
LinkedIn (www.linkedin.com/in/reenasriselvarajan)

...

Reply Reply all Forward

APPENDIX B

PROJECT SPECIFICATION FORM

PROJECT SPECIFICATION/PROJECT BRIEF OF ENGINEERING

FYP Title: Prosthetic Arm with EMG-Based Control and Sensory Feedback

Supervisee Name: Isameldin Ahmed Hussein Ahmed

Supervisee TP Number: TP069561

Programme: Bachelor of Mechatronic Engineering with Honours

Supervisor Name: Ir. Ts. Dr. Yvette Shaan-Li Susiapan

Supervisor Signature:

Date: 20/11/2025

The following should be highlighted in the project specification /brief: (Form attached below)

A. Project Title.

B. Brief description on project background. (.i.e. introduction and problem statement)

C. Brief description of project aim and objectives. (i.e. scope of proposal)

D. Brief description of the system/model/design that will be used in this proposal.

E. Academic research being carried out and other information, techniques being learnt. (i.e. literature - what are the names of books you are going to read / data sets you are going to use)

F. Brief description of the materials/methodologies needed by the proposal. (i.e. data collection methods, motors, sensor and etc.)

G. Brief description of the evaluation and analysis proposed for this project. (i.e. testing, project deliverables and hypothesis, correlation test etc)

H. Illustration of how this project will benefit in the future.

Project Specification Form (Engineering)

Note: The PSF is an online submission. Use this form to discuss your proposal with your supervisor. Please complete all sections before meeting your supervisor so that relevant comments can be furnished.

A. Project Title.

Smart Prosthetic Arm with EMG-Based Control and Sensory Feedback

B. Brief description on project background. (i.e. Introduction about project and problem statement)

The loss of an upper limb can greatly affects an individual's ability to perform basic tasks and go about their daily life. The solution for this would be a prosthesis, however current prosthetic arms usually lack any form of sensory feedback and any form of reflex or involuntary movements. The entry price for prosthetics is also out of reach for the majority of people that need it.

Recent advancements in 3D modelling and 3D-Printing have allowed for the creation of more affordable and practical prosthetics for everyday use. This Project aims to address these issues by developing a low-cost smart prosthetic arm that will be controlled by EMG (Electromyographic) Signals with the added benefit of sensors to allow for feedback given to the user from these sensors. The sensors will provide feedback such as material texture and temperature. This will allow the arm to closely mimic natural human limb behaviour.

C. Brief description of project aim and objectives. (i.e. scope of proposal)

Aim: The Main is to Develop a 3D-printed prosthetic arm that can be controlled by EMG (electromyographic) signals that provides the user with sensory feedback.

Objectives:

1. Process EMG signals to control the five fingers, palm, and elbow with at least 70% accuracy.

2. Develop a feedback system that allows the user to identify surface roughness (smooth, medium, rough) and temperature (hot, warm, cold).

3. Adapt existing prosthetic arm designs to create a low-cost, 3D-printed model using affordable, off-the-shelf components.

4. Evaluate the prosthetic arm's movement accuracy, feedback performance, and task success rate during simple daily activities.

D. Brief description of the system/model/design that will be used in this proposal.

The proposed system will consist of a 3D-printed prosthetic arm that will be controlled by EMG signals from the user's muscles. A microcontroller will read the signals and move the fingers, palm and elbow joint. The arm will also include a simple feedback system that allows the user to feel surface roughness and temperature through vibrations and or sound ques. The design will be based on existing prosthetic models, but they will be adjusted and improved to make the arm low-cost. The system will then be tested to see how well it moves, how well the feedback system works and how well it performs simple daily tasks.

E. Academic research being carried out and other information, techniques being learnt. (i.e. literature - what are the names of books you are going to read / data sets you are going to use)

1. A Comprehensive Review of Myoelectric Prosthesis Control — Mohammad Reza Mohebbian et al., 2021
2. A 3D-Printed Bionic Hand Powered by EMG Signals and Controlled by an Online Neural Network — Karla Avilés-Mendoza et al., 2023
3. Functional Evaluation of a Real-Time EMG-Controlled Prosthetic Hand (ENRICH) — AJ Kalita et al., 2025
4. Electromyography Signals in Embedded Systems — JF Castruita-López et al., 2025
5. Detection, Identification and Removal of Artifacts from sEMG Signals: A Review — M. Ait Yous et al., 2025
6. Impact of 3D printing technology for the construction of a prototype of low-cost robotic arm prostheses — M. Salazar et al., 2024
7. Experimental Evaluation of a Hybrid Sensory Feedback System for Haptic and Kinaesthetic Perception in Hand Prostheses — Emre Sariyildiz et al., 2023
8. Contemporary and Future Development of 3D Printing Technology in the Field of Assistive Technology, Orthotics and Prosthetics — Hassan Beygi et al., 2024
9. The “Federica” Hand — Daniele Esposito et al., 2023
10. Experimental Evaluation of a Hybrid Sensory Feedback System for Haptic and Kinaesthetic Perception in Hand Prostheses — Emre Sariyildiz et al., 2024
11. A low-cost prosthetic arm with real-time haptic feedback — Emre Sariyildiz et al., 2024
12. Open-Source 3D Printing in the Prosthetic Field — The Case of Upper Limb Prostheses — Kevin Wendo et al., 2024
13. NinaPro (Non-Invasive Adaptive Hand Prosthetics Database) as Dataset

F. Brief description of the materials/methodologies needed by the proposal. (i.e. data collection methods, equipment's, testing and etc.)

EMG sensors and electrodes to read muscle signals. Microcontroller (ESP32/Raspberry Pi and Arduino) to process signals and control motors. Motors/actuators to move fingers, palm, and elbow. 3D-printed parts for the arm structure. Vibration motors for feedback on roughness and temperature. OpenEMG (by CGrassin) for signal processing and gesture control.

G. Brief description of the evaluation and analysis proposed for this project. (i.e. testing, project deliverables and hypothesis, correlation test etc)

Test EMG signal response for different gestures. Check tactile feedback for surface roughness (smooth, medium, rough) and temperature (hot, warm, cold). Perform daily activity simulations (grasping, lifting, manipulating objects). Measure motion accuracy, feedback reliability, and task success rate. Analyse data using accuracy, response time, and correlation tests.

H. Illustration of how this project will benefit in the future.

- Provides a low-cost, functional prosthetic arm option for people with limited financial ability to purchase more expensive options.
- Helps users regain hand and arm functionality, improving their independence in doing daily tasks.
- Demonstrates EMG-based control with feedback, which can be expanded in future designs.
- Can contribute to future studies on surface and temperature feedback for prosthetic devices

APPENDIX C

ETHICS FORM

Office Record	Receipt – APU Fast-Track Ethical Approval
Date Received:	Student name: Isameldin Ahmed Hussein Ahmed Student number: TP069561
Received by:	Received by: Date:

APU/APIIT FAST-TRACK ETHICAL APPROVAL FORM (STUDENTS)

Tick one box (level of study):

☐ POSTGRADUATE (PhD / MPhil / Masters)
☐ UNDERGRADUATE (Bachelors degree)
☐ FOUNDATION / DIPLOMA / Other categories

Tick one box (purpose of approval):

☐ Thesis / Dissertation / FYP project
☐ Module assignment
☐ Other: ____________________

Title of Programme on which enrolled …

Tick one box: ☐ Full-Time Study or ☐ Part-Time Study

Title of project / assignment …

Name of student researcher …

Name of supervisor / lecturer… …

Student Researchers- please note that certain professional organisations have ethical guidelines that you may need to consult when completing this form.

Supervisors/Module Lecturers - please seek guidance from the Chair of the School Research Ethics Committee if you are uncertain about any ethical issue arising from this application.

		YES	NO	N/A
1	Will you describe the main procedures to participants in advance, so that they are informed about what to expect?			✓
2	Will you tell participants that their participation is voluntary?			✓
3	Will you obtain written consent for participation?			✓
4	If the research is observational, will you ask participants for their consent to being observed?			✓
5	Will you tell participants that they may withdraw from the research at any time and for any reason?			✓
6	With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?			✓
7	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			✓
8	Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?			✓

If you have ticked No to any of Q1-8**,** you should complete the full Ethics Approval Form.

		YES	NO	N/A
9	Will your project/assignment deliberately mislead participants in any way?		✓	
10	Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort?		✓	
11	Is the nature of the research such that contentious or sensitive issues might be involved? This includes research which could induce psychological stress, anxiety or humiliation, or cause more than minimal pain.		✓	
12	Does your research involve the use of sensitive materials? Eg, records of personal or sensitive confidential information,		✓	

13	Does your research require external agency approval?		✓	
14	Does your research use hazardous or controlled substance?		✓	
15	Does your research require you to visit participants in their home or non-public space?			✓
16	Does your research use genetically modified organisms?		✓	
17	Does your research investigate illegal activities or behaviours?		✓	
18	Does your research involve discussion or collection of information on potentially sensitive, embarrassing or distressing topics, administrative or secure data? This includes research involving respondents through internet where visual images are used, and where sensitive issues are discussed		✓	
19	Does your research involve invasive or potentially intrusive procedures?		✓	
20	Does your research involve administration of substances?		✓	
21	Will your research be involved in the collection/ processing of human tissue samples		✓	
22	Will your participants be receiving financial compensation for participating in your research?			✓
23	Will your research data be used in the future after the conclusion of your project?	✓		
24	Will your research involve in processing sensitive data belonging to an organisation/persons?		✓	
25	Will your research be collecting photographs, videos, and audio recordings of the participants?			✓
26	Will the participants' personal particulars be known to any third party?			✓
27	Will the participants' data confidentiality be made known to the public?			✓
28	Will the research be conducted where the safety of the researchers maybe in question?		✓	
29	Will be the research be conducted outside of the UK and/or Malaysia?		✓	
30	Will your research involve human participants at premises other than those of the University?			✓

If you have ticked Yes to any of Q9 – 30**,** you should complete the full Ethics Approval Form. In relation to question 10 this should include details of what you will tell participants to do if they should experience any problems (e.g. who they can contact for help). You may also need to consider risk assessment issues.

			YES	NO	N/A
31	Does your project/assignment involve work with animals?			✓	
32	Do participants fall into any of the following special groups? **Note that you may also need to obtain satisfactory clearance**	Children (under 18 years of age)			✓
		People with communication or learning difficulties			
		Patients			
		People in custody			
		People who could be regarded as vulnerable or lack capacity to make decision for themselves			
		People engaged in illegal activities (eg drug taking)			
		Groups of people whose relationship among each other allow one to have influence over the other such as: Carers and patients with chronic conditions; teachers			

	from the relevant authorities	and their students; prison authorities and prisoners; employers and employees			
		Deceased person's body parts or other human tissues including bodily fluids (e.g. blood, saliva).			
		groups where permission of a gatekeeper is normally required for initial access to members.			
		Human participants who are off-campus APU staff or students who wish to carry out investigations involving human participants at premises other than those of the University			
33	Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?			✓	

If you have ticked Yes to any Q31-33**,** you should complete the full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU School Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.

STUDENT RESEARCHER

Provide in the boxes below (plus any other appended details) information required in support of your application. THEN SIGN THE FORM.

Please Tick Boxes

I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU School Research Ethics Committee.	✓
I am aware of APU liability policy and will make the necessary arrangement for insurance coverage of all researchers and participants of the project/assignment.	✓
Give a brief description of participants, procedure of recruitment and procedure of data collection (methods, tests used etc) in up to 150 words. N/A	
I also confirm that: i) All key documents e.g. consent form, information sheet, questionnaire/interview, and all material such as emails and posters for the purpose of recruitment of participants are appended to this application.	N/A
Or ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	N/A

Signed… … … … … … … … … … … Print Name… … … … … … … … … … … Date… … … …
(Student Researcher)

Within this document, any variation to the items considered which affects ethical issues of the stated research will require submission of a revised research plan and research methodology details; as a consequence, new ethical consent may need to be sought.

The completed form (and any attachments) should be submitted for consideration by your Supervisor/Module Lecturer

SUPERVISOR/MODULE LECTURER
PLEASE CONFIRM THE FOLLOWING:

	Please Tick Box
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU School Research Ethics Committee	✓
I have checked and approved the key documents required for this proposal (e.g. consent form, information sheet, questionnaire, interview schedule)	N/A

SUPERVISOR AND SECOND ACADEMIC SIGNATORY

STATEMENT OF ETHICAL APPROVAL (please delete as appropriate)

1) **THIS PROJECT/ASSIGNMENT HAS BEEN CONSIDERED USING AGREED APU PROCEDURES AND IS NOW APPROVED**

2) **THIS PROJECT/ASSIGNMENT HAS BEEN APPROVED IN PRINCIPLE AS INVOLVING NO SIGNIFICANT ETHICAL IMPLICATIONS, BUT FINAL APPROVAL FOR DATA COLLECTION IS SUBJECT TO THE SUBMISSION OF KEY DOCUMENTS FOR APPROVAL BY SUPERVISOR (see Appendix A)**

Signed… … … … … … … … … … … … Print Name Dr. Yvette Susiapan … Date 20/11/2025
(Supervisor/Lecturer)

Signed… … … … … … … … … … … … Print Name… … … … … … … … … … … … Date… … … …
(Second Academic Signatory)

Office Record	Receipt – Appendix A (APU Fast-Track Ethics Form)
Date Received: Received by:	Student name: Student number: Received by: Date:

APPENDIX A
AUTHORISATION FOR USE OF KEY DOCUMENTS

Completion of Appendix A is required when for good reasons key documents are not available when a fast track application is approved by the supervisor/module lecturer and second academic signatory.

I have now checked and approved all the key documents associated with this proposal e.g. consent form, information sheet, questionnaire, interview schedule

Title of project/assignment… Prosthetic Arm with EMG-Based Control and Sensory Feedback …

… ……………………

Name of student researcher … Isameldin Ahmed Hussein Ahmed … … … … … … … … ……….

Student ID: …TP069561………   Intake: …APD4F2509ME … … … ……..

Signed… … … … … … … … … … … … Print Name Dr. Yvette Susiapan … … … Date 20/11/2025 ……..
(Supervisor/Lecturer)